Unit 6 Overview: Dilations and Similarity

Unit Narrative

In this unit, students will build on their knowledge of transformations to work with sequence of transformations including rigid and non-rigid transformations. Students are expected to understand the effect of a scale factor on an image and understand the relationship of dilations to proportions. Students will use transformations to discover the definition of similarity and will prove triangle similarity. Students will have the opportunity to explore similarity in mathematical and real-world situations.

In Section 1, students focus on transformations of polygons. In Lesson 1, students build on 8^{th} grade knowledge of dilations and extend their learning to write algebraic descriptions of a dilation. Lesson 2 takes students through describing a sequence of transformations including dilations. In Lesson 3, students discover which transformations preserve distance on a coordinate plane. In Lesson 4 and Lesson 5, students explore how to describe a sequence of transformations and draw sequence of transformations using algebraic and written descriptions.

Section 2 has students connecting the relationships of the corresponding parts of similar polygons and justifying their reasonings through the formal proof of similarity. In Lesson 6, students define similarity using transformations. For Honors Geometry Course, students continue to Lesson H1 and Lesson H2, where they learn that all circles are similar and about proofs by contradiction. Students use proofs by contradiction to prove that any two circles are similar. In Lesson 7, students identify similar triangles and write similarity statements. Students prove triangle similarity in Lesson 8 and Lesson 9. This section concludes with Lesson H3 for Honors Geometry Course where students justify triangle similarity by using the definition of similarity.

Section 3 begins with Lesson 10, where students are introduced to finding missing angles and side lengths based on the definition of similarity. In Lesson 11 through Lesson 13, students use similarity to solve both mathematical and real-world problems.

Unit At-A-Glance

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<tr><td rowspan="4">MA.912.GR.2.1
MA.912.GR.2.2
MA.912.GR.2.3
MA.912.GR.2.5</td><td>Lesson 2</td><td>Sequence of Transformations That Include Dilations</td></tr>
<tr><td>Lesson 3</td><td>Preserving Distance</td></tr>
<tr><td>Lesson 4</td><td>Sequence of Transformations</td></tr>
<tr><td>Lesson 5</td><td>Drawing Transformations</td></tr>
<tr><td rowspan="7">Section 2: Proving Similarity</td><td>Benchmarks Addressed</td><td>Lesson 6</td><td>Similarity in Transformations</td></tr>
<tr><td rowspan="6">MA.912.GR.1.2
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MA.912.GR.2.8
MA.912.GR.2.9 (Honors only)
MA.912.GR.6.5 (Honors only)
MA.912.LT.4.8 (Honors only)</td><td>Lesson H1</td><td>Proving All Circles Are Similar</td></tr>
<tr><td>Lesson H2</td><td>Proofs by Contradiction</td></tr>
<tr><td>Lesson 7</td><td>Triangle Similarity</td></tr>
<tr><td>Lesson 8</td><td>Proving Triangle Similarity – Part 1</td></tr>
<tr><td>Lesson 9</td><td>Proving Triangle Similarity – Part 2</td></tr>
<tr><td>Lesson H3</td><td>Justifying Similarity</td></tr>
<tr><td rowspan="4">Section 3: Solving Using Similarity</td><td>Benchmarks Addressed</td><td>Lesson 10</td><td>Similar Figures</td></tr>
<tr><td rowspan="3">MA.912.GR.1.6</td><td>Lesson 11</td><td>Solving Problems Using Similarity – Part 1</td></tr>
<tr><td>Lesson 12</td><td>Solving Problems Using Similarity – Part 2</td></tr>
<tr><td>Lesson 13</td><td>Similarity in the Real World</td></tr>
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Differentiation

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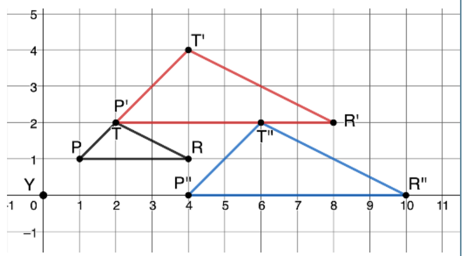
Engagement
The "WHY" of learning

- Throughout the unit, students are exposed to similarity found around them. Whether through art or through real-world examples, the problems provide relevance and recruit interest with authentic and meaningful activities.
- As students progress through 6.6.2 Exploration, they are given an opportunity to foster collaboration and community. Students can engage with the content and explore each question in teams, pairs, or whole-class collaborations. It may be effective to have stations where students physically move to work on each question throughout the lesson. This way, other students who are working on the same thing can offer tips and advice on how to solve parts they already have mastered.

Representation
The "WHAT" of learning

- The graphs of the sequences of transformations along with the algebraic description for each component can help students who need a visual of the graph to assist with thinking through the questions. An auditory element is added when students hear the written representations read aloud while demonstrating each component within each problem's graph. Consider using multiple colors to represent each transformation.
- Lesson 8 is designed to guide information processing and visualization. From the initial activation of prior knowledge to the opportunity for students to develop tools throughout each proof, students are regularly organizing information. Once organized, students are encouraged to share their models and methods of applications with others.



Action & Expression
The "HOW" of learning

- A variety of media can be used to express sequences of transformations throughout the unit. For example, in 6.3.2 Exploration, the use of patty paper is a great visual and kinesthetic way to connect the idea of transformations and preserving distance. You could also provide physical objects and spatial models to convey perspective. Graphing technology could also be introduced to visually show the transformations.
- The students can be engaged kinesthetically, starting with 6.5.2 Guided Instruction, by exploring the questions in a Gallery Walk structure. Additionally, an auditory element is gained from hearing team members reason through and evaluate the accuracy of each step.
- 6.4.2 Guided Instruction is a great opportunity to implement a Stronger Clearer Each Time routine. This will allow students to compose a stronger (better justifications) and clearer (more precise terms and linked, organized, complete sentences) response based on their partners' feedback.. Seeing various ways to express mathematical similarities from their peers will be important as students progress in their mathematical career.

ELL Information

Strongly encourage the use of graph paper and colored pencils for all transformations.

Require tables for organizing preimage coordinates and image coordinates.

Triangle PTR	
P	$(1,1)$
T	$(2,2)$
R	$(4,1)$

Triangle $P'T'R'$	
P'	$(2,2)$
T'	$(4,4)$
R'	$(8,2)$

Triangle $P''T''R''$	
P''	$(4,0)$
T''	$(6,2)$
R''	$(10,0)$

Consider using use three (or more) different colored pencils or markers, one color for each preimage and another for each successive image. This will help signal the preimage and each successive image for each sequence of transformation. (Suggest that the students adopt a similar convention; allow students to choose their own color schema).

Introduce patty paper and websites, such as GeoGebra, to demonstrate transformations.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from "Additional Differentiated Support" below:

- Dilation
- Proportional relationship
- Scale factor
- Similarity

Additional Differentiated Support

Scaffolding Support	On-Ramp Learning Tool
Example Activity: Use Geoboards and Geo Bands to create similar shapes. Example Activity: Give students a field of four shapes, where two shapes are the same shape, different size and two are completely different shapes. Students then must indicate which two shapes are similar. Example activity: Teacher provides a visual of a shape, then student replicates similar shapes using Play-Doh, clay, Geoboards, markers and paper, tape, etc.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Identifying Transformations • Scale Factors • Similar Triangles

Section Coherence

Section 1: Sequence of Transformations (Lessons 1-5)

Benchmarks Addressed

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

- *Clarification 1: Instruction includes the connection of transformations to functions that take points in the plane as inputs and give other points in the plane as outputs.*
- *Clarification 2: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 3: Within the Geometry course, rotations are limited to $90°$, $180°$ and $270°$ counterclockwise or clockwise about the center of rotation, and the centers of rotations and dilations are limited to the origin or a point on the figure.*

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.*

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to $90°$, $180°$ and $270°$ counterclockwise or clockwise about the center of rotation.*
- *Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.*

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Instruction includes two or more transformations.*

Learning Targets

Lesson 1	• I can write an algebraic description using coordinate notation of a dilation on a coordinate plane. • I can describe a dilation by giving the scale factor and center of dilation.
Lesson 2	• I can describe a sequence of transformations that includes a dilation. • I can represent a sequence of transformations using coordinate notation with algebraic descriptors.
Lesson 3	• I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance. • I can draw a sequence of transformations to determine if the transformation does or does not preserve distance. • I can identify transformations that do or do not preserve distance.
Lesson 4	• I can represent a sequence of transformations that will map the preimage onto the image. • I can describe a sequence of transformations that will map the preimage onto the image. • I can identify transformations that do or do not preserve distance. • I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.
Lesson 5	• I can draw the transformed figure when given a geometric figure and a sequence of transformations as a written description or an algebraic description.

<table>
<tr><th colspan="3">Section 2: Proving Similarity (Lessons 6-9, H1, H2, and H3)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="7">MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.
Clarification 1: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.
Clarification 2: Instruction focuses on helping a student choose a method they can use reliably.
MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.
Clarification 1: Instruction includes showing that the corresponding sides are proportional, and the corresponding angles are congruent.
MA.912.GR.2.9 (Honors only) Justify the criteria for triangle similarity using the definition of similarity in terms of nonrigid transformations.

MA.912.GR.6.5 (Honors only) Apply transformations to prove that all circles are similar.

MA.912.LT.4.8 (Honors only) Construct proofs, including proofs by contradiction.
Clarification 1: Within the Geometry course, proofs are limited to geometric statements within the course.</td><td>Lesson 6</td><td>I can map one similar figure onto another and justify that the figures are similar.
I can describe a sequence of transformations that will map the preimage onto the image.</td></tr>
<tr><td>Lesson H1</td><td>I can justify that two circles are similar and conclude that any circle can be mapped to another circle using similarity transformations.</td></tr>
<tr><td>Lesson H2</td><td>I can use proof by contradiction to justify that all circles are similar.</td></tr>
<tr><td>Lesson 7</td><td>I can identify similar triangles and write similarity statements.
I can use given information to show that two triangles are similar.</td></tr>
<tr><td>Lesson 8</td><td>I can use triangle similarity to prove triangles are similar.</td></tr>
<tr><td>Lesson 9</td><td>I can use triangle similarity to prove that triangles are similar.</td></tr>
<tr><td>Lesson H3</td><td>I can use the definition of similarity to justify that SSS similarity, SAS similarity, and AA similarity are sufficient to show that two triangles are similar.</td></tr>
</table>

<table>
<tr><th colspan="3">Section 3: Solving Using Similarity (Lessons 10-13)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="4">MA.912.GR.1.6
Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.
• Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.</td><td>Lesson 10</td><td>• I can solve mathematical problems using corresponding angles and sides of similar figures.</td></tr>
<tr><td>Lesson 11</td><td>• I can solve mathematical problems involving similarity in two-dimensional figures.</td></tr>
<tr><td>Lesson 12</td><td>• I can solve real-world problems involving similarity in two-dimensional figures.</td></tr>
<tr><td>Lesson 13</td><td>• I can solve real-world problems involving similarity in two-dimensional figures.</td></tr>
</table>

Vertical Alignment

Grade 8 Unit 9 Transformations	→	Geometry Unit 6 Dilations and Similarity

Building On	Grade Level	Working Toward
MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles. **MA.8.GR.2.1** Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship. **MA.8.GR.2.2** Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship. **MA.8.GR.2.3** Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.	**MA.912.GR.1.2** Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg. **MA.912.GR.1.6** Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. **MA.912.GR.2.1** Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. **MA.912.GR.2.2** Identify transformations that do or do not preserve distance. **MA.912.GR.2.3** Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. **MA.912.GR.2.5** Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane. **MA.912.GR.2.8** Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar. **MA.912.GR.2.9** *(Honors only)* Justify the criteria for triangle similarity using the definition of similarity in terms of nonrigid transformations. **MA.912.GR.6.5** *(Honors only)* Apply transformations to prove that all circles are similar. **MA.912.LT.4.8** *(Honors only)* Construct proofs, including proofs by contradiction.	

6.1 Dilations

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.</td><td>Learning Targets</td><td>I can write an algebraic description using coordinate notation of a dilation on a coordinate plane.
I can describe a dilation by giving the scale factor and center of dilation.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">What is a scale factor?
What is a center of dilation?
How do you write the algebraic description of a dilation?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, the students will be determining the scale factor of a dilation and writing an algebraic description using the scale factor. Connections will be made between using the points of the preimage and image to find the scale factor, and using the slope and length of sides of preimages and images.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>6.1.1 Warm-Up (~5 min.)</td><td>Drawing using point of perspective (SE p. 285)</td><td>6.1.4 Guided Instruction (10-15 min.)</td><td>Writing the algebraic description of a dilation (SE p. 288)</td></tr>
<tr><td>6.1.2 Refresher (5 min.)</td><td>Features of a dilation (SE p. 286)</td><td>6.1.5 Guided Practice (5-10 min.)</td><td>Practice writing the algebraic description of a dilation (SE p. 289)</td></tr>
<tr><td>6.1.3 Collaboration (20-25 min.)</td><td>Determining the scale factor of a dilation in multiple ways (SE p. 286)</td><td>6.1.6 Wrap-Up (~5 min.)</td><td>Students summarize their learning (SE p. 290)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
Center of Dilation
Scale Factor
Preimage
Image</td><td colspan="2">(Optional) Straightedge</td><td>Throughout this lesson, students will be exposed to multiple methods of determining the scale factor of a dilation. Make sure, as the teacher, you are proficient in each method so you can support students with each method.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

- Students may not know the difference between the image and the preimage.
- Students may incorrectly state the scale factor by swapping the numerator and the denominator of the ratio.
- Students may not correctly determine the slope of lines when using a different scale on the x-axis from the y-axis.

</td><td>

- Consider providing a quick refresher (no more than a couple of minutes) on determining the length of a line segment using the distance formula.
- Determine if your students need the support of all or part of the 6.1.2 Refresher.

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<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>

Anticipate, Monitor, Select, Sequence, Connect

In 6.1.3 Collaboration, students are asked to make connections between determining the scale factor from points on the preimage and image to finding the scale factor from the slope and from the length of a line segment. Implement the monitor and select part of this routine by selecting students for whole-group discussion that will result in students hearing the connections made from multiple sources.

</td><td>

MA.K12.MTR.2.1: Multiple Representations

Throughout the lesson, students are exposed to multiple methods of determining the value of the scale factor for a given dilation. Students are given opportunities to express connections between the methods so that they can make decisions on which method to choose given the context of the problem.

</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">

When determining student pairings for the 6.1.3 Collaboration, consider pairing confident students with students who need additional support. Set the expectation that each student will contribute and guide the other in making connections.

Students are first introduced to dilations and scale factor in Grade 8 Unit 9. Consider using this unit as a resource and additional entry-point for reviewing the content to prepare students for the content of the lesson.

</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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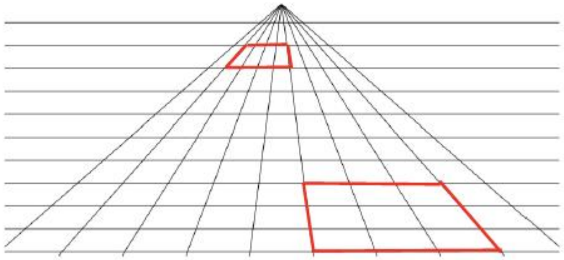
6.1.1 Warm-Up: What Is Your Perspective?

Component Overview: *Students observe how parallel lines in a three-dimensional drawing are represented using point perspective.*

6.1.1 Warm-Up: What Is Your Perspective?

Point perspective is a common method of drawing a three-dimensional object onto a two-dimensional plane. The artist chooses one point from which everything in the drawing is set. It was first discovered during the Renaissance. An example of a painting that uses point perspective is Vincent van Gogh's *Vincent's Bedroom in Arles*.

1. Use the grid to sketch a rectangle that is close to the point of perspective and a rectangle that is farther away from the point of perspective.



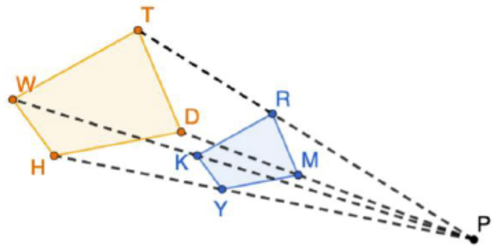
The one point the artist chose to draw everything from is considered the "point of dilation." This question allows for discussion of how point perspective is connected to dilations. If time permits, return to 6.1.1 Warm-Up at the end of the lesson to probe students about this connection.

6.1.2 Refresher: Dilations

Component Overview: *Students are given a word bank to scaffold their current knowledge of dilations and prepare them to apply the algebraic description of dilations. Take time, through discussion of the answers, to assess if students have a solid grasp of this prerequisite knowledge before moving into the lesson content.*

6.1.2 Refresher: Dilations

1. $KRMY$ was created by dilating $WTDH$. The ratio of WT to KR is $\frac{5}{3}$.



a. Use the words in the bank below to complete the statements that follow.

center of dilation pre-image	image scale factor

Point P is the __*center of dilation*___.

$KRMY$ is the _*image*_______________.

$WTDH$ is the __*pre-image*____________.

The ___*scale factor*_____ is $\frac{3}{5}$.

b. Use the diagram to determine each.

What is the ratio of $\frac{WH}{KY}$? $\frac{5}{3}$

What is the ratio of $\frac{PR}{RT}$? $\frac{3}{2}$

Consider asking the following:

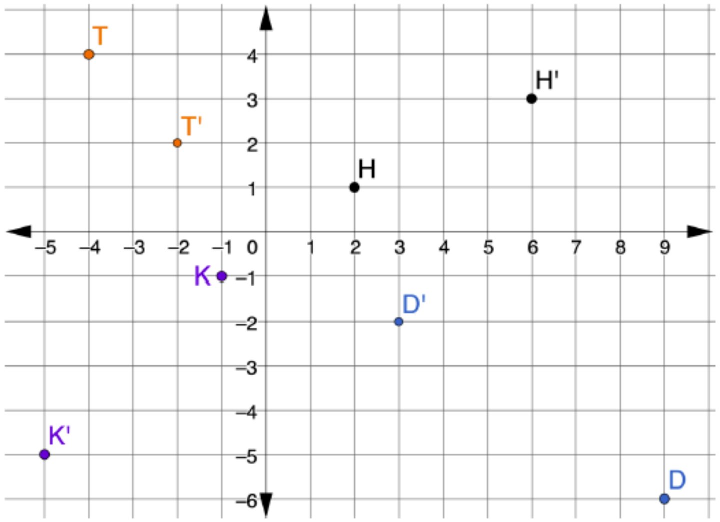
- *Why is it important to know where your center of dilation is?*
- *What is the difference between the image and preimage?*
- *What is a scale factor?*

6.1.3 Collaboration: Algebraic Description of a Dilation

Component Overview: *Students are introduced to the concept of dilations in the coordinate plane. Students will examine the positions of preimage points and image points to determine the scale factor and write an algebraic representation of the dilation.*

6.1.3 Collaboration: Algebraic Description of a Dilation

1. Points D, H, K, and T are shown on the coordinate grid. Each point has been dilated using $(0,0)$ as the center of dilation.



a. Complete the table.

Point	Ordered Pair	Ordered Pair of the Dilation	Scale Factor
D	$(9,-6)$	$(3,-2)$	$\frac{1}{3}$
H	$(2,1)$	$(6,3)$	3
K	$(-1,-1)$	$(-5,-5)$	5
T	$(-4,4)$	$(-2,2)$	$\frac{1}{2}$

In the coordinate plane, the algebraic representation of a dilation centered at the origin is $(x,y) \rightarrow (kx,ky)$, where k is the scale factor.

Begin this component by providing students an opportunity to make observations first, prompting them to make connections between the image and preimage.

- *What do you notice about the ordered pair and the ordered pair of the dilation?*
- *How does this relationship lead to finding the scale factor?*

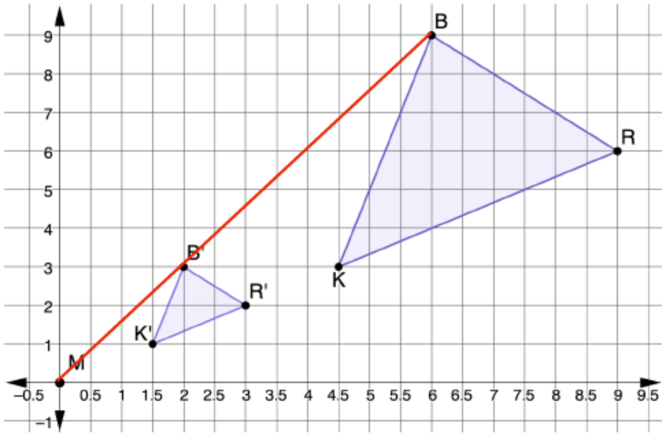
Each of these points are a different representation of a dilation. They are not connected to each other. Make sure to spend time addressing how to write the algebraic representation for each set of points.

6.1.3 Collaboration: Algebraic Description of a Dilation (continued)

b. Write the algebraic representation for each point.

Point	Algebraic Representation of Dilation
D	$(x, y) \rightarrow (\frac{1}{3}x, \frac{1}{3}y)$
H	$(x, y) \rightarrow (3x, 3y)$
K	$(x, y) \rightarrow (5x, 5y)$
T	$(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$

2. $\Delta R'B'K'$ is the result of dilating ΔRBK using the origin as the center of dilation.



a. What is the scale factor of the dilation?
The scale factor is $\frac{1}{3}$

b. Write the algebraic description of the dilation.
$(x, y) \rightarrow \left(\frac{1}{3}x, \frac{1}{3}y\right)$

c. Draw a line segment from point M that passes through B' and ends at B.
See drawing.

Each of these points represents the same dilation of the given figure. While monitoring, record which points students used to determine the scale factor. Call on these students to demonstrate how they determined the scale factor, making the connection that any corresponding points in the preimage and image can be used.

6.1.3 Collaboration: Algebraic Description of a Dilation (continued)

d. The table shows different relationships of the line segments MB, $B'B$, and MB'.

Line Segment	Slope	Distance
MB'	$\frac{3}{2}$	$\sqrt{2^2+3^2}=\sqrt{13}$
$B'B$	$\frac{6}{4}$	$\sqrt{6^2+4^2}=\sqrt{52}=2\sqrt{13}$
MB	$\frac{9}{6}$	$\sqrt{6^2+9^2}=\sqrt{117}=3\sqrt{13}$

e. Discuss with your partner how the scale factor could be determined using slope. Summarize your discussion.
Answers will vary. The scale factor can be found using the ratio of the rise for the slope of MB' to the rise of the slope of MB.

f. Discuss with your partner how the scale factor could be determined using distance.
Answers will vary. To find the scale factor using distance write the ratio of "the distance from the point of dilation to the image point" to "the distance from the point of dilation to the corresponding pre-image point". For example, using Point B' to Point B, this ratio is $\frac{\sqrt{13}}{\sqrt{117}}=\frac{\sqrt{1}}{\sqrt{9}}=\frac{1}{3}$.

Some students may understand a scale factor using the slope or distance. This may be helpful when students are drawing a dilation.

- *Why is using the slope or distance an acceptable method to use to find dilations?*
- *How can you use the slope or distance for finding dilations?*

Students do not need to write a response for question 2f. It is a discussion opportunity. A sample response is included for reference as teachers listen to student discussions.

6.1.3 Collaboration: Algebraic Description of a Dilation (continued)

3. The table shows different relationships among the line segments BR, $B'R'$, KR and $K'R'$.

Line Segment	Slope	Distance
BR	$-\frac{3}{3}$	$\sqrt{3^2+3^2}=\sqrt{18}=3\sqrt{2}$
$B'R'$	$-\frac{1}{1}$	$\sqrt{1^2+1^2}=\sqrt{2}$
KR	$\frac{3}{4.5}$	$\sqrt{3^2+4.5^2}=\sqrt{29.25}$
$K'R'$	$\frac{1}{1.5}$	$\sqrt{1^2+1.5^2}=\sqrt{3.25}$

Discuss with your partner how the scale factor is also found in the relationships between the slopes of BR and $B'R'$, and KR and $K'R'$. Write a summary of the relationships.

When writing the ratio of the numerators (rise) of the slopes in the image to the numerators (rise) of the slopes of the pre-image, you get a ratio of $\frac{1}{3}$. You also get the same ratio when writing the ratio of the denominators (runs). The ratio of the length of each line segment of the image to the length of the corresponding line segment of the pre-image is equal to the scale factor, $\frac{1}{3}$. For example, $\frac{B'R'}{BR}=\frac{\sqrt{2}}{3\sqrt{2}}=\frac{1}{3}$.

In question 2, the students examine the lines drawn from the point of dilation to corresponding points on the preimage and image. In question 3, the students examine the lines between points on a preimage and compare those to the lines between corresponding points on the image. Probe students to make sure they see the similarities and differences between these two questions.

Consider providing graph paper to provide a visual entry-point with question 3.

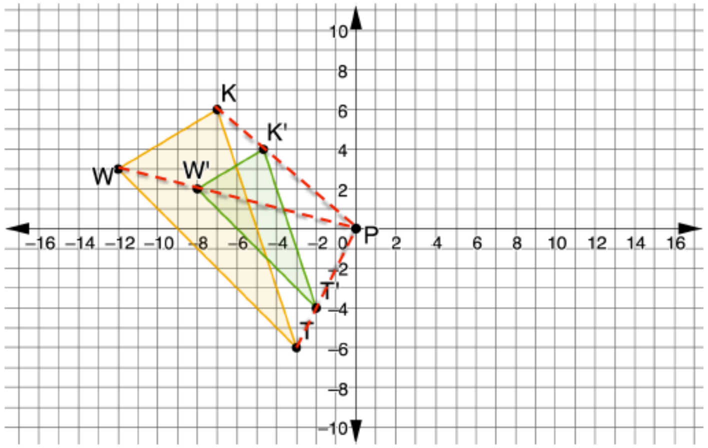
6.1.4 Guided Instruction: Describing Dilations

***Component Overview**: Students are guided through the multiple methods of determining the scale factor of a dilation. Exposing students to different methods equips them to make a choice between methods depending on how dilation information is provided.*

6.1.4 Guided Instruction: Describing Dilations

1. $\Delta K'M'T'$ is the result of dilating ΔKMT using the origin as the center of dilation. The vertices of ΔKMT are located at $K(-8,-6)$, $M(6,7)$, and $T(0,4)$. The vertices of $\Delta K'M'T'$ are located at $K'(-12,-9)$, $M'(9,10.5)$, and $T'(0,6)$. Write the algebraic description of the dilation.
   $(x,y) \rightarrow (\frac{3}{2}x, \frac{3}{2}y)$

2. Triangles TWK and $T'W'K'$ are shown on the coordinate grid, where $\Delta T'W'K'$ is a dilation of ΔTWK and point P is the center of dilation.



   a. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.
   See image.

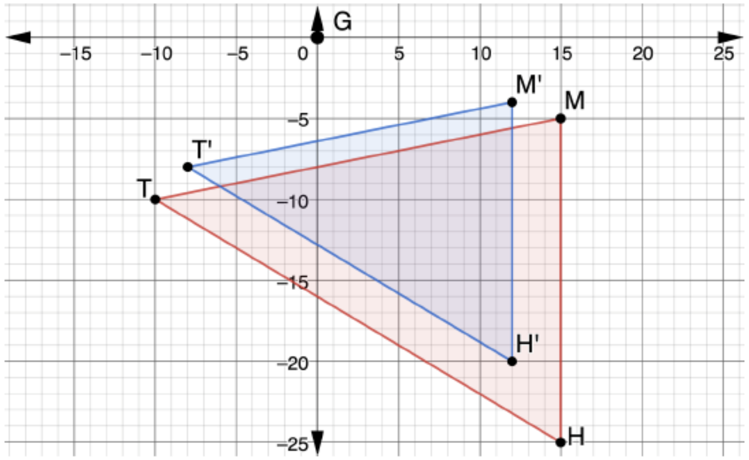
   b. What is the scale factor of the dilation?
   The scale factor is $\frac{2}{3}$.

   c. Write the algebraic description of the dilation.
   $(x,y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)$

Students may prefer one method of determining scale factor over another. For each of these questions, expose students to all methods for them to make decisions based on what they understand and how the information is given.

6.1.4 Guided Instruction: Describing Dilations (continued)

3. Triangles THM and $T'H'M'$ are shown on the coordinate grid, where $\Delta T'H'M''$ is a dilation of ΔTHM.



Write a description of the dilation by giving the center of dilation and the scale factor.

Answers will vary. A dilation of ΔTHM centered at point G (or the origin) using a scale factor of $\frac{4}{5}$ created $\Delta T'H'M'$.

Students can use a variety of ways to determine the scale factor. For students who are struggling to determine coordinates, consider suggesting that the students use GT' and GT. Equipping students with multiple methods of finding a dilation help students flexibly solve a variety of problems.

Explain how the collinearity of corresponding vertices shows the orientation of the shapes is the same. Explain that when the orientation of the shapes is not the same, but the shapes are similar, a transformation was applied before or after a dilation.

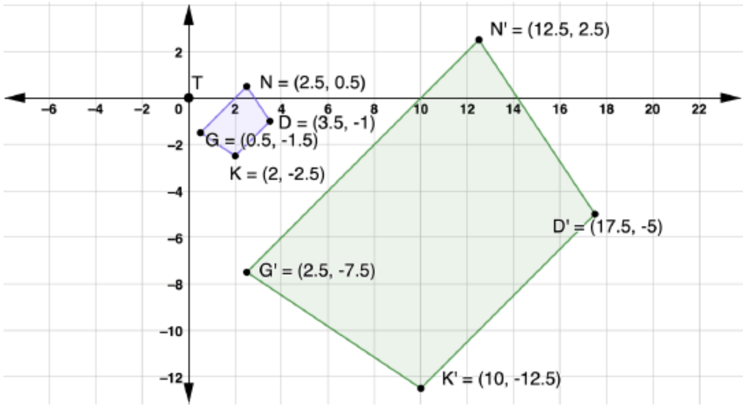
6.1.5 Guided Practice: Describing Dilations

Component Overview: *Students will describe dilations and write the algebraic representation of the dilation. Allow students to choose the method they prefer for determining the scale factor.*

6.1.5 Guided Practice: Describing Dilations

1. $\overline{M'T'}$ has coordinates $M'(-16, 20)$ and $T'(4, -8)$, and is the result of the dilation of $\overline{MT}$ centered at the origin. The coordinates of $\overline{MT}$ are $M(-4, 5)$ and $T(1, -2)$. Write the algebraic description of the dilation.
   $(x, y) \rightarrow (4x, 4y)$

2. $NDKG$ and $N'D'K'G'$ are shown on the coordinate grid, where $N'D'K'G'$ is a dilation of $NDKG$.



   Write a description of the dilation by giving the center of dilation and the scale factor.
   Answers will vary. The quadrilateral $NDKG$ centered at point T (or the origin) using a scale factor of 5 created the quadrilateral $N'D'K'G'$.

While students are working, monitor and make note of which methods students are using. When debriefing these questions, select students who used different methods to demonstrate their work.

To add a challenge, have students demonstrate how to find the scale factor using another method than the one they originally chose. This move will provide students with a way of analyzing the accuracy of their answers.

6.1.6 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

6.1.6 Wrap-Up: Cool Down

1. Explain how dilations are different from rigid transformations, in your own words.
   Answers will vary.

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

6.2 Sequences of Transformations That Include Dilations

Lesson Introduction

Benchmarks	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	Learning Targets	• I can describe a sequence of transformation that includes a dilation. • I can represent a sequence of transformations using coordinate notation with algebraic descriptors.

	Building On	Working Toward
Benchmark Coherence	MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.	N/A

Essential Question	• How do you know if a sequence of transformations is commutative?
Lesson Narrative	In this lesson, students will be combining dilations with other transformations, such as translations and reflections. They will be working with a partner to write algebraic and written expressions for these combinations of transformations. Students will also investigate which transformations are commutative.

Component Summary	6.2.1 Warm-Up (~5 min.)	Determining an error with a dilation *(SE p. 291)*	6.2.3 Guided Practice (10 min.)	Writing an algebraic description for a sequence of transformations *(SE p. 294)*
	6.2.2 Guided Instruction (25-30 min.)	Describing a sequence of transformations *(SE p. 292)*	6.2.4 Your Turn! (5 min.)	Describing a sequence of transformations *(SE p. 294)*
	6.2.5 Wrap-Up (~5 min.)	Ticket Out the Door writing algebraic descriptions of transformations *(SE p. 295)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Transformation • Dilation • Translation • Reflection • Preimage • Image	• (Optional) Graph Paper • (Optional) Cut Outs of Preimages and Images	• N/A

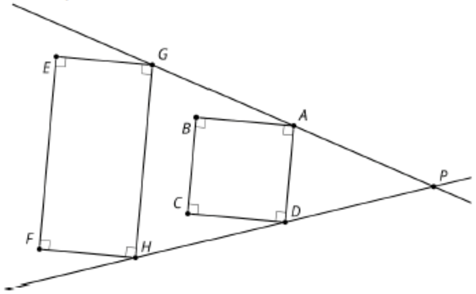
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not realize that some sequences of transformations are communitive and some are not.
Students may struggle getting the correct order of transformations to generate the same image from a preimage.</td><td>Consider spending a few moments reviewing the algebraic descriptions of the transformations learned in Unit 4 by encouraging students to locate those descriptions in their notes.
If time is an issue, then consider assigning 6.2.4 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
6.2.2 Guided Instruction is a natural place to implement this routine. The questions are designed to spark thinking for the individual learner first and then allow each student to make a conjecture about the answers to the questions. Then, students share with a partner to explain their reasoning, that can then be shared whole group.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In 6.2.2 Guided Instruction, students discover additional methods for determining the scale factor of a dilation. Students use structure to understand and connect mathematical concepts and to focus on relevant details within a problem. They also create plans and procedures to logically order events, steps, or ideas to solve problems. This allows them to be able to decompose a complex problem into manageable parts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">To provide a kinesthetic and visual element to the lesson, provide students with cut outs of the preimages and images in each of the components. Students can use the images to demonstrate the final position when following a sequence of transformations and when changing the order of the sequence to determine whether the transformations are commutative.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.2.1 Warm-Up: Dilation Miscalculation

Component Overview: *Students determine if an image demonstrates the characteristics of a dilation and provide a justification of their reasoning.*

6.2.1 Warm-Up: Dilation Miscalculation

1. Cassandra tried to determine if the diagram shown represents a dilation.
$EGHF$, $BADC$, $\overleftrightarrow{PG}$ and $\overleftrightarrow{PH}$ are shown.



Cassandra listed four characteristics of dilations:

Characteristic	Yes/No
Corresponding angle measures are equal	Yes
Corresponding sides are parallel	Yes
Corresponding vertices are collinear with the point of dilation	Yes
Orientation of the shapes are the same	Yes

a. Of the characteristics Cassandra listed, which are not true for $EGHF$ and $BADC$?
Corresponding vertices are collinear with the point of dilation.

b. Explain why the diagram shows that the dilation does not have these characteristics. If necessary, use the diagram to draw.
Not all corresponding vertices are collinear with the point of dilation. Points A and G are collinear with point P, as are points D and H. Points E and B are not collinear with point P and points F and C are not collinear with point P.

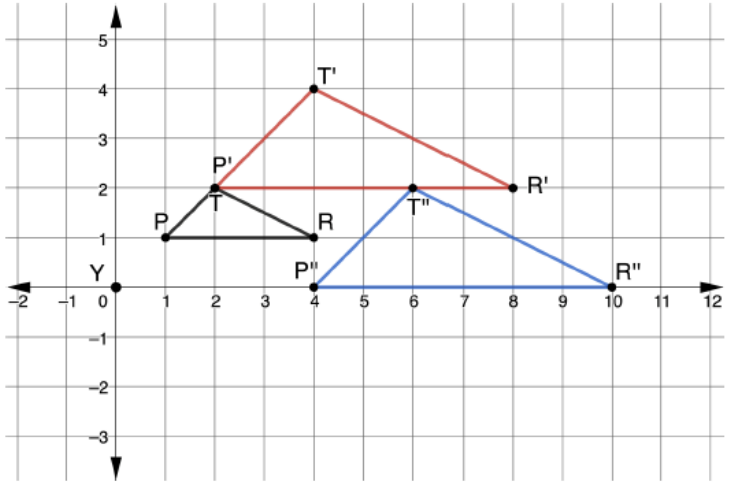
To add a level of complexity, have students make edits to the figures to demonstrate a true dilation.

6.2.2 Guided Instruction: Sequence of Transformations

Component Overview: *Students will describe a sequence of transformations and then determine if the transformations are commutative.*

6.2.2 Guided Instruction: Sequence of Transformations

1. A sequence of transformations of ΔPTR is shown on the coordinate grid. ΔPTR is dilated then translated, where $\Delta P'T'R'$ is the dilation and $\Delta P''T''R''$ is the translation of $\Delta P'T'R'$.



ΔPTR	
P	$(1, 1)$
T	$(2, 2)$
R	$(4, 1)$

$\Delta P'T'R'$	
P'	$(2, 2)$
T'	$(4, 4)$
R'	$(8, 2)$

$\Delta P''T''R''$	
P''	$(4, 0)$
T''	$(6, 2)$
R''	$(10, 0)$

a. Complete the description.

ΔPTR was dilated using a scale factor of _2__ centered at the origin. The resulting polygon, $\Delta P'T'R'$, was translated __*right 2, down 2*_ to create $\Delta P''T''R''$.

Think-Pair-Share
Consider employing the instructional routine Think-Pair-Share to allow students time to first think through each question, discuss their reasonings with their partner, and then bring their ideas to the whole group.

How can you determine if the scale factor will represent a ratio greater than one or less than one?

b. Complete the table.

Transformation	Algebraic Description
Dilation	$(x, y) \rightarrow (2x, 2y)$
Translation	$(x, y) \rightarrow (x + 2, y - 2)$

c. Determine the coordinates of each triangle if ΔPTR is first translated then dilated.

Pre-image

Triangle PTR	
P	$(1, 1)$
T	$(2, 2)$
R	$(4, 1)$

Translation

Triangle $P'T'R'$	
P'	$(3, -1)$
T'	$(4,0)$
R'	$(6, -1)$

Dilation

Triangle $P''T''R''$	
P''	$(6, -2)$
T''	$(8, 0)$
R''	$(12, -2)$

Some sequences of transformations are commutative, which means the transformations can be applied in any order.

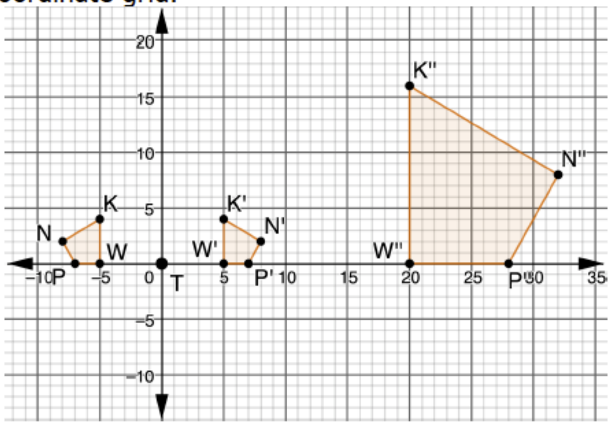
d. Use the tables to determine if the sequence of transformations applied to ΔPTR is commutative. Work with your partner to write an explanation.
The transformations are not commutative. When applying the sequence of transformations in a different order, the vertices for the two final images are not the same, so the images will not coincide.

Notice and Wonder
After completing question 1c, consider utilizing a Notice and Wonder routine to offer students the opportunity to make observations on when the transformations are applied in different orders.

Have students graph the points in the table of values for question 1c to have a visual representation of the resulting image when the transformations changed order.

6.2.2 Guided Instruction: Sequence of Transformations (continued)

2. $KWPN$, $K'W'P'N'$, and $K''W''P''N''$ are shown on the coordinate grid.



$KWPN$	
K	$(-5,4)$
W	$(-5,0)$
P	$(-7,0)$
N	$(-8,2)$

$K'W'P'N'$	
K'	$(5,4)$
W'	$(5,0)$
P'	$(7,0)$
N'	$(8,2)$

$K''W''P''N''$	
K	$(20,16)$
W	$(20,0)$
P	$(28,0)$
N	$(32,8)$

a. Discuss with your partner the first transformation that occurred. Describe how you determined the first transformation.
Answers will vary. The first transformation that occurred was a reflection. I know this because I see that the pre-image and image are reflected images of one another. I also know the algebraic description of a reflection over the y-axis is $(x,y) \rightarrow (-x,y)$.

b. Determine the scale factor in the second transformation.
The scale factor of the second transformation was 4.

Poll the Class
Consider using the instructional routine Poll the Class to begin a discussion on the transformation that occurred first. After the partner discussion, poll the class to determine what transformation occurred first. Call on different students to share their justifications for the transformation they chose.

c. Complete the description of the transformations.

$KWPN$ was _reflected across the y-axis_____ to create $K'W'P'N'$. $K'W'P'N'$ was dilated using a scale factor of _4_ centered at __Point T___ to create $K''W''P''N''$.

d. Complete the table.

Transformation	Algebraic Description
Reflection across y-axis	$(x, y) \to (-x, y)$
Dilation	$(x, y) \to (4x, 4y)$

e. Work with your partner to determine if the sequence of transformations applied to $KWPN$ is commutative. Complete the table by first applying the dilation, and then the other transformation. Then, write a conclusion based on the tables.

$KWPN$		$K'W'P'N'$		$K''W''P''N''$	
K	$(-5, 4)$	K'	$(-20, 16)$	K	$(20, 16)$
W	$(-5, 0)$	W'	$(-20, 0)$	W	$(20, 0)$
P	$(-7, 0)$	P'	$(-28, 0)$	P	$(28, 0)$
N	$(-8, 2)$	N'	$(-32, 8)$	N	$(32, 8)$

The sequence of transformations is commutative.

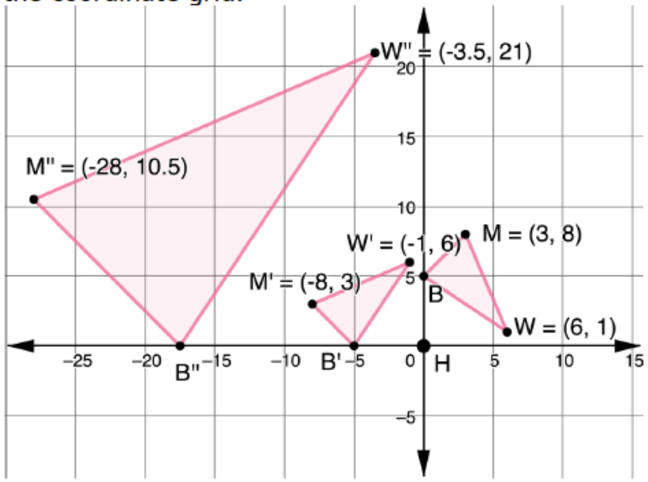
Provide opportunity for meaningful discussion as to if this sequence of transformations is commutative or not. When selecting a particular student to present their work, choose students who took different approaches to the problem. In fact, it may be advantageous to choose incorrect responses to highlight how and why they are incorrect. You can highlight a variety of responses or strategies or show a progression from simple to complex representation. Make sure that over time all students feel they are authors of the mathematical ideas presented.

6.2.3 Guided Practice: Mapping a Dilation

Component Overview: *Students will apply their knowledge of sequences of transformations by interpreting the graph, starting with the preimage, following through to the end of the mapping process, and writing the algebraic description.*

6.2.3 Guided Practice: Mapping a Dilation

1. A sequence of transformations of ΔMWB is shown on the coordinate grid.



a. Complete the table of values for each triangle.

Preimage | **Dilation**

Triangle MWB	
M	$(3, 8)$
W	$(6, 1)$
B	$(0, 5)$

Triangle $M'W'B'$	
M'	$(-8, 3)$
W'	$(-1, 6)$
B'	$(-5, 0)$

Triangle $M''W''B''$	
M''	$(-28, 10.5)$
W''	$(-3.5, 21)$
B''	$(-17.5, 0)$

b. Complete the table.

Transformation	Algebraic Description
Rotation 90° counterclockwise	$(x, y) \rightarrow (-y, x)$
Dilation	$(x, y) \rightarrow (3.5x, 3.5y)$

Allow informal observation throughout the first two components to dictate how "guided" this Guided Practice should be for you and your classroom. Consider doing small-group instruction for some while others attempt independently.

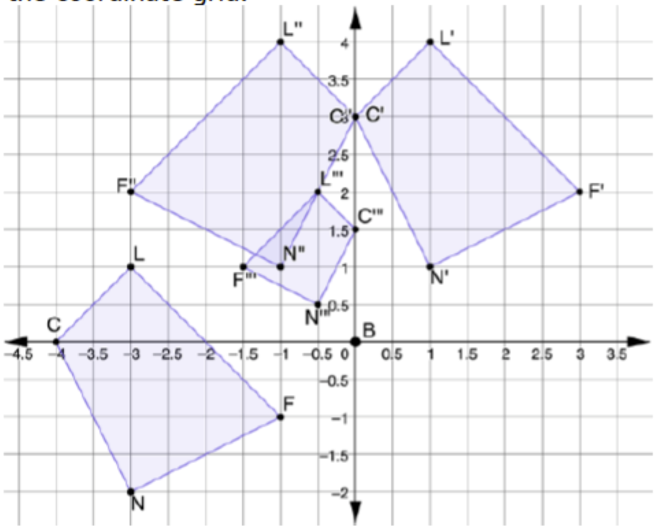
For students who finish early, challenge them to determine if the transformations in question 1 are commutative.

6.2.4 Your Turn!

***Component Overview**: Students will demonstrate their understanding of sequences of transformations by describing the transformations and then writing the algebraic descriptions.*

6.2.4 Your Turn!

1. $LFNC$, $L'F'N'C'$, $L''F''N''C''$, and $L'''F'''N'''C'''$ are shown on the coordinate grid.



   a. Describe the sequence of transformations of $LFNC$ that resulted in $L'''F'''N'''C'''$.
   $LFNC$ was shifted right 4 units and up 3 units to become quadrilateral $L'F'N'C'$. Then $L'F'N'C'$ was reflected across the y-axis to become $L''F''N''C''$. Finally, $L''F''N''C''$ was dilated from point B by a scale factor of $\frac{1}{2}$ to become $L'''F'''N'''C'''$.

   b. Write the algebraic descriptions of the sequence of transformations.

Transformation	Algebraic Description
Translation	$(x, y) \rightarrow (x + 4, y + 3)$
Reflection	$(x, y) \rightarrow (-x, y)$
Dilation	$(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$

Students may be more challenged now that they must apply and describe three transformations. Students might get stuck in getting the correct order of transformation to generate the same image from a preimage. The teacher can help the students by asking them to sketch their descriptions step-by-step using the algebraic description and see if they match the given image.

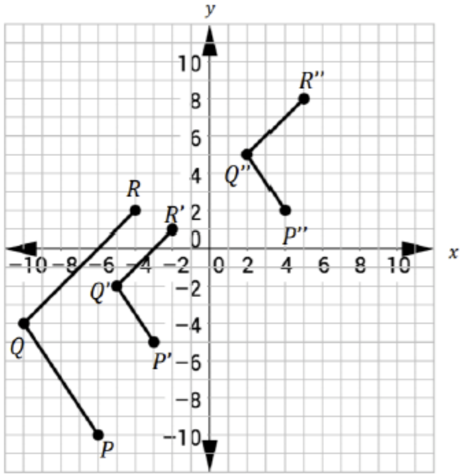
6.2.5 Wrap-Up: Ticket Out the Door

Component Overview: *6.2.5 Wrap-Up assesses students' abilities to describe the transformations and write an algebraic description.*

6.2.5 Wrap-Up: Ticket Out the Door

Consider using this exercise as a formative assessment to gather data on student progress for this lesson.

1. Points $P(-6,-10), Q(-10,-4)$, and $R(-4,2)$ are plotted on the coordinate grid to form figure PQR. Two transformations of figure PQR are shown.



Gladys first transformed figure PQR to $P'Q'R'$ using a dilation centered at the origin. She then transformed figure $P'Q'R'$ to $P''Q''R''$.
Complete the table.

Transformation	Written Description	Algebraic Description
PQR to $P'Q'R'$	*Dilation centered at the origin with a scale factor of* $\frac{1}{2}$	$(x,y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$
$P'Q'R'$ to $P''Q''R''$	*Translation of 7 units up, 7 units to the right*	$(x,y) \rightarrow (x+7, y+7)$

6.3 Preserving Distance

Lesson Introduction

<table>
<tr><td>Benchmarks</td><td>MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.</td><td>Learning Targets</td><td>• I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.
• I can draw a sequence of transformations to determine if the transformation does or does not preserve distance.
• I can identify transformations that do or do not preserve distance.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td>Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td>MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="3">• How do you know if a sequence of transformations does or does not preserve distance?
• What methods might you use to prove that the distance was preserved?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="3">This lesson explores sequences of transformation and whether the transformation preserved distance. Students will work with a partner to explore several quadrilaterals and their behavior under different transformations. They will “bring it together” through discovery of which transformation does or does not preserve distance.</td></tr>
</table>

<table>
<tr><td rowspan="3">Component Summary</td><td>6.3.1 Warm-Up (~5 min.)</td><td>Notice and wonder about a grid drawing (SE p. 296)</td><td>6.3.3 Bring It Together (5 min.)</td><td>Completing statements of transformations preserving distance (SE p. 299)</td></tr>
<tr><td>6.3.2 Exploration (15-20 min.)</td><td>Discovering transformations that preserve distance (SE p. 297)</td><td>6.3.4 Guided Instruction (10-15 min.)</td><td>Determining if sides are congruent after performing transformations (SE p. 299)</td></tr>
<tr><td>6.3.5 Wrap-Up (~5 min.)</td><td colspan="3">Reflection using KWL (SE p. 301)</td></tr>
</table>

<table>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td>Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
• Reflection
• Rotation
• Translation
• Dilation
• Congruent</td><td>• Patty Paper
• (Optional) Graph Paper
• (Optional) Digital Graphing Tool (such as Desmos or GeoGebra)</td><td>• Consider having the quadrilaterals from 6.3.2 Exploration question 3 graphed ahead of time if needed for the discussion.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle recognizing which type of transformations occurred from a grid.
Students may not recognize the correct transformation if presented only the coordinates of the preimage and image.</td><td>Consider using Anchor Charts to illustrate the different types of transformations.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
Before completing 6.3.3 Bring It Together, set students up to take turns in answering the questions. Set the expectation that each partner takes a statement and justifies their choices by referring to where it was discovered in 6.3.2 Exploration. This will build upon the expectation that good mathematicians seek answers in their texts and resources.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will use patterns and structure to help them understand and connect the concept of rigid and nonrigid transformations. They will focus on the relevant details within the problem to determine which are and are not.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The use of patty paper in this lesson is a great visual and kinesthetic way to model preserving distance with transformations. You could also provide physical objects and spatial models to convey perspective or interaction with this lesson as well. Use of graphing technology could additionally be introduced to visually show how the transformation was accomplished.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.3.1 Warm-Up: Notice and Wonder

***Component Overview**: Students are asked to write down things they notice and wonder about the drawing of Mary Hawtrey.*

6.3.1 Warm-Up: Notice and Wonder

Artist John Hoskins sketched a drawing of Mary Hawtrey, also known as Lady Mary Bankes, in 1821. Henry Pierce Bone then used the sketch to create an enamel miniature painting.

Following the death of her husband, Lady Mary Bankes successfully defended the Corfe Castle in Dorset, England during a siege in 1643. Unfortunately, during a second siege in 1646 the castle was destroyed.

1. What do you notice about the sketch of Lady Mary Bankes?
   Answers will vary. Students may notice the signatures below the illustration, the background drawing drawn in proportion to invisible point in the distance (e.g. individual picture is bigger, landscape is smaller).

2. What do you wonder about the sketch of Lady Mary Bankes or her story?
   Answers will vary. I wonder what she is holding in her hand and whether it has any significance.

After the students have had the opportunity to write down things they notice and wonder, ask several students to share their observations. You could record these for all to see. Steer the conversation towards a mathematical discussion. This low-stake activity gets students thinking and responding, leading them into the context of the lesson, and might pique their curiosity.

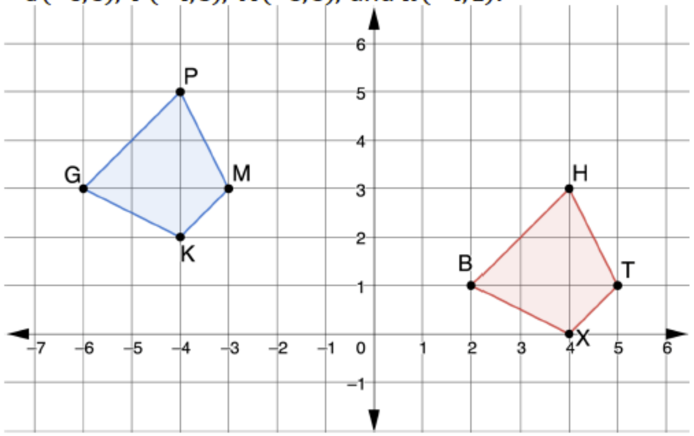
6.3.2 Exploration: Sequence of Transformations

Component Overview: *Students will work with a partner to determine what transformation(s) have happened in the questions. They will use patty paper to help connect the movement of the object and the transformation that was applied to it. They will then form a conjecture about the corresponding sides of the objects. Finally, they will confirm if distance was preserved after performing the sequence of transformations.*

6.3.2 Exploration: Sequence of Transformations

With your partner, answer the following questions.

1. $BHTX$ with vertices $B(2,1)$, $H(4,3)$, $T(5,1)$, and $X(4,0)$ is the result of a transformation of $GPMK$ with vertices $G(-6,3)$, $P(-4,5)$, $M(-3,3)$, and $K(-4,2)$.



   a. What transformation occurred to $GPMK$ that resulted in $BHTX$?
   $BHTX$ is a translation from the original image, it has been shifted 8 units to the right and 2 units down.

   b. Use patty paper to trace $GPMK$, then place $GPMK$ onto $BHTX$.

   c. What do you notice about the distances between the vertices of corresponding sides?
   The distances between the vertices of $BHTX$ are the same as the distances between the vertices of the original quadrilateral $GPMK$.

Students may find the use of the patty paper helpful for question 1a and for 1b. Consider having the image copied so that it can be manipulated to determine the transformation. This will provide a kinesthetic entry-point into the question.

d. What can be concluded about the lengths of the corresponding sides after this transformation occurs?
The side lengths will remain constant if the transformation is a translation.

2. The coordinates of $RWTY$ and $R'W'T'Y'$ are shown in the table, where $RWTY$ is the preimage.

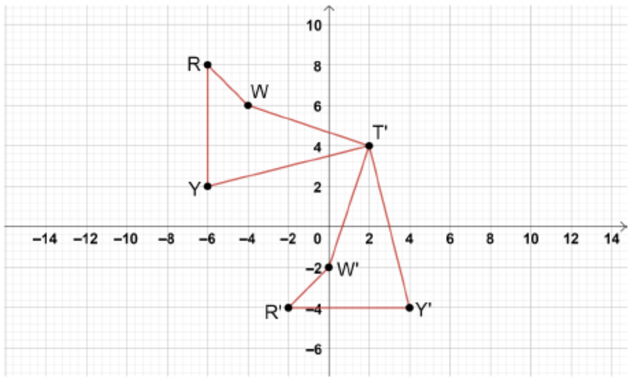
$RWTY$

Vertex	x	y
R	-6	8
W	-4	6
T	2	4
Y	-6	2

$R'W'T'Y'$

Vertex	x	y
R'	-2	-4
W'	0	-2
T'	2	4
Y'	4	-4

a. Use the coordinate grid to graph $RWTY$ and $R'W'T'Y'$.



b. What type of transformation was applied to the preimage, $RWTY$?
$R'W'T'Y'$ is a rotation of the preimage $RWTY$. The image has been rotated about point T $90°$ counterclockwise.

Collect and Display
Consider using the instructional routine Collect and Display after questions 1d, 2c, 3b, and 4b. Have students share their responses with the class, and display responses for all students to see. This will help students draw conclusions prior to 6.3.3 Bring It Together and help them make the connections about which transformations preserve distance.

6.3.2 Exploration: Sequence of Transformations (continued)

c. How could the coordinates of the preimage and image be used to show that the transformation preserved side length?
The coordinates can be used to find the distance of each side. When corresponding sides have the same distance, the transformation preserves side length.

3. The coordinates of $MPRD$ and $SGTW$ are shown in the table, where $MPRD$ is the preimage.

$MPRD$

Vertex	x	y
M	8	-7
P	9	-4
R	4	-2
D	1	-5

$SGTW$

Vertex	x	y
S	8	7
G	9	4
T	4	2
W	1	5

a. What transformation occurred from the preimage to the image?
There was a reflection across the x-axis. The x values have remained the same and the y values have been multiplied by -1.

b. Explain how the coordinates of the preimage and image show that the transformation preserved distance.
Using the distance formula, the coordinates on $SGTW$ demonstrate that the side lengths are the same as on $MPRD$, this shows that the transformation preserved distance.

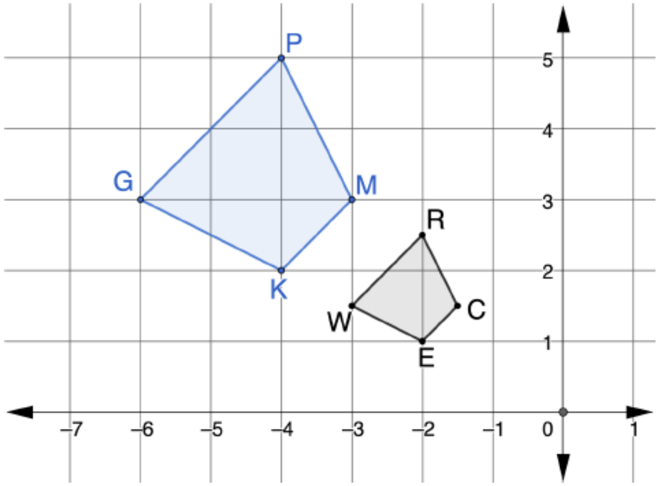
To encourage mathematical reasoning as students complete the transformations, prompt students with questions like:

- *What led you to the transformation you chose for this question?*
- *Could more than one transformation work in this question? Why or why not?*

Students may provide the answer to preserved distance using different words. Lead students to stating that the distance between the points remain the same. As this is a rotation, it would be less likely that students notice directly from the points.

Consider providing students with digital tools like GeoGebra or Desmos to complete question 3. This can provide kinesthetic and visual entry-points.

4. $WRCE$ is the result of a transformation on $GPMK$.



a. What transformation occurred to $GPMK$ that resulted in $WRCE$?
A dilation centered at the origin with a scale factor of $\frac{1}{2}$.

b. What can be concluded about the lengths of each corresponding side after this transformation occurs?
The lengths of corresponding sides are not the same.

Students who finish early could extend their thinking by creating a sequence of transformations of their own and exchanging with a partner to complete each other's problem.

6.3.3 Bring It Together: Preserving Distance

Component Overview: *Students will summarize the properties about transformations preserving distance by completing several statements.*

6.3.3 Bring It Together: Preserving Distance

1. Complete the statements based on the Exploration.

   a. $BHTX$ was created by
      - reflection.
      - translation.
      - rotation.
      - dilation.

      This transformation
      - does
      - does not

      preserve distance between the vertices.

   b. $KMPG$ was created by
      - reflection.
      - translation.
      - rotation.
      - dilation.

      This transformation
      - does
      - does not

      preserve distance between the vertices.

   c. $SGTW$ was created by
      - reflection.
      - translation.
      - rotation.
      - dilation.

      This transformation
      - does
      - does not

      preserve distance between the vertices.

   d. $WRCE$ was created by
      - reflection.
      - translation.
      - rotation.
      - dilation.

      This transformation
      - does
      - does not

      preserve distance between the vertices.

Take Turns
Before completing these statements, set students up to take turns in answering the questions. Set the expectation that each partner takes a statement and justifies their choices by referring to where this was discovered in 6.3.2 Exploration. This will build upon the expectation that good mathematicians seek answers in texts and resources.

This page of the lesson is a phenomenal resource for students going forward with transformations and preserving distance. Consider having your students bookmark this page for quick access.

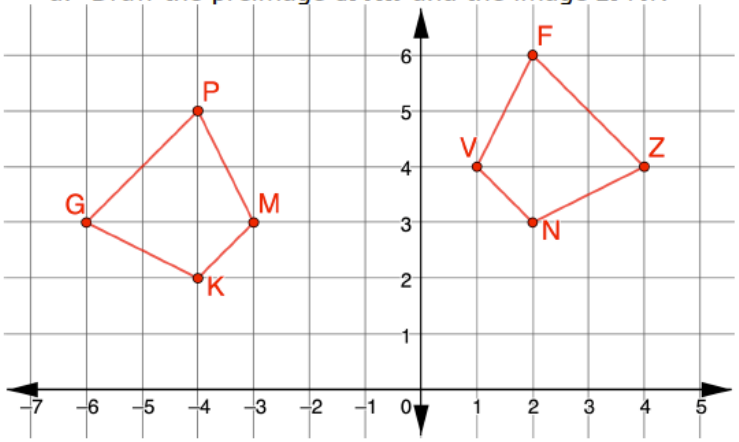
6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation

Component Overview: *Students will graph a preimage and image to determine what transformation has occurred and then justify whether the transformations preserve distance. Encourage students to use a variety of methods for determining congruence, such as patty paper and calculating the lengths of line segments.*

6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation

1. $GPMK$ with vertices $G(-6,3)$, $P(-4,5)$, $M(-3,3)$, and $K(-4,2)$ is reflected over $x=-2$, then translated right 2 units and up 1 unit to create $ZFVN$, respectively.

   a. Draw the preimage $GPMK$ and the image $ZFVN$.



   b. Use patty paper or the coordinates to determine if the corresponding sides of $GPMK$ and $ZFVN$ are congruent.
   The corresponding sides are congruent.

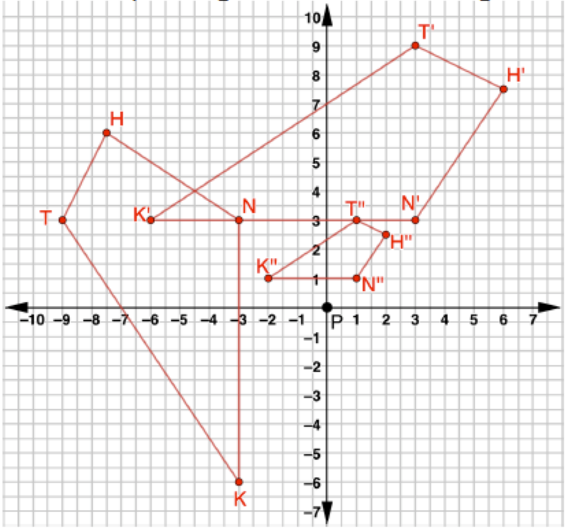
2. $THNK$ with vertices $T(-9,3)$, $H(-7.5,6)$, $N(-3,3)$, and $K(-3,-6)$ is rotated $90°$ clockwise about the origin, then dilated using the scale factor of $\frac{1}{3}$ centered at the origin to create $T''H''N''K''$, respectively.

Consider asking the following:

- *If you swapped the order of the transformations in question 1, would the result be the same image?*
- *What method did you use to determine if the corresponding sides were congruent? Why did you choose this method?*

6.3.4 Guided Instruction: Sequence of Transformations and Distance Preservation (continued)

a. Draw the preimage $THNK$ and the image $T''H''N''K''$.



b. Determine if the corresponding sides of $THNK$ and $T''H''N''K''$ are congruent.
No. The sides are not congruent.

c. Explain how you could use the description of a sequence of transformations to determine if the transformation does or does not preserve distance.
Answers will vary. You could use the description of the sequence to determine if distance would be preserved by using your knowledge of the types of transformations. The only transformation that changes side lengths is dilation.

d. How would your response differ if you were given the coordinates of a sequence of transformations?
Answers will vary. If I were given the coordinates of the transformations, I could use the distance between sides to determine if the distance was preserved. I also could use the points to draw the images and determine which transformations had occurred.

If students finish early, challenge them to determine if this sequence of transformations is commutative.

6.3.5 Wrap-Up: Reflection

***Component Overview**: Students complete a KWL self-reflection on the lesson.*

6.3.5 Wrap-Up: Reflection

1. Complete the table.
   Answers will vary.

The terms and information used in the lesson that I knew were …	Something I wondered during the lesson was …	Something I learned during the lesson was …

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.

6.4 Sequence of Transformations

Lesson Introduction

Benchmarks	MA.912.GR.2.2 Identify transformations that do or do not preserve distance. MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	Learning Targets	• I can represent a sequence of transformations that will map the preimage onto the image. • I can describe a sequence of transformations that will map the preimage onto the image. • I can identify transformations that do or do not preserve distance. • I can use a description of a sequence of transformations to determine if the transformation does or does not preserve distance.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.	N/A

Essential Questions	• What transformations occurred to map one shape onto another? • What transformations included in a sequence of transformations don't preserve distance? • How can the orientation of the vertex points on a figure and on the plane help determine the transformations that took place?
Lesson Narrative	This lesson continues exploring sequences of transformations that preserve distance. Students will continue to identify and describe sequences of transformations through written explanations and algebraic descriptions.

Component Summary				
	6.4.1 Warm-Up (~5 min.)	Would You Rather choice between driving speeds *(SE p. 302)*	6.4.3 Guided Practice (10–15 min.)	Error analysis for a sequence of transformations *(SE p. 305)*
	6.4.2 Guided Instruction (10–15 min.)	Describing and analyzing a sequence of transformations *(SE p. 302)*	6.4.4 Your Turn! (10–15 min.)	Describing and writing a sequence of transformations *(SE p. 306)*
	6.4.5 Wrap-Up (~5 min.)	Determining the sequence of transformations *(SE p. 306)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Transformation • Rotation • Reflection • Translation • Dilation	• (Optional) Patty Paper • (Optional) Copies of Preimages and Images	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may compare the rates in the 6.4.1 Warm-Up without writing them in common units.
Students may write the wrong algebraic description for given transformations.
Students may swap the order of the sequence of transformations because they think they are commutative.
Students may write a dilation as an enlargement by using a number greater than one instead of as a reduction by using a number less than one or vice versa.</td><td>Students may provide an algebraic description that matches an incorrect written description. Be sure to pay close attention to each response to determine remediation needs.
Some questions may have more than one answer. Analyze each student’s response for correct sequences.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Stronger and Clearer Each Time
6.4.2 Guided Instruction is an opportunity for students to revise and refine their ideas and their written output. Using a structured pairing strategy provides opportunities to clarify their responses and revise their original written response.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
6.4.3 Guided Practice is a collaboration opportunity for students. Students get to work together to review pre-created fictional student thinking. They identify and correct errors and justify their thinking comparatively, gaining important practice in this Math Thinking and Reasoning Standard.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Provide kinesthetic and visual entry-points by having students first take a copy of the preimage and physically demonstrate what sequence of transformations will result in the final image. Then students can move on to providing a written description and an algebraic description of the sequence of transformations.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.4.1 Warm-Up: Would You Rather?

Component Overview: *Students are given an opportunity to select between driving at two different rates. Regardless of which they choose, make sure students justify their choice with mathematics.*

6.4.1 Warm-Up: Would You Rather?

1. Would you rather drive a car at a rate of 40 miles per hour, or drive a car at a rate of 50 feet per second? Remember: There are 5280 feet in a mile.

   Explain your choice using mathematics.
   Answers will vary. I would rather drive 40 miles per hour. If I were driving 50 feet per second, I would be driving approximately 34 miles per hour, which would be slower.

Consider asking students if they would prefer to take a ride traveling 60 kilometers per hour? Sixty kilometers per hour is about 37 miles per hour. This could lead to a discussion about why we have different units of measure in different parts of the world and for different applications.

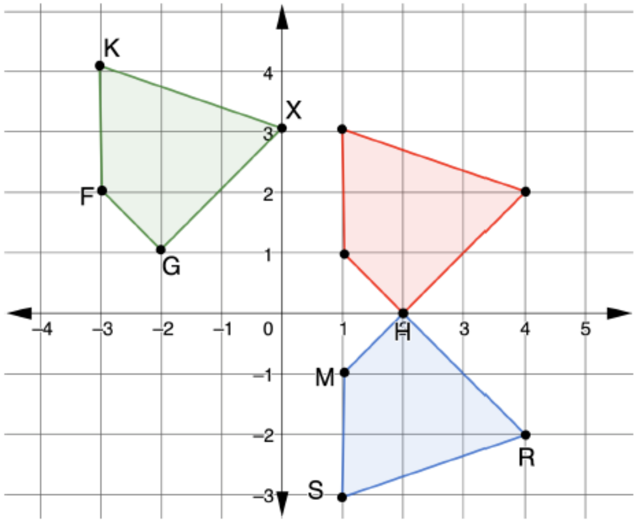
- *What would happen if you traveled to another country and a speed limit sign showed 70 with no units? What problems could this cause?*

6.4.2 Guided Instruction: Sequence of Transformations

***Component Overview**: Students identify and describe three different sequences of transformations to determine if distance was preserved in each. The questions are scaffolded to gradually move away from providing each individual transformation in the sequence. Encourage students to continue to look back at their notes for algebraic descriptions of the different transformations if they cannot fluently recall them.*

6.4.2 Guided Instruction: Sequence of Transformations

1. $XGFK$ is the result of a sequence of transformations of $RSMH$. The **red** quadrilateral is the first transformation.



a. Describe the first transformation of $RSMH$.
A reflection across the x-axis

b. Write the transformation algebraically.
$(x, y) \rightarrow (x, -y)$

c. Describe the second transformation of $RSMH$ that results in $XGFK$.
A translation left 4 units, up 1 unit

d. Write the transformation algebraically.
$(x, y) \rightarrow (x - 4, y + 1)$

Before moving to questions 1a-1e, have students read the question stem and discuss which figure is the preimage and which are the first and second transformations. This move will assist any students with reading difficulties or ELL students with determining where the sequence starts and where it ends. Students may also benefit from labeling the points in both transformed images with the same letters in the preimage to help determine which transformations occurred in the sequence.

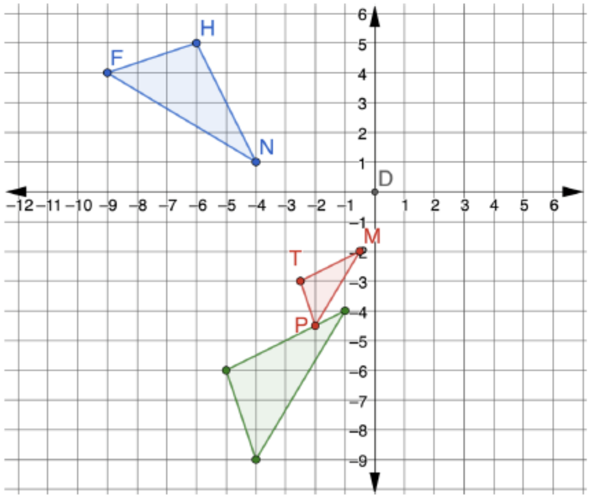
For question 1a and 1b, consider asking the following:
- *What clues in the image helped you determine which transformation occurred? (Possible answer: the vertex points $RSMH$ are clockwise in the preimage but counterclockwise in the image, indicating a reflection occurred.)*
- *What strategy helps you to remember the algebraic description of this transformation?*

For question 1c and 1d, consider asking the following:
- *What clues in the image helped you determine which transformation occurred? (Possible answer: the orientation of the vertex points $RSMH$ on the image and the orientation of the image on the plane did not change from the previous image, indicating a translation.)*
- *What strategy helps you to remember the algebraic description of this transformation?*

6.4.2 Guided Instruction: Sequence of Transformations (continued)

e. Explain if the sequence of transformations preserves distance.
Both the transformations in the sequence are rigid motions, so they preserve distance.

2. ΔMTP is the result of a sequence of transformations of ΔHNF.



a. Determine the sequence of transformations that will map ΔHNF onto ΔMTP. Complete the table.

	ΔHNF **Transformed to** $\Delta H'N'F'$	$\Delta H'N'F'$ **Transformed to** ΔMTP
Written Description	*Rotated 90° counterclockwise about the origin.*	*Dilated through point D with a scale factor of $\frac{1}{2}$*
Algebraic Description	$(x, y) \rightarrow (-y, x)$	$(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$

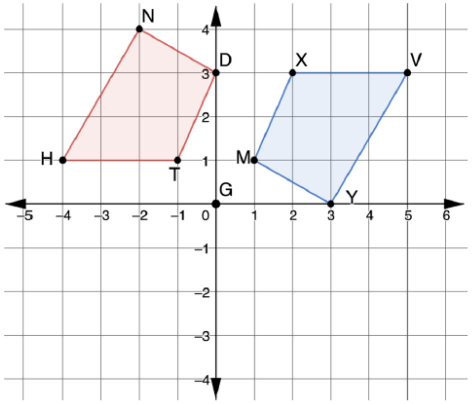
Stronger and Clearer Each Time
For a different strategy, consider putting students in groups of four. Without talking, each student writes the answer to question 1a. When finished, the students pass their answers to the student on their right. The students read the answer for question 1a, write or ask any clarifying questions, and pass it back. The students revise their answer to question 1a and then answer question 1b. The students then pass their answer to the person across from them. This student checks their work, writes or asks clarifying questions, and then passes it back. The students again revise or make any edits and then move to question 1c. This continues until all questions are answered, reviewed by a group member, and then edited/revised if needed.

For question 2a, consider asking the following:

- *What clues in the image helped you determine which transformations occurred? (Possible answer: for the first transformation, the orientation of the vertex points on the image stayed the same, but the orientation of the image on the plane changed from the previous image, indicating a rotation. For the second transformation, the length of the lines change or the image changed size, so I knew it was a dilation.)*
- *What strategy helps you to remember the algebraic description of these transformations?*

b. Explain how to determine if this sequence of transformations preserves distance.
The sequence did not preserve distance. You can tell because there was a dilation that reduced each side by a scale factor of $\frac{1}{2}$. This can also be verified by using the distance formula.

3. $TDNH$ is the result of a sequence of transformations of $XMYV$.



a. Determine a sequence of transformations that will map $XMYV$ onto $TDNH$. Complete the table.
Answers will vary.

	First Transformation	Second Transformation
Written Description	*Rotated $180°$ clockwise about point G.*	*Translated up 4 units and to the right 1 unit.*
Algebraic Description	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (x + 1, y + 4)$

b. Which transformations preserve distance?
Both transformations preserve distance. The rotation and translation both maintain side lengths equal to the pre-image.

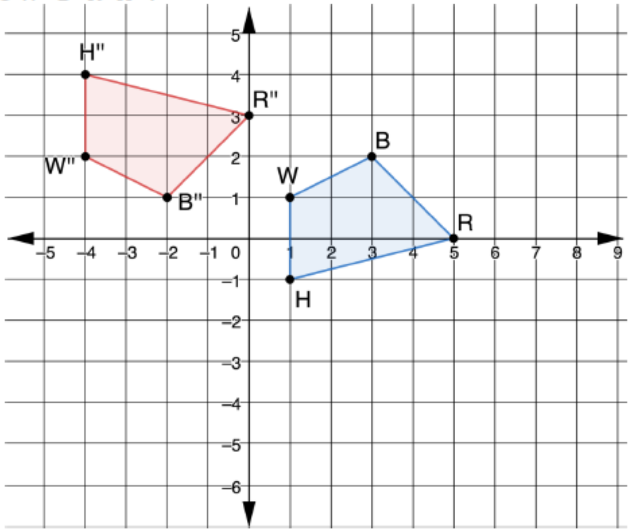
Encourage students to provide a more detailed justification if they respond, "because it is a dilation." Guide the conversation to include the vocabulary "preserved distance" and how using the distance formula or patty paper can verify this.

6.4.3 Guided Practice: Mapping a Given Preimage onto an Image

Component Overview: *Students consider a transformation and review two sample student works to determine which student determined the correct sequence of transformations that results in an image.*

6.4.3 Guided Practice: Mapping a Given Preimage onto an Image

$WBRH$ was transformed by a sequence of transformations to create $W''B''R''H''$.



Esmeralda and Wesley were determining the sequence of transformations and wrote their answers in the table.

	$WBRH$ **Transformed to** $W'B'R'H'$	$W'B'R'H'$ **Transformed to** $W''B''R''H''$
Esmeralda's Answer	Translated 5 units to the left and 3 units down	Reflected over the x-axis
Wesley's Answer	Translated 3 units down	Rotated 180° about the origin

1. Determine whose sequence of transformations is correct. Explain your reasoning.
   Esmeralda is correct because her transformations map WBRH onto W"B"R"H".

Students may use different methods to determine which of the two answers are correct. Some students may use patty paper and a copy of the preimage to manipulate. Others may test the vertex points to determine if they map onto the image. Have students who used different methods share their work with the class. Students benefit from seeing multiple methods and hearing the thinking of others.

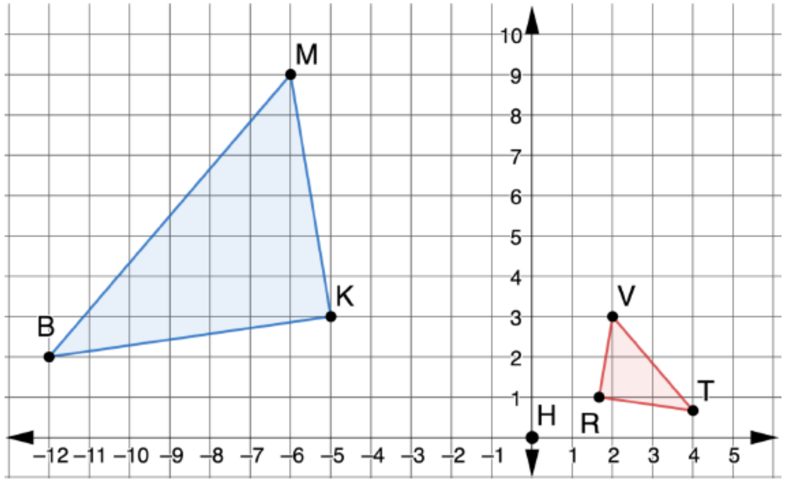
MA.K12.MTR.4.1: Discussion and Reflection
This is an opportunity for students to work collaboratively and explore the mathematical thinking and reasoning of others. Offering for students to work through question 1 with a partner or small group will allow them to explore potential errors and misunderstandings in a safe and effective way.

6.4.4 Your Turn!

Component Overview: *Students describe and write the algebraic description for a sequence of transformations given the preimage and final image.*

6.4.4 Your Turn!

1. ΔMKB was transformed to create ΔVRT.



a. Describe a sequence of transformations that will map ΔMKB onto ΔVRT.
Answers will vary. ΔMKB was reflected across the y-axis and then dilated by a scale factor of $\frac{1}{3}$ from the origin.

b. Write the sequence of transformations using algebraic descriptions.
Answers will vary.
First transformation: $(x, y) \rightarrow (-x, y)$
Second transformation: $(x, y) \rightarrow (\frac{1}{3}x, \frac{1}{3}y)$

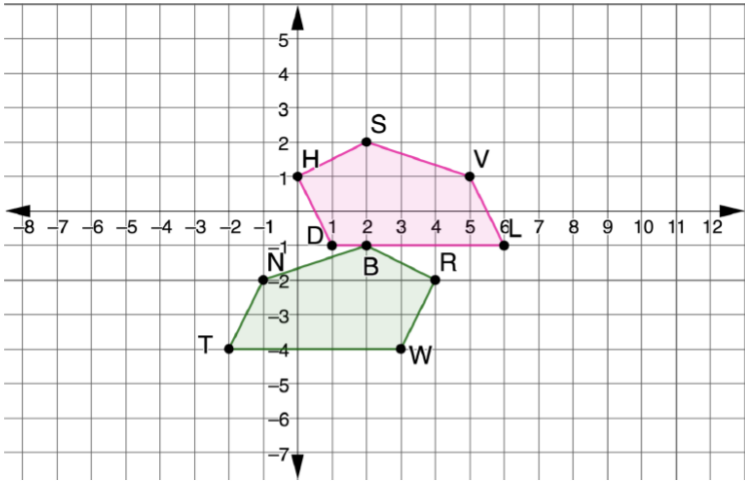
Consider having students record their work on a separate piece of paper without a name. When students are done, mix up the papers and redistribute to the class. The students should comment about the work on the paper. Comments should include both positive (I like the way you…) and constructive (While your first transformation is correct, your second transformation; You are correct in recognizing a dilation, but the scale factor you chose…). Post the results in a Gallery Walk style so anyone can walk by and gain advice without being identified as the one who made a mistake.

6.4.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students determine the sequence used to map a preimage onto an image in both written form and as an algebraic description.*

6.4.5 Wrap-Up: Ticket Out the Door

1. A sequence of transformations can map $NBRWT$ onto $VSHDL$.



Determine a sequence of transformations that will map $NBRWT$ onto $VSHDL$. Complete the table.
Answers will vary.

	First Transformation	Second Transformation
Written Description	*Reflected over the y-axis*	*Translated up 3 and right 4*
Algebraic Description	$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x + 4, y + 3)$

Use the answers to question 1 to determine remediation steps. If an incorrect written description is given, a student may need more practice with identifying transformations. If an algebraic description doesn't match the written description, students may need strategies (i.e. Foldable with key terms) for remembering the algebraic descriptions of each transformation.

Students may choose a different line of reflection, which will determine the units translated. Be sure to analyze each student's answer as a sequence of moves.

6.5 Drawing Transformations

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformation on a coordinate plane.</td><td>Learning Target</td><td>• I can draw the transformed figure when given a geometric figure and a sequence of transformations as a written description or an algebraic description.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">• How is applying transformations using a written description similar or different from using an algebraic description?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">This lesson builds on the idea that we can follow and perform a variety of transformations. Students will be asked to apply sequence of transformations to a given preimage by graphing the images using the written and/or algebraic descriptions given.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>6.5.1 Warm-Up (~5 min.)</td><td>Error analysis of a description of a sequence of transformations (SE p. 307)</td><td>6.5.3 Guided Practice (5-10 min.)</td><td>Applying a sequence of transformations given as a written description (SE p. 309)</td></tr>
<tr><td>6.5.2 Guided Instruction (25-30 min.)</td><td>Applying transformations given as written and algebraic descriptions (SE p. 308)</td><td>6.5.4 Your Turn! (5-10 min.)</td><td>Applying a sequence of transformations given as an algebraic description (SE p. 310)</td></tr>
<tr><td>6.5.5 Wrap-Up (~5 min.)</td><td colspan="3">Students summarize their learning (SE p. 310)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td>Materials</td><td>Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
• Transformation
• Dilation
• Rotation
• Reflection
• Dilation</td><td>• (Optional) Posters
• (Optional) Markers</td><td>• If using the Gallery Walk, posters may need to be made of each question ahead of time.</td></tr>
</table>

Teacher Tips

Common Misconceptions	Things to Consider
• Student may apply the rotation in 6.5.2 Guided Instruction around the origin instead of Point F. • Students may apply only two of the three transformations in 6.5.3 Guided Practice.	• For all transformations described using algebraic descriptions, consider having students name the transformation that will occur prior to applying the transformation.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Take Turns 6.5.2 Guided Instruction questions 1-3 can be presented as a Gallery Walk. During the work at the posters, students take turns in the activity by choosing a different group member to do all of the writing based on the group's discussion. This move requires all members to participate and listen to the discussion and prevents one student from taking over.	**MA.K12.MTR 2.1: Multiple Representations** The transformations in 6.5.2 Guided Instruction are stated in word description and in algebraic description. Including both of these allows students to make connections between the concepts and representations in both forms and the coordinate plane.

Differentiate Instruction

In 6.5.3 Guided Practice, students are applying a sequence of three transformations. Consider differentiating the product by eliminating one of the transformations or by breaking the question into two questions, with the second question beginning with the image produced by the first or second transformation.

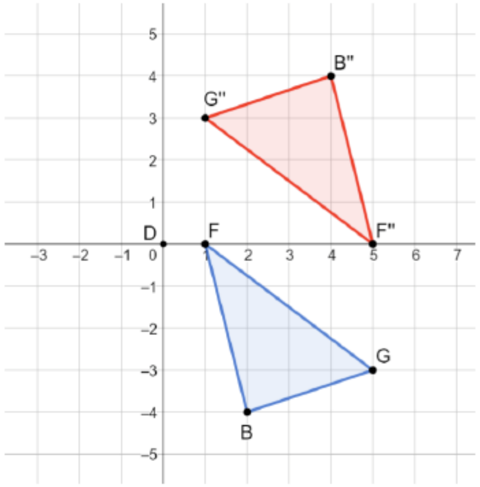
Lesson Notes:

6.5.1 Warm-Up: Error Analysis

Component Overview: *Two sample student responses to determine a sequence of transformations applied to a figure are given. Students will determine and justify their reasoning for which response accurately describes the sequence.*

6.5.1 Warm-Up: Error Analysis

1. ΔBGF was transformed by a sequence of transformations to create $\Delta B''G''F''$.



Brian and Emily were determining the sequence of transformations and wrote their answers in the table.

	ΔBGF Transformed to $\Delta B'G'F'$	$\Delta B'G'F'$ Transformed to $\Delta B''G''F''$
Brian's Answer	Rotated 180° about the origin	Translated right 6 units
Emily's Answer	Reflected over the x-axis	Translated right 4 units

Determine whose sequence of transformations is correct. Explain your reasoning.

Brian is correct because his sequence of transformations maps ΔBGF to $\Delta B''G''F''$.

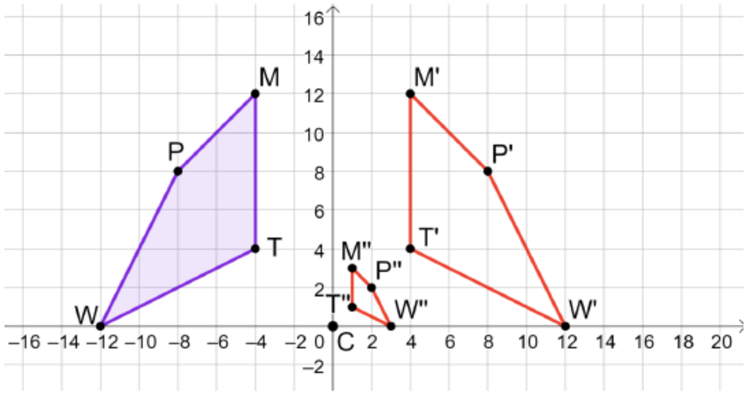
This component can provide valuable data for which students still need remediation on describing transformations. Monitor students' responses as they work to determine who may need extra support during the day's lesson.

6.5.2 Guided Instruction: Drawing Transformations

***Component Overview**: Students apply a sequence of transformations to a given figure by graphing the image.*

6.5.2 Guided Instruction: Drawing Transformations

1. $MTWP$ and point C are shown on the coordinate plane.



a. Draw the quadrilateral that is a reflection of $MTWP$ over the y-axis on the coordinate plane above. Label the quadrilateral $M'T'W'P'$.
Image is drawn on the graph.

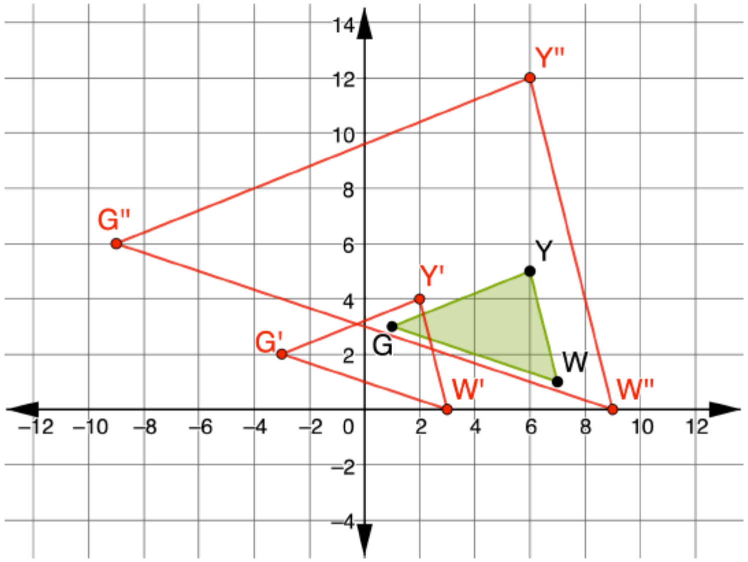
b. Draw the quadrilateral that is a dilation of $M'T'W'P'$ centered at point C using a scale factor of $\frac{1}{4}$ on the coordinate plane above. Label the quadrilateral $M''T''W''P''$.
The image is drawn on the graph.

c. Compare your drawings with your partner. Make any corrections if needed.

Gallery Walk
Post each question around the room on a separate poster. For large classes, post three different sets of the questions. Place students in groups of two or three. At each poster rotation, utilize the Take Turns routine by having a different group member take the marker and be responsible for any writing or drawing done.

- *Each group begins at a poster, discussing and applying the first transformation.*
- *Groups rotate to the next poster and check that the first transformation was completed correctly.*
    - *If so, apply the second transformation.*
    - *If not, provide written feedback for why the first transformation was not correct and then draw the correct transformation.*
- *Groups rotate to the next poster and check the work.*
    - *If both transformations were done correctly, write a positive statement about the work or a suggestion of another strategy that could have been used.*
    - *If any of the work is incorrect, provide written feedback and then draw the correct transformations.*
- *Groups rotate back to their original poster and review all comments and/or edits.*

2. ΔGYW is shown on the coordinate plane.



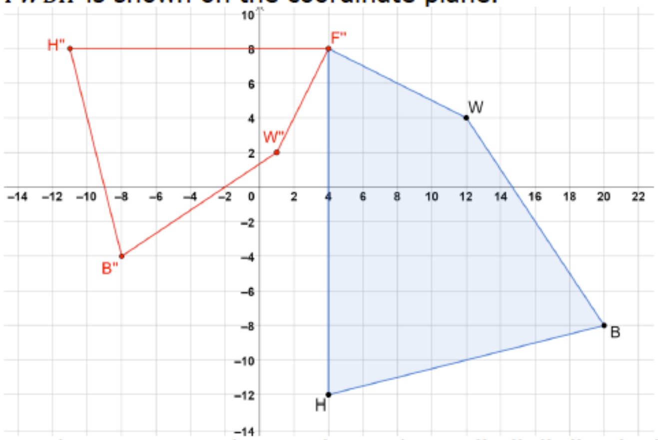
a. Draw $\Delta G'Y'W'$ as a result of $(x, y) \rightarrow (x - 4, y - 1)$ on the coordinate plane.
The image is drawn on the graph.

b. Draw $\Delta G''Y''W''$ as a result of $(x, y) \rightarrow (3x, 3y)$ on the coordinate plane.
The image is drawn on the graph.

Students may incorrectly apply the dilation as a reduction by dividing by 3 instead of an enlargement by multiplying by 3.

6.5.2 Guided Instruction: Drawing Transformations (continued)

3. $FWBH$ is shown on the coordinate plane.



On the same coordinate plane, draw $F''W''B''H''$, which is a result of dilating $FWBH$ using a scale factor of $\frac{3}{4}$, with the center at point F, then rotating the quadrilateral $90°$ clockwise about point F.
The image is drawn on the graph.

Students may incorrectly rotate around the origin instead of around point F.

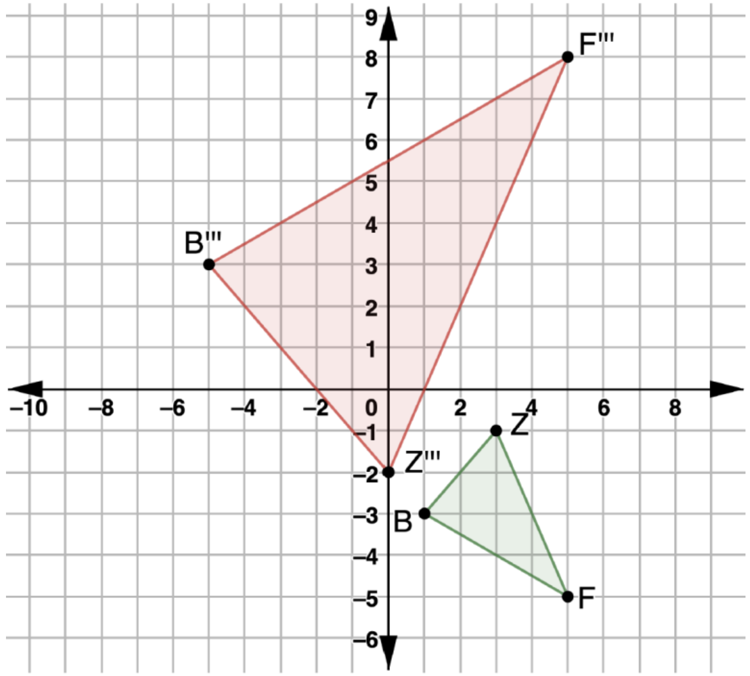
If students use algebraic descriptions for determining the placement of rotated points, ask them why this cannot be used on question 3. Guide students for choosing another method by asking questions like:

- *What kind of lines intersect in a $90°$ angle?*
- *What do we know about the slope of perpendicular lines?*
- *How can I use slope and distance to determine where to draw a line segment rotated $90°$?*

***Component Overview**: Students are challenged to perform a sequence of three transformations.*

6.5.3 Guided Practice: Sequence of Transformations

1. ΔFZB is shown on the coordinate plane.



On the same coordinate plane draw, $\Delta F'''Z'''B'''$, which is the result of a reflection over the x-axis, then translated left 3 units and down 3 units, then dilated with a scale factor of $\frac{5}{2}$ using point Z'' as the center of dilation.

Image is drawn on the graph.

Question 1 is a sequence of three transformations. Students may benefit from working with a partner. Consider breaking students into pairs by using musical pairings. Have students all raise a hand when the music begins and walk randomly around the room. Once the music stops, they high-five the person nearest them who then becomes their partner.

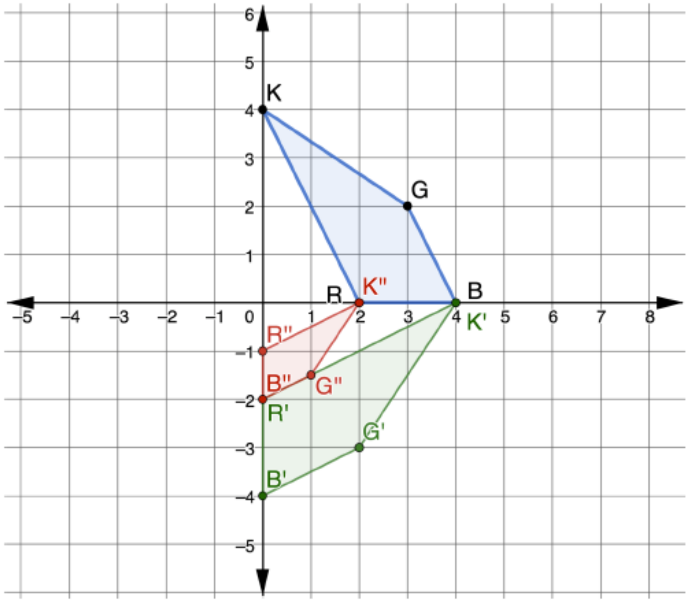
Consider differentiating the question by eliminating one of the transformations or by breaking the question into two questions, with the second question beginning with the image produced by the first or second transformation.

6.5.4 Your Turn!

Component Overview: *Students apply a sequence of transformations presented as an algebraic description.*

6.5.4 Your Turn!

1. $KGBR$ is shown on the coordinate plane.



On the same coordinate plane, draw $K''G''B''R''$, which is the result of $(x, y) \rightarrow (y, -x)$, then $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$.

Image is drawn on the graph.

Before students begin applying the transformations, encourage them to anticipate what transformations will occur based on the algebraic descriptions.

6.5.5 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

6.5.5 Wrap-Up: Cool Down

1. Why do you think it is important to use the prime notation when working with transformations?

   Answers will vary.

6.6 Similarity in Transformations

Lesson Introduction

Benchmarks	MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar. MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	Learning Targets	• I can map one similar figure onto another and justify that the figures are similar. • I can describe a sequence of transformations that will map the preimage onto the image.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• What is true about the corresponding angles of similar figures? • What is true about the corresponding sides of similar figures? • What criteria must a figure have to be similar to another?
Lesson Narrative	This lesson guides students through exploring the definition of similar figures by discovering that similar figures have congruent angles and proportional sides. Students then use this criterion to determine if given images are similar.

Component Summary	6.6.1 Warm-Up (~5 min.)	Dilating a polygon on an isometric grid *(SE p. 311)*	6.6.3 Guided Practice (5-10 min.)	Identifying and justifying whether given figures are similar *(SE p. 314)*
	6.6.2 Exploration (15-20 min.)	Discovering the qualities of similar figures *(SE p. 312)*	6.6.4 Your Turn! (5-10 min.)	Justifying why figures are similar *(SE p. 316)*
	6.6.5 Wrap-Up (~5 min.)	Summarizing the lesson by writing a note to a friend *(SE p. 317)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Similarity <u>Additional Terms:</u> N/A	• Patty Paper • (Optional) Graph Paper • (Optional) Rulers • (Optional) Cut Outs of the Images and Preimages	• If using stations to encourage collaboration, print out multiple copies of each question to have them available at each station.

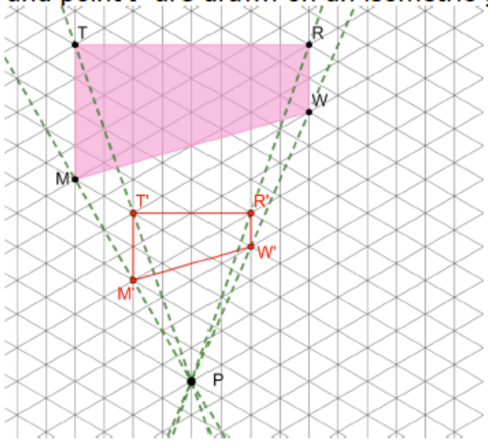
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may reflect the preimage in 6.6.2 Exploration question 1c instead of rotating it.
Students may dilate using a center at the origin instead of at point C' in 6.6.4 Your Turn! question 1.</td><td>Students may struggle with 6.6.4 Your Turn! question 1a. Consider creating a handout with scaffolded questions to guide students in determining the position of the final image.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Anticipate, Monitor, Select, Sequence, Connect
Throughout the lesson, students may write sequences of transformations that are not commonly used by the majority of the class. While monitoring student work, make note of any different sequences, and select those students to share with the whole class. This purposeful selecting of students will result in students staying engaged, because the selections were different, and not repetitive of what they have already seen or done.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In the 6.6.1 Warm-Up, students apply previously learned concepts to a new situation, developing an ability to transfer what they have learned to the world around them. In 6.6.2 Exploration, students relate the previously learned concepts of mapping, lengths of segments, and proportions to connect and discover the criteria needed to state two figures are similar.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">As students progress through 6.6.2 Exploration, students are given an opportunity to foster collaboration and build a mathematical community. Students can engage with the content and explore each question as teams, pairs, or whole-class collaborations. It may be effective to have stations where students physically go to work on each question throughout the lesson. This way, other students who are working on the same thing can offer tips and advice on how to solve parts they already have mastered.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.6.1 Warm-Up: Dilation on an Isometric Graph

Component Overview: *Students apply their skills of drawing the dilation of a figure on a coordinate plane to dilating a figure on an isometric grid.*

6.6.1 Warm-Up: Dilation on an Isometric Graph

1. $TRWM$ and point P are drawn on an isometric graph.



   a. Draw a dilation of $TRWM$ centered at point P using a scale factor of $\frac{1}{2}$.
   The drawing is completed on the graph in red.

   b. Explain what features on the graph were most useful in helping you determine where the dilation of $TRWM$ should be drawn.
   Answers will vary. One thing that helped me was the intersections of the triangles. By counting from point P to each vertex, the new quadrilateral vertex was found by using the midpoints.

This question is a great example of applying something previously learned to a new situation. If time permits, allow students to share their thinking for question 1b. Hearing how others worked through the question can help students make connections that they did not previously see.

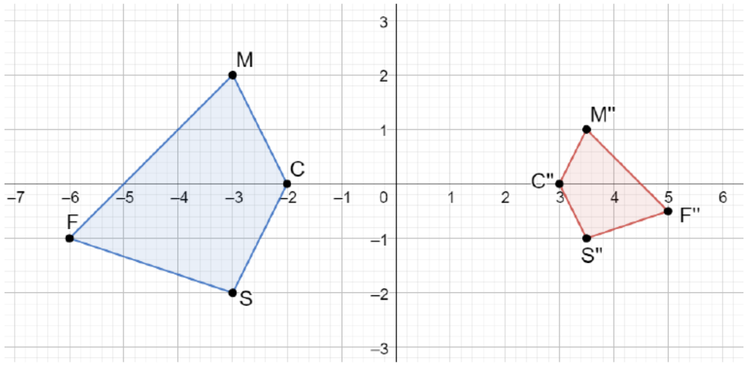
6.6.2 Exploration: Similarity

Component Overview: *Students explore the corresponding parts of an image that were created from a preimage using a sequence of transformations. Using patty paper, the students determine that corresponding angles are congruent. Using side lengths, the students determine that the corresponding sides have proportional lengths. Connections are made that similar figures are formed when a sequence of transformations includes a dilation and are verified with these two criteria.*

6.6.2 Exploration: Similarity

With your partner, complete the following.

1. $M''C''S''F''$ was created by applying a sequence of transformations to $MCSF$.



   a. With patty paper, trace $\angle FMC$ in $MCSF$.

   b. Place the tracing over $\angle F''M''C''$ in $M''C''S''F''$.

   c. What do you notice about $\angle FMC$ and $\angle F''M''C''$?
   The angles are congruent.

   d. Repeat the tracing for $\angle C$, $\angle S$, and $\angle F$. What do you notice about each angle and their corresponding angles?
   All of the corresponding angles are congruent.

   e. Complete the table.

Angles	Comparison Between Angles
$\angle FMC$ and $\angle F''M''C''$	*Congruent*
$\angle MCS$ and $\angle M''C''S''$	*Congruent*
$\angle CSF$ and $\angle C''S''F''$	*Congruent*
$\angle SFM$ and $\angle S''F''M''$	*Congruent*

Notice and Wonder
Consider utilizing the instructional routine “Notice and Wonder” prior to students answering question 1. This will allow students to make observations about the figures and understand the context in a low-stakes environment.

6.6.2 Exploration: Similarity (continued)

f. Determine a sequence of transformations that can be used to map $MCSF$ to $M''C''S''F''$.
Answers may vary. Reflected across the line $x = \frac{1}{2}$ and then dilated by a factor of $\frac{1}{2}$ with point C' as the center of dilation.

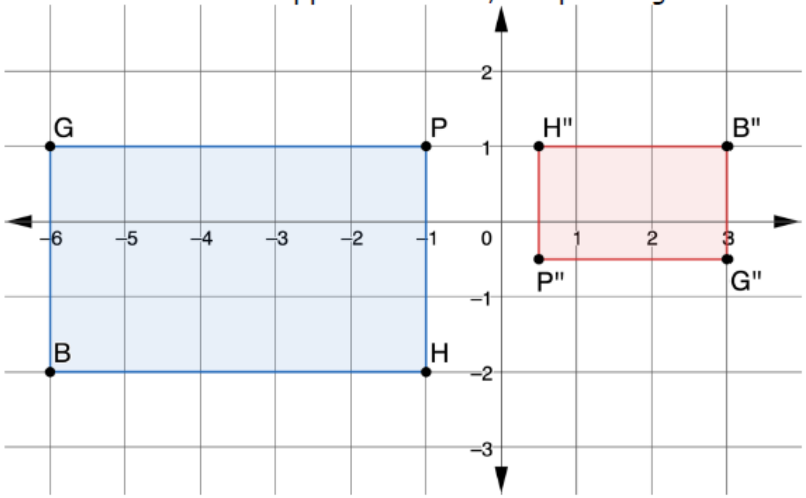
g. Ask two classmates the transformations they determined. Record their transformations and write a summary of their reasoning. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary was correct. Remind your classmates to include any details on the center of rotation or dilation and any scale factor.
Answers will vary.

Transformations	My Summary of Their Reason	Initials

h. Work with your partner to verify your classmates' transformations are correct.

While monitoring student work, make note of the different sequences that students determined. Have select students share any sequences that were not commonly used by the class to highlight different ideas.

2. $H''B''G''P''$ was created using a sequence of transformations applied to $HBGP$, the preimage.



a. Complete the table.

Preimage	Side Length	Image	Side Length	Ratio of Preimage to Image Lengths
$\overline{GP}$	5 *units*	$\overline{G"P"}$	2.5 *units*	$\frac{2}{1}$
$\overline{PH}$	3 *units*	$\overline{P"H"}$	1.5 *units*	$\frac{2}{1}$
$\overline{HB}$	5 *units*	$\overline{H"B"}$	2.5 *units*	$\frac{2}{1}$
$\overline{BG}$	3 *units*	$\overline{B"G"}$	1.5 *units*	$\frac{2}{1}$

b. Describe the relationship between the ratios of the preimage to the image.
All of the ratios of a side length to the corresponding side length are the same, $2{:}1$.

Students may choose to write the ratio as 2 *to* 1, $\frac{2}{1}$, *or as* $2{:}1$. *All are acceptable. Continue to use different representations to keep students familiar with all forms.*

6.6.2 Exploration: Similarity (continued)

c. Determine the sequence of transformations that was used to create $H''B''G''P''$ from $HBGP$.
The figure was rotated $180°$ about the origin and then dilated a scale factor of $\frac{1}{2}$ with the center of dilation at the origin.

d. Discuss with your partner how to describe how $MCSF$ is related to $M''C''S''F''$ and how $HBGP$ is related to $H''B''G''P''$. Summarize your discussion.
Answers will vary. In both sets of quadrilaterals, the corresponding angles in the preimage and image are congruent but the corresponding sides are not congruent.

3. What qualities of similar figures do these quadrilaterals have?
They have corresponding angles that are congruent and corresponding sides that are proportional.

Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation or a sequence of transformations that includes a dilation.

4. Complete the statements using words from the bank.

Word Bank			
congruent	dilation	proportional	similar

When a sequence of transformations includes a *dilation*, the resulting image of the sequence of transformations is *similar* to the preimage. All of the angles of the resulting image are *congruent* to the corresponding angles of the preimage. The corresponding sides of the image to the preimage are *proportional*.

Students may incorrectly reflect the preimage instead of rotating it since this will produce the same shape. Probe students about the position of the corresponding vertices and their alignment to the image if a reflection is performed.

The key word "similarity" will be essential throughout the rest of Geometry. Make sure students take time to learn this definition.

Consider discussing how the ratio of the corresponding lengths relates to the scale factor of the dilation after students have completed 6.6.2 Exploration.

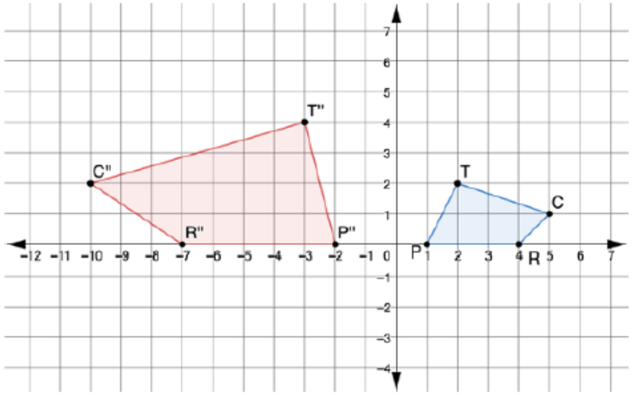
6.6.3 Guided Practice: Similarity

Component Overview: *Students observe three preimages and images to determine similarity. If they are not similar, a justification is required.*

6.6.3 Guided Practice: Similarity

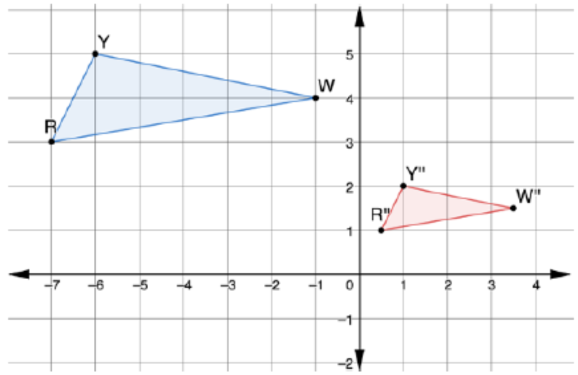
For each pair of figures, write the sequence of transformations that would show the figures are similar. If the figures are not similar, explain why.

1.



The figures are not similar. The ratio of PR to $P''R''$, $3{:}5$, is not equal to the ratio of TP to $T''P''$, $\sqrt{5}{:}\sqrt{17}$.

2.



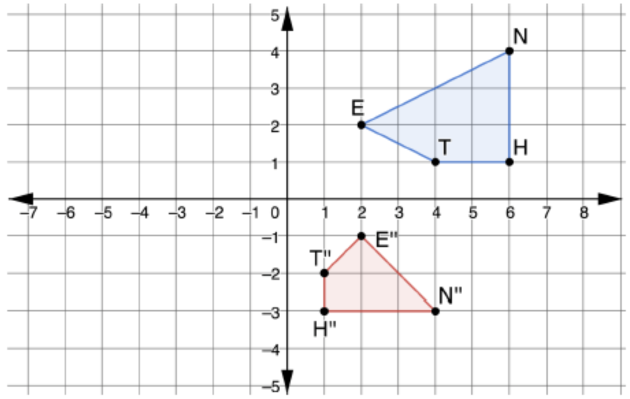
These figures are similar.
Answers will vary. The preimage has been dilated by a scale factor of $\frac{1}{2}$ with the center of dilation at the origin and then translated right 4 and down 0.5.

To add a challenge, have students determine the least number of points that can be moved to make any figures that are not similar become similar figures. (Sample answer for question 1: two moves. Move R'' to $(-8, 0)$ and move T'' to $(-4, 4)$.)

For question 2, students may choose a different sequence of transformations. While monitoring, make note of different methods and highlight those during the review of this question.

6.6.3 Guided Practice: Similarity (continued)

3.



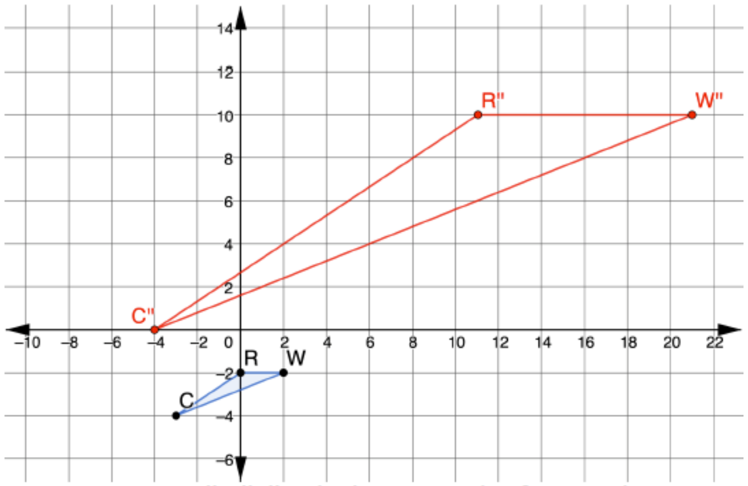
The figures are not similar. The ratio of HT to $H''T''$ is $2:1$, but the ratio of HN to $H''N''$ is $1:1$.

For students who may be struggling, consider providing students with the image after the first transformations. This can help students see the two transformations that were performed to help them determine if the shapes are similar.

Component Overview: *Students justify whether two figures are similar by two different methods, one by a sequence of transformations and the other by proportionality of sides.*

6.6.4 Your Turn!

1. ΔRWC is shown on the coordinate plane.



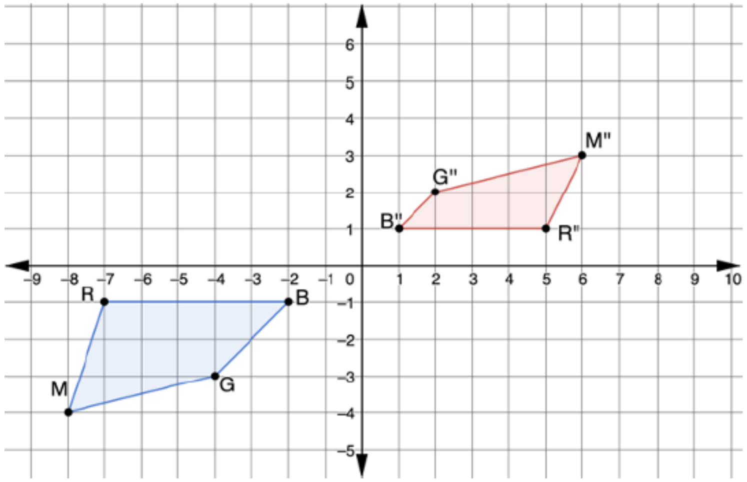
a. Draw $\Delta R''W''C''$, which is a result of a translation up 4 units and left 1 unit, then dilated using a scale factor of 5 centered at C'.
Image is drawn on graph.

b. Explain how ΔRWC is similar to $\Delta R''W''C''$.
The triangles are similar because the final image is the result of a translation, which is a rigid transformation, and a dilation, which preserves angle measure and the corresponding sides from preimage to image have the same ratio.

Students may mistakenly dilate the image centered at the origin instead of point C'. If this happens, have the student re-read the directions aloud to determine their error.

6.6.4 Your Turn! (continued)

2. Carter and Isabella are asked to determine if the quadrilaterals are similar. Their responses are recorded in the table.



Carter's Response	Isabella's Response
Yes, the quadrilaterals are similar because all the corresponding angles are congruent.	No, the quadrilaterals are not similar because the corresponding sides are not proportional.

Is Carter's or Isabella's response correct? Explain your decision.

Answers will vary. Isabella's response is correct. The corresponding sides are not proportional. The ratio of BR to $B'R'$ is $5:4$ but the ratio of RM to $R''M''$ is $\sqrt{10}:\sqrt{5}$.

This would be a great opportunity to develop students' abilities to appropriately communicate with their peers and offer help without providing answers. Consider having the students write a note to Carter that includes one positive statement and one constructive statement.

6.6.5 Wrap-Up: Cool Down

Component Overview: *Students summarize the day by writing a note to a friend explaining the lesson.*

6.6.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand how to determine if a sequence of transformations results in similar figures.
   Answers will vary.

Consider offering students the opportunity to create a picture that summarizes the day's lesson. This may be a comic book or a puzzle.

6.H1 Proving All Circles Are Similar

Lesson Introduction

Benchmark	MA.912.GR.6.5 Apply transformations to prove that all circles are similar.	Learning Target	• I can justify that two circles are similar and conclude that any circle can be mapped to another circle using similarity transformations.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How do rigid and nonrigid transformations prove that all circles have similarity? • Why does a circle follow the same rules as other shapes regarding rigid transformation?
Lesson Narrative	This lesson explores rigid and nonrigid transformations of circles. Students explore how one circle can be mapped onto other circles that are either congruent or the result of an enlargement or reduction. Conclusions are drawn that all circles are similar but not necessarily congruent.

Component Summary

6.H1.1 Warm-Up (~5 min.)	Notice and Wonder about overlapping circles *(SE p. 318)*	6.H1.3 Your Turn! (10-15 min.)	Transforming a circle to create a similar circle *(SE p. 320)*
6.H1.2 Exploration (15-20 min.)	Mapping a circle onto three other circles to determine similarity *(SE p. 318)*	6.H1.4 Wrap-Up (~5 min.)	Reflecting on the interest and understanding of the lesson *(SE p. 321)*

Resources

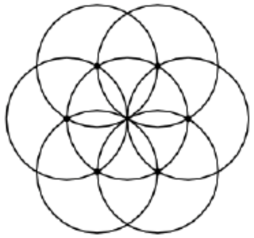
Glossary	Materials	Additional Preparation
<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Similar • Congruent	• (Optional) Masking Tape • (Optional) Circular Objects	• This is an honors lesson. Students in regular Geometry do not need to complete this lesson. • For 6.H1.2 Exploration or 6.H1.3 Your Turn!, if you are considering using the classroom floor as a grid, set this up prior to class. Consider using a ruler to place sticky notes one foot from a wall, then roll the tape out, and repeat this process measuring one foot from the last piece of tape.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may think that congruent circles are not similar.</td><td>Consider creating a large grid on the floor and using yarn, hula-hoops, or other round objects to recreate the circles in 6.H1.2 Exploration or 6.H1.3 Your Turn! Then, students could physically move the circles as suggested.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Stronger and Clearer Each Time
For 6.H1.3 Your Turn! question 1b, consider implementing this routine. This structure provides a purpose for student conversations as well as fortifies student output. Throughout the process, students should be pressed to create a stronger (better justifications and examples) and clearer (with more precise terms and linked, organized, complete sentences) response.</td><td>MA.K12.MTR.5.1: Patterns and Structure
This lesson is designed to guide students through relating previously learned concepts to the new concept of circles. From 6.H1.2 Exploration, students are using the same vocabulary of mapping circles as they did when mapping polygons earlier in Unit 6.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Students will be able to activate background knowledge at 6.H1.2 Exploration. This initially occurs when students are asked to “map” one circle to another. This is the same terminology as earlier in the unit, and the cadence of this lesson mimics previous lessons, when students learned to map and identify transformations.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.H1.1 Warm-Up: Notice and Wonder

Component Overview: *Students notice and wonder about overlapping circles and how they could create unique and artistic patterns and shapes.*

6.H1.1 Warm-Up: Notice and Wonder



1. What do you notice?
   Answers will vary. I see 7 overlapping circles.

2. What do you wonder?
   Answers will vary. I wonder how somebody thought to arrange circles in this artistic way.

Consider asking the following probing questions:

- *If we remove or color in parts of this image, what naturally occurring shapes may you find? Do you see a flower? Do you see a sand dollar? Do you see a star fish?*

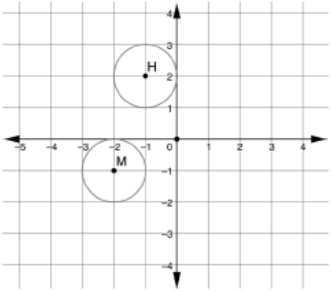
6.H1.2 Exploration: Are Circles Similar?

***Component Overview**: Students map a given circle onto three different circles using rigid and nonrigid transformations to determine that all circles are similar but not necessarily congruent.*

6.H1.2 Exploration: Are Circles Similar?

Answer the following questions with your partner.

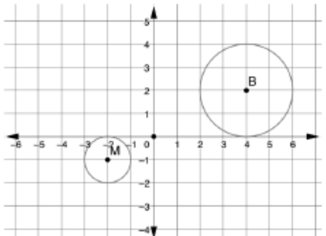
1. Circle M and Circle H are shown on the coordinate plane.



   a. How can Circle M be mapped onto Circle H?
   Answers will vary. Translate Circle M right 1 unit and up 3 units.

   b. What can you conclude about Circle M and Circle H?
   Answers will vary. All points on Circle M map to all points on Circle H. Therefore, the circles are congruent.

2. Circle M and Circle B are shown on the coordinate plane.



   a. How can Circle M be mapped onto Circle B?
   Answers will vary. Circle M can be mapped onto Circle B by translating it right 6 units and up 3 units, then dilating the result by a scale factor of 2 centered at $(4,2)$.

Consider asking the following while monitoring student work on question 1.

- *What is another way to verify that two circles are congruent? (Answer: they have the same radius.)*
- *Are there other ways to map Circle M onto Circle H? (Possible answer other than the listed answer: students may also rotate Circle M $90°$ clockwise about the origin.)*

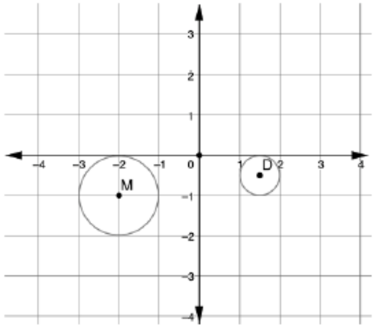
Consider asking the following while monitoring student work on question 2.

- *How do I know that the two circles are not congruent? (Answer: they don't have the same radius.)*
- *Are there other ways to map Circle M onto Circle P? (Possible answer other than the listed answer: students may also rotate Circle M $180°$ about the origin and then dilate by 2 about the origin.)*

6.H1.2 Exploration: Are Circles Similar? (continued)

b. What can you conclude about Circle M and Circle B?
The circles are similar but not congruent.

3. Circle M and Circle D are shown on the coordinate plane.



a. How can Circle M be mapped onto Circle D?
Answers will vary. Circle M is translated right 3.5 units, up 0.5 unit. The result is then dilated using a scale factor of 0.5 with Point D as the center of dilation.

b. What can you conclude about Circle M and Circle D?
They are similar, but not congruent.

4. What can you conclude about Circle M, Circle H, Circle B, and Circle D?
Because Circle M can be mapped to all three circles, we can conclude that they are all similar.

5. Complete the statement.

○ All ○ Some ○ None	circles are	○ congruent ○ similar	because any

circle can be mapped to another circle using similarity transformations.

Consider asking the following while monitoring student work on question 5.

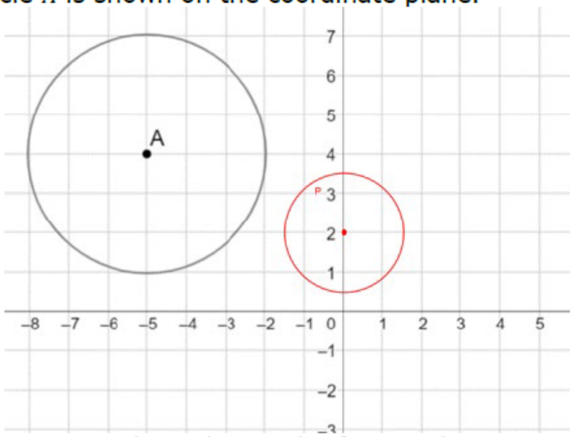
- *What transformations are considered “congruence” transformations? (Translations, reflection, rotation.)*
- *What transformations are considered “similarity” transformations? (Translation, reflection, rotation, and dilation.)*

6.H1.3 Your Turn!

Component Overview: *Students draw a similar circle to a given circle using rigid and nonrigid transformations and explain why they are similar using the word "transformation."*

6.H1.3 Your Turn!

1. Circle A is shown on the coordinate plane.



a. Draw Circle P, the result of a translation 5 units right and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle P.

b. Explain how Circle A and Circle P are similar. Include the word "transformation" in your explanation.
Circle A and Circle P are similar because a dilation is a transformation that maintains proportionality.

2. Lana and Henry are writing notes from today's lesson.

Lana's Notes	Henry's Notes
All circles are similar; therefore, all circles are also congruent.	All circles are similar, but not all circles are congruent.

Whose notes do you agree with? Why?
Answers will vary. I agree with Henry because, while all circles are similar, not all circles have the same radii and therefore are not congruent.

For a kinesthetic entry-point, consider creating a large coordinate grid on the floor and using yarn, Hula-Hoops, or other circular objects to recreate the circles in this component and have students physically move the circles as suggested.

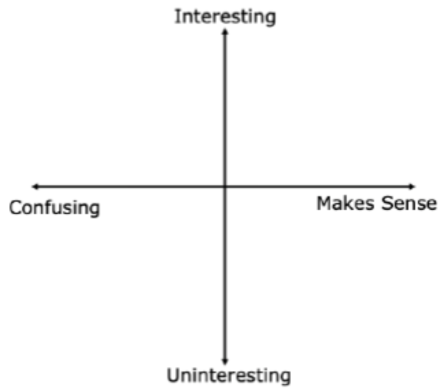
Stronger and Clearer Each Time
For question 1b, consider implementing this routine. After students have written a response to the question, pair them with another student to refine and clarify their response through conversation. Then, pair students differently and repeat. After several pairings, all students should have a correctly revised response.

6.H1.4 Wrap-Up: Reflection

***Component Overview**: Students plot a point on the grid that best represents their interest and confidence on the lesson.*

6.H1.4 Wrap-Up: Reflection

1. Plot a point on the gird below to indicate your level of interest and understanding for today's learning target.
   Answers will vary.



Explain the location of your point.
Answers will vary.

Consider using the point a student placed the grid as a jumping-off point for individual conferences about their progress and understanding.

6.H2 Proofs by Contradiction

Benchmark	MA.912.LT.4.8 Construct proofs, including proofs by contradiction.	Learning Target	• I can use proof by contradiction to justify that all circles are similar.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• What is a proof by contradiction? • Is there a relationship between the radius of circles and their circumference?
Lesson Narrative	This lesson introduces the idea of constructing a proof by contradiction by relating how an unlikely conclusion of a series of conditional statements can be proved as false using contradiction. Students then explore the relationships of similar circles to support student learning as they are guided through a proof by contradiction to prove that all circles are similar.

Component Summary	6.H2.1 Warm-Up (~5-10 min.)	Creating outrageous conditional statements *(SE p. 322)*	6.H2.3 Exploration (10 min.)	Discovering that corresponding parts of circles are proportional *(SE p. 323)*
	6.H2.2 Exploration (10 min.)	Writing contradiction statements *(SE p. 322)*	6.H2.4 Guided Instruction (15 min.)	Proving circles are similar using a proof by contradiction *(SE p. 324)*
	6.H2.5 Wrap-Up (~5 min.)	Reflection on understanding of the lesson *(SE p. 325)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Proof by Contradiction <u>Additional Terms:</u> • Corresponding • Proportional	• Commercial Video(s) • (Optional) Index Cards for 6.H2.5 Wrap-Up	• 6.H2.1 Warm-Up contains a video to accompany the question prompt.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may write a contradiction as the inverse or converse.
Students may not understand that a proof by contradiction requires one to assume that a statement is true to then prove that the assumed statement is false.</td><td>Students have learned that side lengths of corresponding similar figures are proportional but have not yet learned that other measures, like circumference of circles, share a proportional relationship.
Consider debriefing the 6.H2.2 Exploration before moving on to 6.H2.3 Exploration. This will allow the opportunity to ensure students were able to correctly write the negation statements in 6.H2.2 Exploration.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematica; Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
6.H2.3 Exploration questions 1c and 1e already have the notice part of this routine. Consider implementing this routine by asking what the students may wonder. Making this move may get their curiosity piqued for other relationships of similar figures.</td><td>MA.K12.MTR.3.1: Mathematical Fluency
This lesson is designed to lean on the students' mathematical fluency throughout the lesson. 6.H2.2 Exploration relies on students' ability to complete tasks accurately and with confidence while adapting procedures to a new context. 6.H2.3 Exploration relies on the students' abilities to maintain flexibility and accuracy while finding parts of a circle and ratios of parts of two different circles. Teachers are encouraged to provide opportunities for students to reflect on the method they used and determine if a more efficient method could have been used.</td></tr>
<tr><td>Differentiate Instruction</td><td>The statements in the 6.H2.5 Wrap-Up provide an opportunity for students to monitor their own progress and guide their own effort and practice. Use responses to these prompts to work with students in developing a plan for moving toward mastery of these concepts.</td><td>☐ I understand how to write a contradiction.
☐ I need support on how to write a contradiction.
☐ I understand that all circles are similar.
☐ I need support on understanding why all circles are similar.
☐ I understand how to prove that all circles are similar by contradiction.
☐ I need support on how to prove that all circles are similar by contradiction.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.H2.1 Warm-Up: Critical Thinking

Component Overview: *Play the video of the commercial for the entire class. After students watch the video (31 seconds), prompt them to write five conditional statements.*

6.H2.1 Warm-Up: Critical Thinking

1. DirecTV commercials are entertaining because the conditional statements are outrageous. Write five conditional statements with an outrageous conclusion that could be used for a DirecTV commercial.
   Answers will vary.
   1. *If your cable company has hidden fees, then you will end up swimming with sharks.*
   2. *If your cable company charges for slowly increases their rates, then you will get locked out of your car.*
   3. *If your cable company charges you more for less channels, then you will get charged by a rhinoceros.*
   4. *If your cable company charges you a DVR service fee, then you will drink licorice flavored soda.*
   5. *If your cable company has terrible customer service, then you will be chased by a heard of sheep.*

Allow students time to be creative with their five conditional statements. Establish the acceptable parameters so that students do not break any social boundaries in their statements. This is a great opportunity to engage students. However, it can easily take more than the designated amount of time. Consider assigning a student as a timekeeper to allow students to take ownership of the pacing of this lesson.

6.H2.2 Exploration: Contradictions

***Component Overview**: Students write contradiction statements to explore how contradictions change the outcome of conditional statements.*

6.H2.2 Exploration: Contradictions

In the DirecTV commercial, there were seven conditional statements before the final conclusion.

Order	Statement
1	If your cable company puts you on hold, then you get angry.
2	If you get angry, then you go blow off steam.
3	If you go blow off steam, then an accident happens.
4	If an accident happens, then you get an eye patch from the doctor.
5	If you get an eye patch, then people think you are tough.
6	If people think you are tough, then they want to fight you.
7	If people want to fight you, then you get hurt.
8	You will end up in a roadside ditch.

1. Assume that the conclusion is not true. Write a contradiction to the conclusion.
   You will not end up in a roadside ditch.

2. Share your contradiction with your partner.

6.H2.2 Exploration is scaffolded to allow students to engage with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 6.H2.4 Guided Instruction. Students may revise their thinking throughout as they engage with each new question.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share with the whole group before 6.H2.4 Guided Instruction.

6.H2.2 Exploration and 6.H2.3 Exploration pair to set up students' understanding of how to use a proof by contradiction to prove that all circles are similar.

6.H2.2 Exploration: Contradictions (continued)

3. Use your contradiction to write a "rewind" for the commercial with statements that would lead to the contradiction.

Order	Statement
8	*You will not end up in a roadside ditch.*
7	*If people will want to fight you, then you will not get hurt.*
6	*If people think you are tough, then they will not want to fight you.*
5	*If you get an eyepatch, then people will not think you are tough.*
4	*If an accident happens, then you do not get an eyepatch from the doctor.*
3	*If you go blow off steam, then an accident will not happen.*
2	*If you get angry, then you do not go to blow off steam.*
1	*If your cable company puts you on hold, then you do not get angry.*

4. Explain how writing the contradiction would change the commercial.
Answers will vary. The commercial will not be very effective. If nothing bad happens when you get put on hold, like ending up in a ditch, then why should you care that you got put on hold?

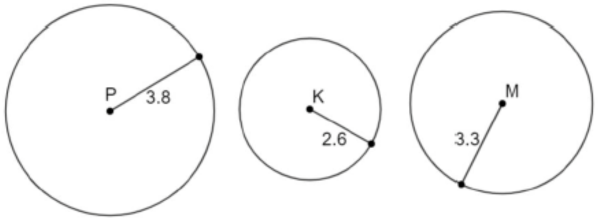
Students may have a hard time following the reverse flow of these steps. The important thing here to focus on is the negation of each statement. Effectively writing negations is an important first step in writing proofs by contradiction later in the lesson.

6.H2.3 Exploration: Circles Are Similar

Component Overview: *Students will use the provided measurements to determine the circumference of the circles. They will then compare the ratio of the radius and circumference between the circles and write a conjecture based on their findings.*

6.H2.3 Exploration: Circles are Similar

1. Circles P, K, and M are shown. The radius of Circle P is 3.8 units long. The radius of Circle K is 2.6 units long. The radius of Circle M is 3.3 units long.



a. With your partner, fill in the table. *Round to nearest thousandth. Use the pi button on your calculator.*

Circle	Radius	Circumference
P	3.8	23.876
K	2.6	16.336
M	3.3	20.735

b. Complete the table.

Circle P and K	
Ratio of Radius of Circle P and Radius of Circle K	$\frac{3.8}{2.6} \approx 1.462$
Ratio of Circumference of Circle P and Circumference of Circle K	$\frac{23.876}{16.336} \approx 1.462$

c. What do you notice about the ratios?
They are equivalent.

Consider assigning this as a group activity with groups of four.

- *Assign groups roles.*
- *Complete question 1a as a group.*
- *Split groups into pairs.*
- *One pair completes questions 1b and 1c, the other completes questions 1d and 1e.*
- *Both pairs come together, share results, and work on question 2.*

d. Complete the table.

Circle K and M	
Ratio of Radius of Circle K and Radius of Circle M	$\frac{2.6}{3.3} \approx 0.788$
Ratio of Circumference of Circle K and Circumference of Circle M	$\frac{16.336}{20.735} \approx 0.788$

e. What do you notice about the ratios?
They are equivalent.

2. Make a conjecture about the ratios of radii and circumferences of circles.
Answers will vary. I think the ratios of the radii and the ratios of the circumferences of two circles will be equivalent to each other.

Notice and Wonder
Questions 1c and 1e already have the notice part of this routine. Before moving to question 2, consider asking what the students may wonder. Take a moment to explore any named wonderings about other ratios of corresponding parts of circles, like their areas.

6.H2.4 Guided Instruction: Proof by Contradiction

***Component Overview**: Students are introduced to a new form of proof called proof by contradiction. Using proof by contradiction, students determine that circles are similar.*

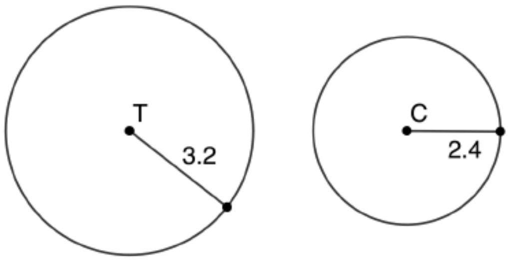
6.H2.4 Guided Instruction: Proof by Contradiction

A way to write a proof is by using a contradiction.

Proof by Contradiction

Assume the statement to be proven is false, and work to show its falsity until the result of that assumption is a contradiction.

1. Circles T and C are shown. The radius of Circle T is 3.2 units long. The radius of Circle C is 2.4 units long.



Prove that Circle T is similar to Circle C.

a. Write an assumption that the statement is false.
Assume that Circle T is not similar to Circle C.

b. Find the ratio of the radius of Circle T to the radius of Circle C.
$\frac{3.2}{2.4} \approx 1.333$

c. Find the ratio of the circumference of Circle T to the circumfernce of Circle C.
$\frac{6.4\pi}{4.8\pi} \approx 1.333$

Students naturally understand that all circles are similar so proving that two circles are similar may appear to be pointless. This is why proving that two circles are similar is a nice avenue for proof by contradiction.

Consider extending this component to include radii represented as variables and writing the circumference in terms of π. Prompt students to make connections between the ratios formed by the radii and the ratios formed by the circumference in this form. Then have students write the ratios of the areas of the circles and make a conjecture.

d. Explain how the ratios show the assumption is not true.
The radius and the circumference are proportional, so the circles are similar.

e. Complete the proof.

Assume that Circle T is *not similar* to Circle C. The ratio of the radius of Circle T to the radius of Circle C is $\frac{4}{3}$. The ratio of the circumference of Circle T to the circumference of Circle C is $\frac{4}{3}$. The ratio of the radii and the ratio of the circumferences are *equivalent*, so corresponding parts of Circle T and Circle C are proportional. Since the corresponding parts of Circle T and Circle C are proportional, the circles are *similar*. This is a contradiction, so the assumption that Circle T is not similar to Circle C is not true.

Provide an opportunity for students to make connections between the contradiction and the original proof statement by asking, "Because the assumption that Circle T is not similar to Circle C is not true, what have we proved?" (Sample answer: that Circle T is similar to Circle C.)

6.H2.5 Wrap-Up: Reflection

***Component Overview**: Students complete a self-reflection on their level of understanding of the content of this lesson.*

6.H2.5 Wrap-Up: Reflection

1. Check below all the statements that apply to your understanding.
   Answers will vary.

   - ☐ I understand how to write a contradiction.
   - ☐ I need support on how to write a contradiction.
   - ☐ I understand that all circles are similar.
   - ☐ I need support on understanding why all circles are similar.
   - ☐ I understand how to prove that all circles are similar by contradiction.
   - ☐ I need support on how to prove that all circles are similar by contradiction.

Consider asking students to write their answers on a note card and drop them in a bucket on the way out of the class. Tally how many checks are given to each statement to determine where the most support is needed. Consider statements that have a large number of checks as a possible station on another day.

6.7 Triangle Similarity

Lesson Introduction

Benchmark	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.	Learning Targets	• I can identify similar triangles and write similarity statements. • I can use given information to show that two triangles are similar.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What makes two triangles similar? • How can you determine if two triangles are similar if you do not know any angle measures? • How can you determine if two triangles are similar if you do not know any side lengths?
Lesson Narrative	This lesson offers students the opportunity to explore different ways of proving two triangles similar. Students will explore this through a sequence of transformations as well as by creating triangles following a given description. The formalized names for the theorems that demonstrate similarity are identified and connections are made to the drawings.

Component Summary				
	6.7.1 Warm-Up (~5 min.)	Describing a sequence of transformations *(SE p. 326)*	6.7.3 Your Turn! (5 min.)	Determining similarity *(SE p. 328)*
	6.7.2 Exploration (10 min.)	Describing a sequence of transformations that show similarity *(SE p. 327)*	6.7.4 Activity (20-25 min.)	Creating triangles given a description to determine similarity *(SE p. 329)*
	6.7.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 331)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Angle-Angle Similarity (AA~) Theorem • Side-Side-Side (SSS~) Theorem • Side-Angle-Side (SAS~) Theorem <u>Additional Terms:</u> • Similarity • Congruent • Proportional	• Centimeter Rulers • Protractor • (Optional) Patty Paper • (Optional) Chart Paper	• 6.7.4 Activity includes a Gallery Walk that requires chart paper for each station. Print the instructions and post at each station. • Consider verifying the measurements for 6.7.2 Exploration.

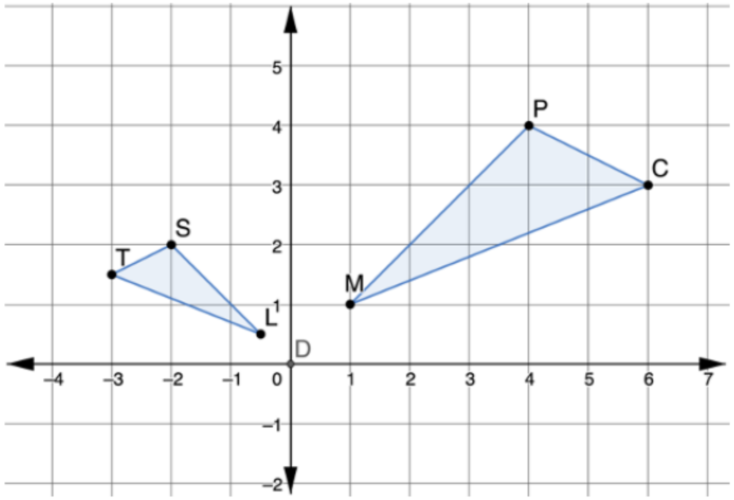
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not state that 6.7.4 Activity descriptions D and F do not create triangles.
Students may not understand that if the scale factor is greater than 1, then the dilated image is larger than the original figure and if the scale factor is between 0 and 1, then the dilated image is smaller than the original figure.</td><td>For the 6.7.4 Activity, consider adding a step of writing similarity statements for each pair of triangles that are similar. Students will first need to label the vertices of the triangles prior to doing this.
Consider only giving descriptions A-E to all students for 6.7.4 Activity and using F and G for those groups who finish early.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Group Presentations
6.7.4 Activity provides an opportunity for students to work in partner pairs or small groups to draw triangle pairs based on given descriptions. For question 3, student groups can present one of the drawings to the whole group and provide justification of why they are or are not similar and what theorem matches the description. Ask class members to restate their understanding of the group’s reasoning.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
In each component of this lesson, students discuss and reflect on mathematical topics and ideas. 6.7.2 Exploration provides students an opportunity to explore similar triangles. 6.7.4 Activity also encourages students to work through generating similar figures using given information. Teachers should consider highlighting common misconceptions and revelations students had to further deepen student understanding.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">6.7.4 Activity is a great opportunity for providing multiple entry-points to the mathematical language and concepts. In question 1, students first draw the pairs of triangles for a kinesthetic entry-point. In question 2, students can visually see the possibilities of similarity by observing their drawings and those of their partner and/or other partner pairs. Then, in question 3 students connect the provided theorems to what they have observed in the triangle relationships.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.7.1 Warm-Up: Bell Work

Component Overview: *Students describe a sequence of transformations that will map one triangle onto the other.*

6.7.1 Warm-Up: Bell Work

1. ΔMPC and ΔTSL are shown on the coordinate grid, where ΔMPC is the pre-image.



Describe a sequence of two transformations that will map the triangles.

Answers will vary. Dilate ΔMPC using a scale factor of $\frac{1}{2}$ centered at point D. Then, reflect $\Delta M'P'C'$ across the y-axis.

Are the transformations in your sequence commutative for mapping this preimage onto the image?

Consider providing students with patty paper to help them determine the rigid motion(s) in the sequence.

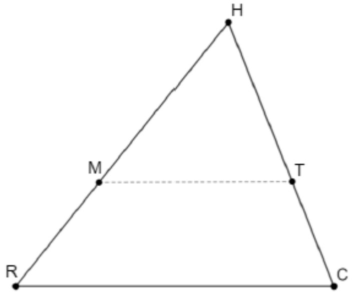
6.7.2 Exploration: Similar Triangles

Component Overview*: Students will explore the connection between the ratio of side lengths in similar triangles. Once it is determined that the provided triangles are similar, they list all transformations that will result in similar triangles.*

6.7.2 Exploration: Similar Triangles

With your partner, complete the following.

ΔHCR is shown, where $m\angle HCR = 68°$, $m\angle HRC = 51°$, and $m\angle RHC = 61°$. Point M is on $\overline{HR}$ and point T is on $\overline{HC}$. $\overline{MT}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.
   Answers will vary for length measurements depending on the version used in print.

Measurements of ΔRHC		Measurements of ΔMHT	
$RH = 7.7$ cm	$m\angle HRC = 51°$	$MH = 4.2$ cm	$m\angle HMT = 51°$
$HC = 6.5$ cm	$m\angle RHC = 61°$	$HT = 3.9$ cm	$m\angle MHT = 61°$
$RC = 7.2$ cm	$m\angle HCR = 68°$	$MT = 4.3$ cm	$m\angle HTM = 68°$

2. What do you notice about the corresponding angles of ΔRHC and ΔMHT?
   These angles are the same measure.

Throughout 6.7.2 Exploration, listen carefully for key words like "ratio," "proportion," "similar," "congruent," and "angles" to ensure that mathematical discussion is taking place and that students are reflecting on the questions.

Consider giving students a larger triangle on a separate sheet of paper created using GeoGebra or Desmos. Use the provided angle measurements to create the triangle.

3. Discuss with your partner what relationship could have been given so that it would not have been necessary to use a protractor to find $m\angle HMT$ and $m\angle HTM$. Summarize your discussion.
Answers will vary. If we had been given that $\overline{RC} \parallel \overline{MT}$ then the angles would have been congruent by corresponding angles.

4. What do you notice about the corresponding sides of ΔRHC and ΔMHT?
Answers will vary. The corresponding sides are proportional.

5. Discuss with your partner how ΔRHC and ΔMHT are related. Summarize your findings.
Answers will vary. All the corresponding angles of the triangles are congruent and all of the correponding sides are proportional.

6. Complete the statement that shows the relationship.
ΔRHC is similar to $\underline{\Delta MHT}$.

7. Complete each relationship for ΔRHC and ΔMHT.

$$\frac{RH}{MH} = \frac{RC}{TM}$$

$$\frac{HT}{HC} = \frac{TM}{CR}$$

8. Discuss with your partner a sequence of transformations that can be used to map ΔRHC onto ΔMHT. Summarize your discussion by describing the sequence.
Answers will vary. Dilate ΔRHC using a scale factor of $\frac{3}{5}$ centered at point H.

Question 3 offers a moment for spiraling back to Unit 3. While monitoring students, look and listen for use of the vocabulary like "parallel lines" and "corresponding angles." For students not using the vocabulary, ask probing questions to guide them back to this vocabulary.

When describing the relationship between the two triangles, students may write the reciprocal of the scale factor needed depending on the wording of their answer. Question students that do this by asking what happens to values when multiplying by fractions greater than 1.

6.7.2 Exploration: Similar Triangles (continued)

9. Compare your sequence of transformations with another pair of students.
   Ask two classmates for their sequence of transformations. Record their sequence and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary is correct.
   Answers will vary.

Sequence	My Summary of Their Reason	Initials

10. What transformation ensures that the sequence of transformations will result in similar triangles?
    The dilation ensures that the result is similar triangles.

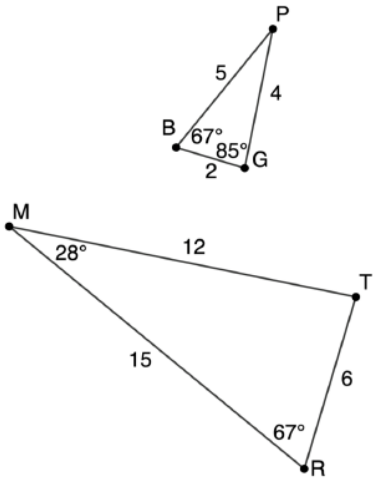
Observe and listen as students work with their partners. Make note of student ideas that you can ask to be shared in the whole group before beginning 6.7.3 Your Turn! Consider highlighting student voices to share rich discussions or comments overheard while you circulated or to address common misconceptions that were evident among the majority of the class.

6.7.3 Your Turn!

***Component Overview**: Students determine if two triangles are similar. If they are, students write a similarity statement. If they are not, students provide an explanation of why not.*

6.7.3 Your Turn!

1. ΔPGB and ΔMTR are shown. Use the given information to determine if ΔPGB is similar to ΔMTR. If they are similar, write a similarity statement. If they are not, explain why.



Yes. $\Delta PGB \sim \Delta MTR$.

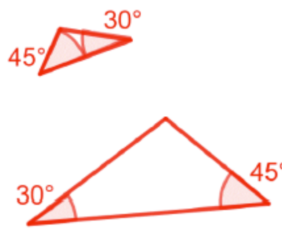
While monitoring students, make note of different methods of determining similarity. When discussing this question as a group, call on students to share their different strategies of thinking. Hearing different methods from peers can help resolve misconceptions that some students have.

6.7.4 Activity: Stations

Component Overview: *Students will draw pairs of triangles from given descriptions. Provide an opportunity for students to observe the drawings of other partner groups to make connections to which descriptions produced similar triangles. Students will then name the similarity theorems for all descriptions that produced similar images.*

6.7.4 Activity: Stations

1. With your partner, create two triangles using the description provided. Try to make the two triangles as different as possible.

Description Letter	Drawing
A	
B	

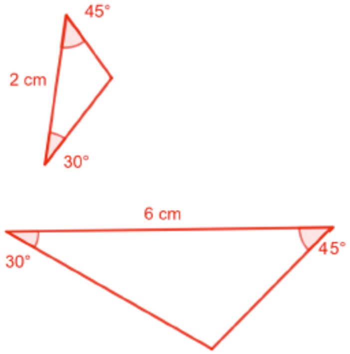
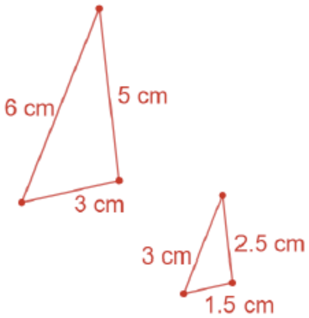
When drawing the triangles, if the side measures are not given in the description, students should draw their triangles with sides no shorter than 2 cm.

After the triangles are drawn using the descriptions, students should then label all of the sides and angles with their measurements.

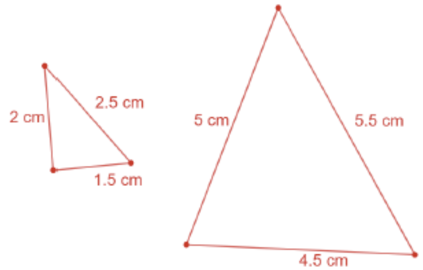
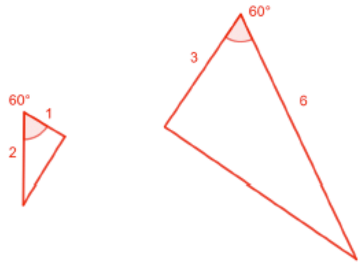
6.7.4 Activity can be done on large poster or chart paper, as stations that partner groups move through, or with activity cards that are passed from partner group to partner group. Consider printing multiple sets of descriptions and dividing the class into multiple partner groups.

Consider only giving descriptions A-E to all students for 6.7.4 Activity and using F and G for those groups who finish early.

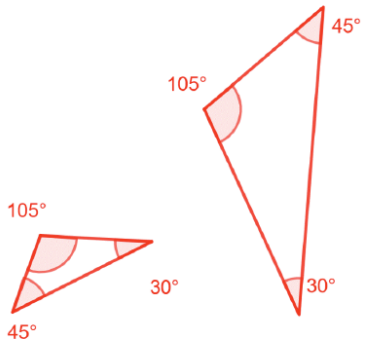
Students will need a ruler with centimeters and a protractor.

Description Letter	Drawing
C	
D	

6.7.4 Activity: Stations (continued)

Description Letter	Drawing
E	
F	

6.7.4 Activity: Stations (continued)

Description Letter	Drawing
G	

2. Which descriptions resulted in similar triangles?
   B, C, D, F, and G are similar figures.

Creating triangles with only angles can be difficult. If students struggle, consider suggesting they draw the three angles separately on patty paper, cut them out, and then slide the pieces together to create the triangles.

6.7.4 Activity: Stations (continued)

Angle-Angle (AA~) Theorem

If two angles of one triangle are congruent to two angles of another triangle, then the two triangles are similar.

Side-Side-Side Similarity (SSS~) Theorem

If the lengths of the corresponding sides of two triangles are proportional, then the triangles are similar.

Side-Angle-Side Similarity (SAS~) Theorem

If the lengths of two sides are proportional and their included angles are congruent on two different triangles, then the triangles are similar.

3. Match each description with the postulate or theorem that describes it. If none describe it, then write "none."

Description Letter	Theorem
A	*None*
B	*Angle-Angle Similarity*
C	*Angle-Angle Similarity*
D	*Side-Side-Side Similarity*
E	*None*
F	*Side-Angle-Side Similarity*
G	*Angle-Angle Similarity*

Consider having students develop a Foldable, Frayer Model, or Word Wall that includes all these terms. Possibly categorize these theorems with others they have learned throughout the course so far.

What similarities and differences do you notice between these similarity theorems and the congruence theorems learned in Unit 5?

To provide a challenge, for the descriptions that did not result in similar triangles, have students write what additional information that, if provided, would have resulted in similar triangles.

6.7.5 Wrap-Up: Cool Down

Component Overview*: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

6.7.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.
   Answers will vary.

This would be a great opportunity to explore more about what the students may still have questions about or predict where this topic may go next. Have students predict questions they may see in the next lesson or on the next assessment.

6.8 Proving Triangle Similarity – Part 1

Lesson Introduction

Benchmark	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.	Learning Target	• I can use triangle similarity to prove triangles are similar.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How do you prove two triangles are similar? • Why isn't there an AAA, AAS, or an ASA similarity theorem?
Lesson Narrative	This lesson brings together the information in Lessons 6 and 7 to write formal proofs to conclude that two triangles are similar. Students see that sometimes proofs can be written using different means of justification. Students also explore why there is no AAA, AAS, or ASA similarity theorem.

Component Summary				
	6.8.1 Warm-Up (~5 min.)	Writing equivalent ratios for triangular garden *(SE p. 332)*	6.8.3 Guided Instruction (15-20 min.)	Proving triangles similar using a two-column proof *(SE p. 334)*
	6.8.2 Collaboration (10 min.)	Writing similarity statements and identifying proportional and congruent parts *(SE p. 333)*	6.8.4 Try It (10 min.)	Practice proving triangles similar using a two-column proof *(SE p. 336)*
	6.8.5 Wrap-Up (~5 min.)	Ticket Out the Door completing two-column proof *(SE p. 337)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Similarity • Congruent • Proportional	• (Optional) Patty Paper • (Optional) Rulers • (Optional) Protractors	• Consider providing a bank of statements and reasons for students to select from when they are completing two-column proofs.

Teacher Tips

Common Misconceptions	Things to Consider
• Students may write similarity statements incorrectly by not having the corresponding vertices in the same order. • Students may mistake naming triangles with similarity statements where the order of corresponding parts is necessary.	• Students may use transitive or substitution properties of equality interchangeably. Clarify your expectations with your students. • 6.8.4 Try It is the first place that the Side-Side-Side Similarity Theorem is used. This was done to help students transfer learning. Use observational data to determine whether to make the questions in 6.8.4 Try It part of 6.8.3 Guided Instruction. If doing this, then consider how to monitor class time. • If time is an issue, then consider assigning 6.8.4 Try It for homework.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share 6.8.3 Guided Instruction is a great place to offer students an opportunity to work together to process their thinking. This is where students will be able to share different strategies to reach the same conclusion. Students should first work alone before pairing with others. Then have students share with the whole class. This move builds students' confidence before sharing with the entire class.	**MA.K12.MTR.4.1: Discussion and Reflection** 6.8.2 Collaboration offers students many opportunities to work together and discuss key mathematical concepts and practices for this lesson. The partnered approach continues for 6.8.3 Guided Instruction. While students complete question 1d look for opportunities to develop students' ability to justify methods and compare their response to the responses of their peers. Offer praise and support to students, especially to those whose confidence level may not be as high as other students.

Differentiate Instruction

Students who struggle with writing proofs will benefit from the scaffolding provided in the proofs of this lesson. Scaffolding is seen in the two-column proofs with the inclusion of some of the statements and reasons. Some students are ready for a challenge and can be given the proofs on separate paper without the scaffolds built in.

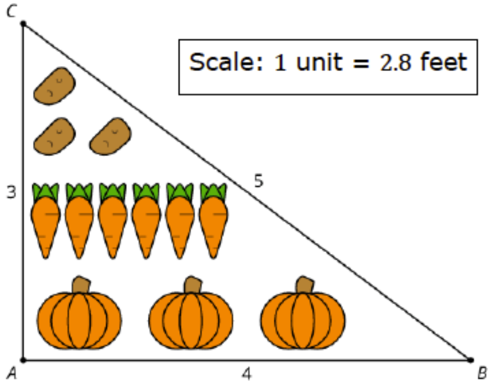
Lesson Notes:

6.8.1 Warm-Up: Vegetable Garden

Component Overview: *Students write equivalent ratios after making observations about a vegetable garden.*

6.8.1 Warm-Up: Vegetable Garden

1. ΔABC is used to create a plan for a vegetable garden.



   a. What do you notice about the plan?
   Answers will vary. I notice the plan is a triangle. I notice that they could plant more by changing how they are arranged.

   b. Write at least three equivalent ratios or equations using lengths from both the diagram and the full-size garden.
   Answers will vary.
   $\frac{3}{8.4} = \frac{5}{14}; \frac{4}{11.2} = \frac{5}{14}; \frac{11.2}{4} = \frac{8.4}{3}$

As time allows, have a variety of students share their responses and reasoning with the whole class. Consider asking students if they have any personal experience planting gardens.

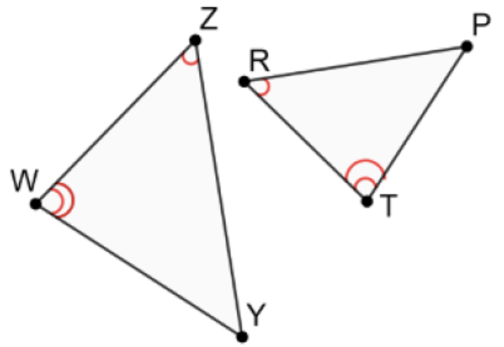
6.8.2 Collaboration: Proving Triangle Similarity

Component Overview: *After marking congruent parts on two triangles, students write a similarity statement and identify other proportional and congruent parts.*

6.8.2 Collaboration: Proving Triangle Similarity

With your partner, answer the following questions.

1. ΔWZY and ΔRPT are shown. $\angle WZY \cong \angle TRP$ and $\angle ZWY \cong \angle RTP$.



   a. Mark the figure using the given information.

   b. What similarity theorem can be applied to show that the triangles are similar?
   Angle-Angle Similarity

   c. Write the similarity statement.
   $\Delta WZY \sim \Delta TRP$

   d. Complete each.

   $\frac{WZ}{TR} = \frac{YW}{PT}$ $\qquad$ $\frac{ZY}{RP} = \frac{ZW}{RT}$ $\qquad$ $\angle WYZ \cong \angle TPR$

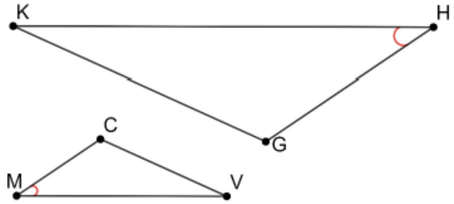
For students who need support in understanding similarity, consider providing these students with patty paper, rulers, and protractors to further investigate the two triangles.

When writing a similarity statement students may use the order of the vertices given in the stem. Help students understand that when just naming triangles order of correspondence is not necessary, but when writing congruence or similarity statements it is necessary.

Why is it possible for your similarity statement to not match your partner's similarity statement and for both to still be correct?

6.8.2 Collaboration: Proving Triangle Similarity (continued)

2. ΔKHG and ΔMVC are shown. $\angle KHG \cong \angle VMC$ and $\frac{KH}{VM} = \frac{GH}{CM}$.



a. Mark the figure using the given information.

b. What similarity theorem can be applied to show that the triangles are similar?
Side-Angle-Side Similarity Theorem

c. Write the similarity statement.
$\Delta KHG \sim \Delta VMC$

d. Complete each.

$\frac{KG}{CV} = \frac{GH}{CM}$ $\angle HGK \cong \angle MCV$ $\angle CVM \cong \angle GKH$

How can these proportional statements be rewritten and still be true?

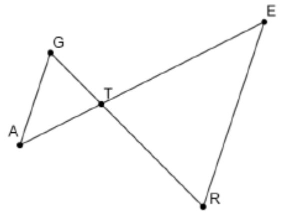
Students should regularly be encouraged to develop and share models and tools being used to identify the correct theorem throughout this lesson. Students have many opportunities to work with partners and many tools to pick from. A well-organized "tool box" may help them approach each question with confidence.

6.8.3 Guided Instruction: Proving Triangle Similarity

Component Overview: *Students begin reasoning through a two-column proof with a partner, discussing why different methods of proving the conclusion can be used. Students discuss why proving similarity does not need all three angles and how this connects to there being no ASA or AAS similarity theorems. The component concludes with writing a two-column proof to prove the similarity of two triangles.*

6.8.3 Guided Instruction: Proving Triangle Similarity

1. ΔGTA and ΔTER are shown, and $\overline{GA} \parallel \overline{ER}$.



   a. With your partner, discuss what conclusions can be drawn using the given information. Summarize your discussion and explain how you reached each conclusion.
   Answers will vary. We concluded that $\angle AGT \cong \angle ERT$ by alternate interior angles formed by parallel segments $\overline{GA}$ and $\overline{ER}$ and transversal $\overline{GR}$. We concluded that $\angle GAT \cong \angle RET$ by alternate interior angles formed by parallel segments $\overline{GA}$ and $\overline{ER}$ and transversal $\overline{AE}$. We concluded that $\angle GTA \cong \angle RTE$ by the vertical angles theorem and $\Delta GTA \sim \Delta RTE$ by AA similarity.

   b. Which similarity postulates should be used to prove $\Delta GTA \sim \Delta RTE$?
   Angle-Angle Similarity

   c. Complete the two-column proof. *Answers may vary.*

Statement	Reason
1. $\overline{GA} \parallel \overline{ER}$	1. *Given*
2. $\angle AGT \cong \angle ERT$	2. If two parallel lines are intercepted by a transversal, then alternate interior angles are congruent.
3. $\angle GAT \cong \angle TER$	3. If two parallel lines are intercepted by a transversal, then alternate interior angles are congruent.
4. $\Delta GTA \sim \Delta RTE$	4. *Angle-Angle Similarity*

   d. Ask two classmates about their proof. Record their step 3 statement and reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
   Answers will vary.

Consider allowing students to work with their partners for all of questions 1a-1e before bringing them together for whole group discussion. This will give students an opportunity to refine their thinking before sharing with the whole group.

Collect and Display
Consider utilizing this language routing when discussing question 1a with the whole class. Direct students to justify their conclusions using formal mathematical language. While the students share their conclusions, display their responses for all to see. Probe students to include which transversal results in congruent alternate interior angles.

6.8.3 Guided Instruction: Proving Triangle Similarity (continued)

Step 3 Statement	Step 3 Reason	Initials

e. Discuss with your partner the two different ways that $\Delta GTA \sim \Delta RTE$ can be proved.
One method is stating that $\angle GAT \cong \angle TER$ and $\angle AGT \cong \angle ERT$ using alternate interior angles and then stating the triangles similar by AA similarity. Another method is stating either $\angle GAT \cong \angle TER$ or $\angle AGT \cong \angle ERT$ using alternate interior angles and stating that $\angle ATG \cong \angle RTE$ by vertical angles and then stating the triangles similar by AA similarity.

f. Explain why you do not need to state three congruent angles in your two-column proof.
If two angles in a triangle are congruent to two angles in another triangle then the third must also be congruent. Therefore, only two angles are needed to be shown congruent for the triangles to be similar.

g. Arian and Isabella were discussing why there is no Angle-Angle-Angle Similarity Theorem.

Arian	Isabella
I think once you know two pairs of angles in a triangle are congruent, then you know the last pair are congruent since the sum of all of the angles of a triangle are $180°$.	I think once you know two pairs of angles in a triangle are congruent, then you know all of the sides are proportional.

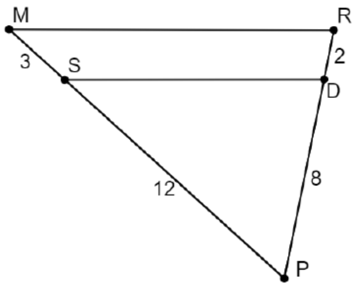
Discuss with your partner why Arian and Isabella's arguments could also be applied to show why Angle-Side-Angle similarity and Angle-Angle-Side similarity are not needed to prove similarity. Summarize your discussion.
Having two pairs of congruent angles is sufficient to prove triangles similar. It is not necessary to have any of the side measures in this case to prove similarity.

Monitor while students are discussing and sharing their step 3 statements and reasons. Look for opportunities to develop students' ability to justify methods and compare their responses to the responses of their peers. Offer praise and support to students, especially to those whose confidence level may not be as high as other students.

Question 1e is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine students' thinking to highlight or their misconceptions to address.

2. ΔSPD and ΔMPR are shown.

Given: $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$
Prove: $\Delta SDP \sim \Delta MRP$



Statement	Reason
1. $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$	1. *Given*
2. $MS + SP = PM$ $RD + DP = RP$	2. *Segment Addition Postulate*
3. $PM = 15$ $RP = 10$	3. Substitution Property of Equality
4. $\frac{PS}{PM} = \frac{12}{15}$ $\frac{PD}{PR} = \frac{8}{10}$	4. *Division Property of Equality*
5. $\frac{PS}{PM} = \frac{PD}{PR}$	5. *Substitution Property of Equality*
6. $\angle P \cong \angle P$	6. *Reflexive Property of Equality*
7. $\Delta SDP \sim \Delta MRP$	7. *Side-Angle-Side Similarity*

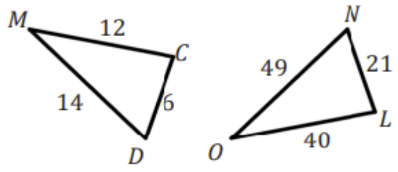
Consider implementing a silent board game for this proof. Display the proof at the front of the room and have students volunteer to go to the board to complete one line of the proof. Students do not talk during this activity. This game allows students who don't see all the reasons in the entire proof an entry-point, as they may know some of the reasons.

6.8.4 Try It: Triangle Similarity

Component Overview: *Students consider reasoning to determine if two triangles are similar given only the side lengths before completing a two-column proof proving that two triangles are similar.*

6.8.4 Try It: Triangle Similarity

1. Amanda and Tyler are determining if $\Delta MCD \sim \Delta OLN$ is true. Their reasonings are shown.

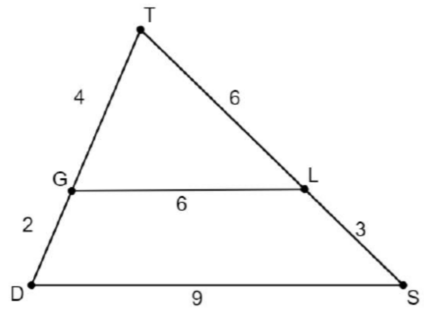


Amanda's Answer	Tyler's Answer
No, the triangles are not similar because the corresponding sides are not proportional.	Yes, the triangles are similar because the corresponding sides are proportional.

Who do you agree with? Explain your reasoning.
I agree with Amanda because when I divided the side lengths of ΔMCD by the corresponding side lengths of ΔOLN, they are not equal.

During this component is the first time students will use SSS similarity in the lesson. Listen as students work through question 1 with their partner to identify misconceptions about what's needed to prove similarity given the side lengths before students move on to the proof in question 2.

2. ΔTLG and ΔTSD are shown.
   Given: $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$
   Prove: $\Delta TLG \sim \Delta TSD$



Statement	Reason
1. $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$	1. *Given*
2. $TG + GD = TD$ $TL + LS = TS$	2. *Segment Addition Postulate*
3. $TD = 6,\ TS = 9$	3. Substitution Property of Equality
4. $\frac{TG}{TD} = \frac{4}{6}, \frac{TL}{TS} = \frac{6}{9}, \frac{GL}{DS} = \frac{6}{9}$	4. *Division Property of Equality*
5. $\frac{TG}{TD} = \frac{TL}{TS} = \frac{GL}{DS}$	5. *Substitution Property of Equality*
6. $\Delta TLG \sim \Delta TSD$	6. *Side-Side-Side similarity*

Students may have a hard time knowing what reasons and statements to fill in. Consider providing an answer choice bank for students to choose from for this component.

For Reason 3, students may also use the substitution property of equality.

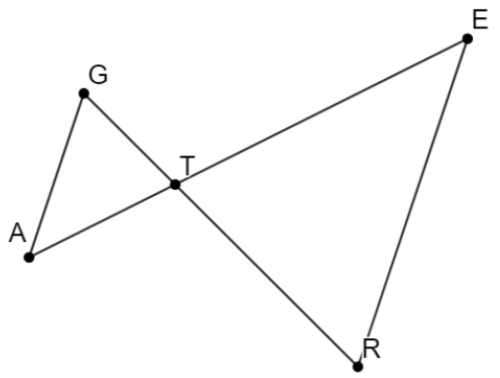
Question 2 is the first place where students encounter the Side-Side-Side Similarity Theorem used in a proof. This was done to help them transfer learning. If your students are not quite ready for such a challenge, then consider making this question more guided.

6.8.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students fill in statements and reasons in a two-column proof to demonstrate that the given triangles are similar.*

6.8.5 Wrap-Up: Ticket Out the Door

1. ΔGTA and ΔTER are shown.



Given: $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$
Prove: $\Delta GTA \sim \Delta RTE$

Complete the two-column proof.

Statement	Reason
1. $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$	1. Given
2. $\frac{TA}{TE} = \frac{2}{3}$, $\frac{TG}{RT} = \frac{2}{3}$	*2. Division Property of Equality*
3. $\frac{TA}{TE} = \frac{TG}{RT}$	3. Transitive Property of Equality
4. $\angle GTA \cong \angle RTE$	*4. Vertical angles are congruent*
5. $\Delta GTA \sim \Delta RTE$	*5. Side-Angle-Side Similarity*

Consider using the responses here to determine remediation needs on proving triangles are similar.

6.9 Proving Triangle Similarity – Part 2

Lesson Introduction

Benchmark	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse Leg.	Learning Target	• I can use triangle similarity to prove that triangles are similar.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• What criteria are needed to prove two triangles are similar? • Do you need to know the measure of all sides or all angles to prove triangles similar?
Lesson Narrative	This lesson is an opportunity for students to apply the criteria for proving two triangles similar in writing a two-column proofs. These proofs are focused on overlapping and shared side triangles. Students will be challenged to use any reasoning learned in previous lessons and units to draw the conclusion.

Component Summary	6.9.1 Warm-Up (~5 min.)	Choosing between two future forms of transportation *(SE p. 338)*	6.9.2 Activity (35-40 min.)	Writing proofs of triangle similarity *(SE p. 338)*
	6.9.3 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 342)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Similarity • Congruent • Proportional	• (Optional) Chart Paper • (Optional) Markers • (Optional) Task Cards	• Consider making multiple sets of the proofs regardless of implementing a Gallery Walk, stations, task cards, or other structure.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may think there is only one way to order the statements in a proof.</td><td>Consider the different ways of grouping/pairing students for this lesson based on students’ strengths and level of understanding. Depending on the choice of structure for 6.9.2 Activity, consider whether it is better to group the students homogeneously or heterogeneously. Consider also making a small teacher-led group if needed.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Correct
When reviewing the completed proofs from 6.9.2 Activity, provide students with an opportunity to analyze the completed work of other students. The intent is to prompt student reflection on the work of others to fortify their own reasoning and to engage students in meaningful conversation.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
6.9.2 Activity is designed for students to discuss and reflect on a variety of proofs. Students will analyze the mathematical thinking of others and compare the efficiency of their process. Through this discussion, students develop skills in communicating mathematical ideas, methods, and vocabulary.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">6.9.2 Activity is an opportunity for students to access prior knowledge and practice the skill of writing proofs using different approaches. For students who struggle with proofs, consider assigning only proofs A-D or providing some of the placement of reasons and statements on proofs B, C, and F.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.9.1 Warm-Up: Would You Rather?

Component Overview: *Students choose between two different options of futuristic vehicles and justify their choice.*

6.9.1 Warm-Up: Would You Rather?

1. In 2021, General Motors introduced the eVTOL air-taxi and the PAV (personal autonomous vehicle), a self-driving car. Both are concepts with no known timeline for production.

   a. Do you think a drone-like personal vehicle like the eVTOL air-taxi will be reality in your lifetime?
   Answers will vary. Yes, people are making lots of cool stuff and this should be a great idea.

   b. Which would you rather have as your vehicle, an air-taxi or a self-driving car, in the future?
   Answers will vary. I would rather have a self-driving car.

   c. Explain your choice.
   Answers will vary. I think that self-driving vehicles are safer.

Consider doing research on the latest news on air-taxis, PAVs, and self-driving cars. Use what you learn to further engage students in conversations about the future of travel.

6.9.2 Activity: Proving Triangle Similarity

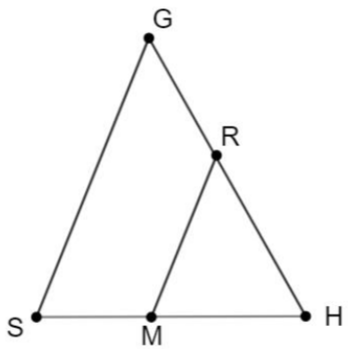
***Component Overview**: Students write formal two-column proofs justifying why two triangles are similar from given information.*

6.9.2 Activity: Proving Triangle Similarity

With your partner, sort through the statement and reason cards. Then, write each proof in the tables provided.

Proof A

Given: $\overline{GS} \parallel \overline{RM}$
Prove: $\Delta RHM \sim \Delta GHS$

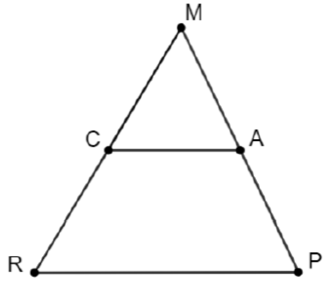


Statement	Reason
1. $\overline{GS} \parallel \overline{RM}$	1. *Given*
2. $\angle H \cong \angle H$	2. *Reflexive Property of Congruence*
3. $\angle GSH \cong \angle RMH$	3. *If two parallel lines are intersected by a transversal, the corresponding angles are congruent*
4. $\Delta RHM \sim \Delta GHS$	4. *Angle-Angle Similarity Theorem*

Consider using one of the following implementation options for this component:

- *Stations – The proofs are stationed around the room with the accompanying statement and reason cards. Groups rotate through the stations, place the cards in the correct position, and then record their answers in their workbooks. Don't forget to remind the students to mix the cards up when leaving the station.*
- *Gallery Walk – The proofs are posted on poster paper around the room along with tape. ALL statement and reason cards are cut up, mixed up, and distributed to the student groups/pairs. For example, there are 48 cards, so a class of 24 students placed in pairs will be assigned four cards. Students rotate around the room and "build" the proofs by adding statements and reasons to the appropriate proof and on the appropriate lines.*
- *Partner Pair Share – Groups of two students are assigned three proofs, either Proofs A-C or Proofs D-F. After the pairs complete their proofs, they pair with another pair who completed the other set of proofs and explain to each other their assigned proofs.*
- *Partner Switch – Pairs of students work together to complete some or all of the proofs. Students then switch partners and verify each other's proofs.*

Any proofs not completed can be assigned for homework or as a station on another day.

Proof B	
Given: $MC = \frac{2}{5}MR$ and $MA = \frac{2}{5}MP$ Prove: $\Delta MCA \sim \Delta MRP$	
Statement	**Reason**
1. $MC = \frac{2}{5}MR$ and $MA = \frac{2}{5}MP$	1. *Given*
2. $\frac{MC}{MR} = \frac{2}{5}, \frac{MA}{MP} = \frac{2}{5}$	2. *Division Property of Equality*
3. $\frac{MC}{MR} = \frac{MA}{MP}$	3. *Substitution Property*
4. $\angle M \cong \angle M$	4. *Reflexive Property of Congruence*
5. $\Delta MCA \sim \Delta MRP$	5. *Side-Angle-Side Similarity Theorem*

Students may choose where to put statements showing the reflexive property anywhere in the proof prior to the conclusion statement.

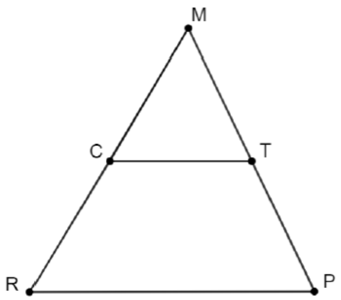
Students may interchange the substitution or transitive property reason statements in Proofs B, C, E, F, and G.

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof C

Given: $\angle RPT$ and $\angle CTP$ are supplementary

Prove: $\Delta MCT \sim \Delta MRP$



Statement	Reason
1. $\angle RPT$ and $\angle CTP$ are supplementary	1. Given
2. $m\angle RPT + m\angle CTP = 180°$	2. Definition of Supplementary Angles
3. $\angle CTM$ and $\angle CTP$ are supplementary	3. If two angles form a linear pair, they are supplementary
4. $m\angle CTM + m\angle CTP = 180°$	4. Definition of Supplementary Angles
5. $m\angle CTM + m\angle CTP = m\angle RPT + m\angle CTP$	5. Transitive Property of Equality
6. $m\angle CTM = m\angle RPT$	6. Subtraction Property of Equality
7. $\angle CTM \cong \angle RPT$	7. Definition of Congruence
8. $\angle M \cong \angle M$	8. Reflexive Property of Congruence
9. $\Delta MCT \sim \Delta MRP$	9. Angle-Angle Similarity Theorem

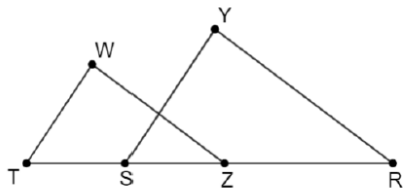
As an extension problem, have students write this proof using a different set of statements and reasons. For example, this proof can also be written by proving $\overline{CT}$ is parallel to $\overline{RP}$ and then using angle pair relationships to establish congruence.

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof D

Given: $\overline{WT} \parallel \overline{YS}$,
$\overline{WZ} \parallel \overline{YR}$

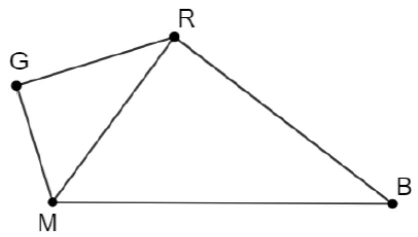
Prove: $\Delta WZT \sim \Delta YRS$



Statement	Reason
1. $\overline{WT} \parallel \overline{YS}$	1. *Given*
2. $\angle WTZ \cong \angle YSR$	2. *If two parallel lines are intersected by a transversal, then corresponding angles are congruent*
3. $\overline{WZ} \parallel \overline{YR}$	3. *Given*
4. $\angle WZT \cong \angle YRS$	4. *If two parallel lines are intersected by a transversal, then corresponding angles are congruent*
5. $\Delta WZT \sim \Delta YRS$	5. *Angle-Angle Similarity Theorem*

Encourage students to think outside the box and draw the lines of each side well outside of the shape. This could help students see how $\overline{TR}$ is a transversal.

6.9.2 Activity: Proving Triangle Similarity (continued)

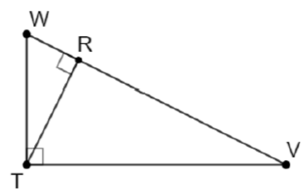
Proof E	
Given: : $GM = \frac{3}{5}RM$, $RG = \frac{3}{5}BR$, $RM = \frac{3}{5}BM$ Prove: $\Delta MGR \sim \Delta MRB$	
Statement	**Reason**
1. $GM = \frac{3}{5}RM$, $RG = \frac{3}{5}BR$ $RM = \frac{3}{5}BM$	1. *Given*
2. $\frac{GM}{RM} = \frac{3}{5}, \frac{RG}{BR} = \frac{3}{5}, \frac{RM}{BM} = \frac{3}{5}$	2. *Division Property of Equality*
3. $\frac{GM}{RM} = \frac{RG}{BR} = \frac{RM}{BM}$	3. *Substitution Property*
4. $\Delta MGR \sim \Delta MRB$	4. *Side-Side-Side Similarity Theorem*

Although this is a short proof, students may need more support on this proof because the statements are algebraically based.

Proof F

Given: $\overline{WT} \perp \overline{VT}$ and $\overline{WV} \perp \overline{RT}$

Prove: $\Delta\, RTW \sim \Delta TVW$



Statement	Reason
1. $\overline{WT} \perp \overline{VT}$	1. *Given*
2. $\angle VTW$ *is a right angle*	2. *Definition of Perpendicular Lines*
3. $m\angle VTW = 90°$	3. *Definition of Right Angle*
4. $\overline{WV} \perp \overline{RT}$	4. *Given*
5. $\angle TRW$ *is a right angle*	5. *Definition of Perpendicular Lines*
6. $m\angle TRW = 90°$	6. *Definition of Right Angle*
7. $m\angle TRW = m\angle VTW$	7. *Substitution Property*
8. $\angle TRW \cong \angle VTW$	8. *Definition of Congruence*
9. $\angle W \cong \angle W$	9. *Reflexive Property of Congruence*
10. $\Delta\, RTW \sim \Delta TVW$	10. *Angle-Angle Similarity Theorem*

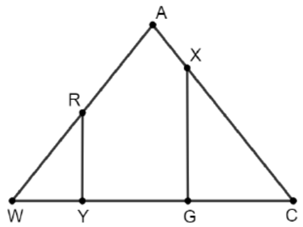
The students may struggle with the overlapping of the triangles in this image. This proof may be more manageable if students separate the triangles by copying each onto another piece of paper.

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof G

Given: $\overline{WA} \cong \overline{CA}$, $\overline{RY} \perp \overline{WC}$, $\overline{XG} \perp \overline{WC}$

Prove: $\Delta RYW \sim \Delta XGC$



Statement	Reason
1. $\overline{WA} \cong \overline{CA}$	1. *Given*
2. $WA = CA$	2. *Definition of Congruence*
3. ΔWAC *is Isosceles*	3. *Definition of Isosceles Triangles*
4. $m\angle AWC = m\angle ACW$	4. *Definition of Isosceles Triangles*
5. $\angle AWC \cong \angle ACW$	5. *Definition of Congruence*
6. $\overline{RY} \perp \overline{WC}$	6. *Given*
7. $\angle RYW$ *is a right angle*	7. *Definition of Perpendicular Lines*
8. $m\angle RYW = 90°$	8. *Definition of Right Angle*
9. $\overline{XG} \perp \overline{WC}$	9. *Given*
10. $\angle XGC$ *is a right angle*	10. *Definition of Perpendicular Lines*
11. $m\angle XGC = 90°$	11. *Definition of Right Angle*
12. $m\angle RYW = m\angle XGC$	12. *Substitution Property*
13. $\angle RYW \cong \angle XGC$	13. *Definition of Congruence*
14. $\Delta RYW \sim \Delta XGC$	14. *Angle-Angle Similarity Theorem*

While students may choose to write all the given statements at the beginning of the proof, encourage them to write only one given statement and then the statements that can be concluded from that one. This move will help students make connections if they still see proofs as disconnected phrases.

For students who finish early, ask them to name the transformations that would show the two triangles are similar. For Honors Geometry, this would be a good segue into Lesson H3 in Unit 6.

6.9.3 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

6.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about proving triangle similarity.
   Answers will vary.

6.H3 Justifying Similarity

Lesson Introduction

Benchmark	MA.912.GR.2.9 Justify the criteria for triangle similarity using the definition of similarity in terms of non-rigid transformations.	Learning Target	• I can use the definition of similarity to justify that SSS similarity, SAS similarity, and AA similarity are sufficient to show that two triangles are similar.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• What makes a figure similar to another? • How can you justify that two figures are similar through transformations?
Lesson Narrative	This lesson guides students to justify the definition of similarity. Through preimages and images, students identify that the figures are similar if a sequence of transformations occurs that includes a dilation.

Component Summary				
	6.H3.1 Warm-Up (~5 min.)	Choosing which image does not belong *(SE p. 343)*	6.H3.2 Collaboration (35-40 min.)	Justifying similarity through transformations *(SE p. 344)*
	6.H3.3 Wrap-Up (~5 min.)	Ticket Out the Door drawing similar figures *(SE p. 348)*		

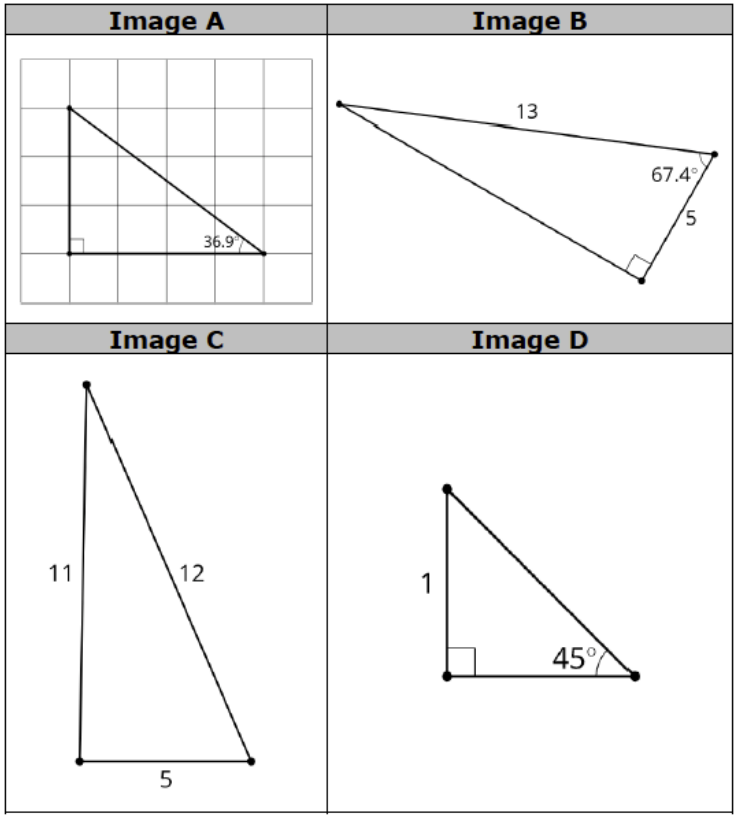
Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Similarity in Terms of Non-Rigid Transformations • Similarity <u>Additional Terms:</u> • Dilation • Translation • Reflection • Rotation • Congruent • Corresponding	• Patty Paper	• Consider creating a rubric to evaluate students' responses to 6.H3.3 Wrap-Up.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• Students may think SSA is enough to justify similarity.</td><td>• Consider prompting students who may have a hard time understanding the motion of each vertex to draw the mapping for each transformation.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
6.H3.2 Collaboration question 1a directly asks students to record what they notice. Allow students the opportunity to express some of their wonderings before moving on. Questioning students about what they wonder can pique their curiosity and increase engagement in the lesson.</td><td>MA.K12.MTR.2.1: Multiple Representations
The transformations in this lesson include visuals, written descriptions, and algebraic descriptions of the same transformations. Providing these multiple representations encourages students to demonstrate understanding and make connections.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The structure of this lesson highlights the critical features of transformations that produce similar figures. As students progress from identifying transformations to interpreting the algebraic representations of the transformations, they continually refer back to the fundamental similarity definitions and theorems. In the middle of this lesson, students are given the opportunity to incorporate patty paper to connect these transformations into a more kinesthetic learning process.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.H3.1 Warm-Up: Which One Doesn’t Belong?

***Component Overview**: Students select the image that they believe does not belong with the other three images and justify why.*

6.H3.1 Warm-Up: Which One Doesn’t Belong?



1. Choose an image that does not belong and explain your choice.
   Answers will vary. A is different because it has a grid. B is different because it shows the lengths of two sides. C is different because it is not a right triangle. D is different because it is a $45^\circ, 45^\circ, 90^\circ$.

Students are asked to find which image doesn’t belong. However, each shape may not belong for different reasons. This question is more about student’s reasoning as to why the images do not belong.

Students may enjoy the opportunity to “convince” other students that they are correct.

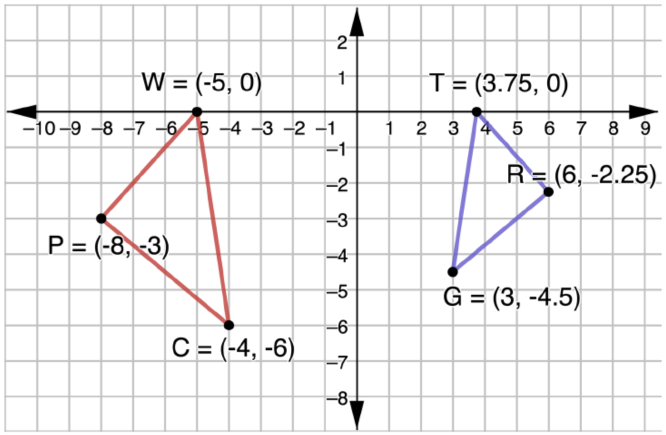
- *Have a few students explain what they selected and why.*
- *Pair up students who had different answers.*
- *Each student should make two to three solid points as to why they are correct (not why the other student is wrong). About $20-30$ seconds.*
- *Students can have “open discussion” for 30 seconds.*
- *Each student can decide to stay with their original answer or change. If they change, they should explain why.*

6.H3.2 Collaboration: Justifying Similarity

Component Overview: *Students analyze a pair of triangles and determine what transformations map one to the other and what the relationship is between the corresponding parts. Through mapping and copying figures, students are led to justifying the definition of similarity in terms of non-rigid motions.*

6.H3.2 Collaboration: Justifying Similarity

1. ΔPWC and ΔRTG are shown.



   a. What do you notice about the two triangles?
   Answers will very. I notice ΔTRG is smaller and pointing in the other direction.

   b. What transformations can be applied so that ΔWPC is mapped to ΔTRG?
   Answers will vary. A reflection across the y-axis and then a dilation centered at the origin with a scale factor of $\frac{3}{4}$.

   c. What can you conclude from the transformations?
   ΔTRG is similar, but not congruent to ΔWPC.

Omario's and Ernest's answers to the question "If ΔPWC and ΔRTG are similar by angle-angle similarity, what information would need to be given?" are shown.

Omario	Ernest
$\angle WPC \cong \angle TRG$ and $\angle WCP \cong \angle TGR$	$\angle PWC \cong \angle RTG$ and $\angle PCW \cong \angle RGT$

Notice and Wonder
Before releasing students to complete this component through collaboration, consider presenting question 1a whole-group and extending it to include "What do you wonder?" Steer the conversation to statements wondering about whether the triangles are similar and how this could be verified.

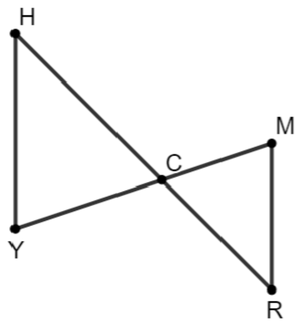
This component is purposefully scaffolded to build students' understanding and confidence as they progress through.

6.H3.2 Collaboration: Justifying Similarity (continued)

d. What do you notice about their answers?
I notice that both students are correct.

e. Explain why both students correctly answered the question.
Both students identified two pairs of congruent corresponding angles. Any two pairs of congruent corresponding angles will be enough to prove the triangles are similar.

2. ΔMRC and ΔYHC are shown.



a. Complete the table.

Given Information	True or False? $\Delta MRC \sim \Delta YHC$ by AA Similarity
$\angle HCY \cong \angle RCM$ $\angle H \cong \angle R$	○ True ○ False
$\angle HCY \cong \angle RCM$ $\frac{MR}{YH} = \frac{MC}{YC}$	○ True ○ False
$\angle H \cong \angle R$ $\angle Y \cong \angle M$	○ True ○ False

Students may think that SSA is justification for similarity. Remind students that when two sides are proportional and a pair of angles are congruent, the angle must be the included angle between the proportional sides. Consider showing a counterexample of two triangles that have SSA, but are not similar.

6.H3.2 Collaboration: Justifying Similarity (continued)

b. Select all of the transformations in a sequence that could be applied so that ΔMRC coincides with ΔYHC:
- ☐ rotate ΔMRC about point C
- ☐ rotate ΔMRC about point
- ☐ reflect ΔMRC across $\overline{MR}$
- ☐ reflect ΔMRC across $\overline{YH}$
- ☐ translate ΔMRC left and up
- ☐ dilate ΔMRC using point C as the center
- ☐ dilate ΔMRC using point R as the center

c. Discuss with your partner what is true about the triangles when all of the transformations have been applied.
Answers will vary. The triangles are congruent after all transformations are applied.

Similarity in terms of non-rigid transformations is one or more rigid transformations (reflection, rotation, translation) combined with a dilation that leads to two figures being similar.

Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.

d. Use the definition of similarity in terms of non-rigid motions to justify that ΔMRC and ΔYHC are similar.
By rotating ΔMRC about point C and then dilating it using point C as the center of dilation, ΔMRC is mapped onto ΔYHC. Since the sequence of transformations includes a dilation, the definition of similarity in terms of non-rigid motions applies.

In addition to the answer shown, another sequence of transformations that could be applied is a rotation about point R, a translation left and up, and then a dilation using point R as the center.

While monitoring discussions, listen for the use of the vocabulary 'similar'.

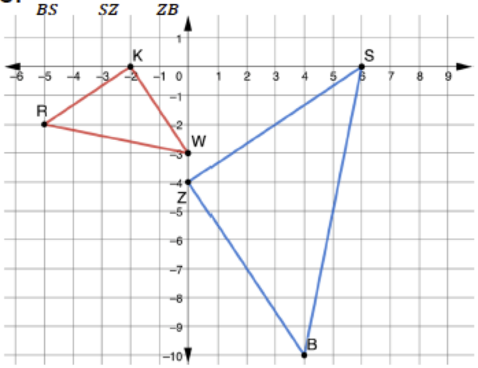
6.H3.2 Collaboration: Justifying Similarity (continued)

3. Consider the following statement.

If it is possible to find one or more transformations that will transform a figure such that all corresponding points coincide with the other figure, then the figures will be similar.

Explain what this statement means in your own words.
Answers will vary. If rigid or non-rigid motions are used to map one figure onto another then the shapes are similar.

4. ΔRKW and ΔBZS are shown.
Assume: $\frac{RW}{BS} = \frac{WK}{SZ} = \frac{KR}{ZB}$



a. Explain how you can use patty paper to verify $\Delta RKW \sim \Delta BZS$.
I can trace one of the triangles and then place each angle one at a time on top of the corresponding angle in the other triangle to determine that the angle measures are the same. Therefore, the triangles will be similar by AA similarity.

b. Write a sequence of transformations that will map ΔRKW onto ΔBZS so that $\Delta RKW \sim \Delta BZS$.
Answers will vary. Rotate ΔRKW 90° counterclockwise about the origin and then dilate the result using a scale factor of 2 with the origin as the center of dilation.

Three Reads
Consider implementing this routine by having the students read the statement three times and then discussing what the statement means prior to them writing an explanation of the meaning.

Read #1: Shared Reading centered around "what is this situation about?" (1 minute)

Read #2: Individual, Pairs, or Shared Reading centered around "what can be counted?" (3–5 minutes)

Read #3: Individual, Pairs, or Shared Reading centered around "how could we solve the problem?" (1–2 minutes)

As students are given the opportunity to develop the concept of transformations with a kinesthetic approach, the patty paper will confirm many students' suspicions about similar figures.

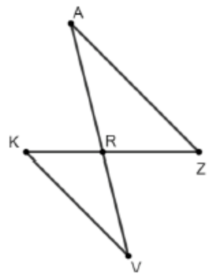
c. If it is given that ΔBZS is a result of $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (2x, 2y)$ to ΔRKW, are the triangles congruent?
No, they are not congruent.

d. If it is given that ΔBZS is a result of $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (2x, 2y)$ to ΔRKW, are the triangles similar?
Yes, they are similar.

e. Discuss with your partner what criteria would have to be given so that SAS similarity would be shown as sufficient to justify triangles are similar. Summarize your discussion.
To justify triangles are similar using SAS similarity, two pairs of corresponding sides with an included angle would need to be given. The ratios of the pairs of corresponding sides would need to be equivalent and the included angles would need to be congruent.

While monitoring students, prompt them to give an explanation for why the triangles are congruent or similar in questions 4c and 4d along with their answer.

5. Katelyn, Diego, and Amelia are writing conclusions about proving similarity in triangles. They used ΔKRV and ΔZRA to write their conclusions. Their conclusions are shown.



Assume: $\angle VKR \cong \angle AZR$ and $\angle VRK \cong \angle ARZ$

6.H3.2 Collaboration: Justifying Similarity (continued)

Katelyn's Conclusion	Diego's Conclusion	Amelia's Conclusion
To prove similarity of two triangles, apply the definition of congruence in terms of rigid motions to show that two angles that are assumed to be congruent are congruent. This is sufficient to prove similarity of two triangles.	Show that two triangles are congruent using rigid motions and dilation. Then, apply the definition of similarity in terms of non-rigid transformations. This is sufficient to prove the triangles are similar.	By applying a series of rigid transformations and a dilation, it could be shown that the triangles are similar by showing that one set of sides are proportional and assumed two angles are congruent.

a. Whose conclusion do you agree with?
Diego is correct.

b. Explain why you agree with that student.
Answers will vary. Diego correctly uses the definition of similarity in terms of non-rigid transformations.

c. Pick one student whose conclusion is incorrect. Write a note to them to help them understand their error.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

Allow the students to elaborate about Amelia's mistake. Provide sentence stems, such as:

________ made a mistake when he/she _________ because_________.

_________ should have ___________ instead of ______________.

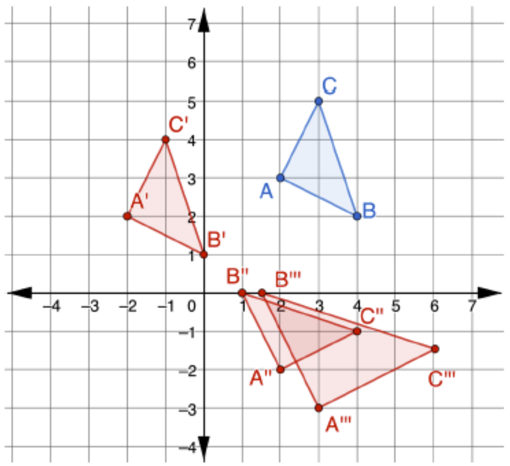
The next time _______ sees ______ he/she should ___________________.

6.H3.3 Wrap-Up: Cool Down

Component Overview: *Students perform a series of transformations and determine if the resulting figure is similar or congruent.*

6.H3.3 Wrap-Up: Cool Down

1. ΔABC is shown.



A sequence of three transformations is shown.

- $(x, y) \rightarrow (x - 4, y - 1)$
- $(x, y) \rightarrow (y, x)$
- $(x, y) \rightarrow (1.5x, 1.5y)$

a. Draw the result of each transformation.

b. Complete the table by noting if the preimage and image are congruent or similar.

Transformation	Relationship of Preimage to Image
$(x, y) \rightarrow (x - 4, y - 1)$	Congruent
$(x, y) \rightarrow (y, x)$	Congruent
$(x, y) \rightarrow (1.5x, 1.5y)$	Similar

Consider making a list of misconceptions to look for when reviewing student responses. Use these to help formulate a rubric for student responses. For example, students may be able to answer 1b correctly but make a mistake on question 1a. A student who accidently miscounts spaces on the first transformation but has all other aspects correct is not showing a lack of knowledge of the benchmark whereas a student work that shows the second transformation as a reflection across the x-axis is showing a misunderstanding.

Consider establishing your rubric and then scoring a few student answers and revising the rubric as necessary. Once your rubric is established choose a few different student responses and delete any identifying factors. Give students the rubric and 3-5 student responses and ask the students to use the rubric to provide a score for each response. Students enjoy evaluating work and using a rubric may help them consider what errors they made. If there is time, then have students share their scores for the common papers to see if students "graded" them the same.

This can also be used as a review activity at the end of the unit.

Use the data collected in 6.H3.3 Wrap-Up to determine any remediation steps needed.

6.10 Similar Figures

Lesson Introduction

Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.	Learning Target	• I can solve mathematical problems using corresponding angles and sides of similar figures.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.	N/A

Essential Questions	• How can similar figures be used to find the measures of angles? • How can similar figures be used to find the measures of sides?
Lesson Narrative	This lesson will apply the relationships of similar triangles and corresponding components from Lessons 1-9 to other similar polygons. Students will determine the measures of angles through congruence of corresponding parts and the measures of side lengths through proportional reasoning.

Component Summary				
	6.10.1 Warm-Up (~5 min.)	Proving two triangles similar *(SE p. 349)*	6.10.3 Your Turn! (15-20 min.)	Determining the measures of corresponding parts of similar figures *(SE p. 351)*
	6.10.2 Collaboration (15-20 min.)	Connecting relationships of similar triangles to similar polygons *(SE p. 349)*	6.10.4 Wrap-Up (~5 min.)	Reflection on the lesson understanding *(SE p. 353)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Similarity • Congruent • Proportional • Corresponding	• N/A	• (Optional) Consider cutting out the images and preimages from 6.10.2 Collaboration and 6.10.3 Your Turn! for students who may need the differentiated support of being able to manipulate the figures when determining corresponding sides and angles.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may match the corresponding parts in the similarity statement incorrectly.
Students may name the scale factor of ΔHRA to ΔVKM instead of from ΔVKM to ΔHRA in 6.10.2 Collaboration.</td><td>Consider offering 6.10.3 Your Turn! as a partner or individual component.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
6.10.2 Collaboration is organized to be completed with a partner. Begin this activity individually, connect with the partner, and determine the best answer. Finally share with the class what has been agreed upon. You can continue to use the Think-Pair-Share routine into 6.10.3 Your Turn!, if desired.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
6.10.2 Collaboration and 6.10.3 Your Turn! both require students to participate fully in effortful learning both with partners and independently. Students must analyze the problems and ask questions when they are in small groups. Students also will build perseverance and remain engaged while working independently.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Students may struggle with identifying the corresponding parts of similar figures. Consider providing cut out images of the figures that students can manipulate to determine the corresponding parts. After the figures have been oriented with each other, have the students complete or make a list of the corresponding sides and angles.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

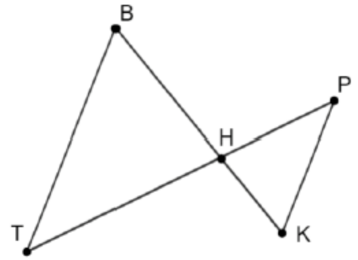
6.10.1 Warm-Up: Bell Work

Component Overview: *Students prove that two triangles are similar by completing the missing statements and reasons in a partially completed two-column proof.*

6.10.1 Warm-Up: Bell Work

1. Complete the proof.

Given: $\overline{BT} \parallel \overline{PK}$
Prove: $\Delta BHT \sim \Delta KHP$



Statement	Reason
1. $\overline{BT} \parallel \overline{PK}$	1. *Given*
2. $\angle TBH \cong \angle PKH$	2. *Alternate interior angles are congruent*
3. $\angle BTH \cong \angle KPH$	3. *Alternate interior angles are congruent*
4. $\Delta BHT \sim \Delta KHP$	4. *Angle-Angle Similarity*

Students may choose to use one of the alternate interior angle relationships and the vertical angles relationship. Both justifications are correct. Students should be encouraged to use any valid reasoning.

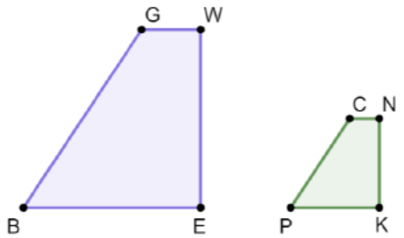
6.10.2 Collaboration: Similar Figures

Component Overview: *Students evaluate similar figures and determine unknown corresponding angles and corresponding sides.*

6.10.2 Collaboration: Similar Figures

With your partner, answer the following questions.

1. $GWEB$ and $CNKP$ are shown.



a. If $GWEB \sim CNKP$, what is true about the corresponding angles?
Corresponding angles are congruent.

b. If $GWEB \sim CNKP$, what is true about the corresponding sides?
Corresponding sides are proportional.

c. Fill in the table given that $GWEB \sim CNKP$.

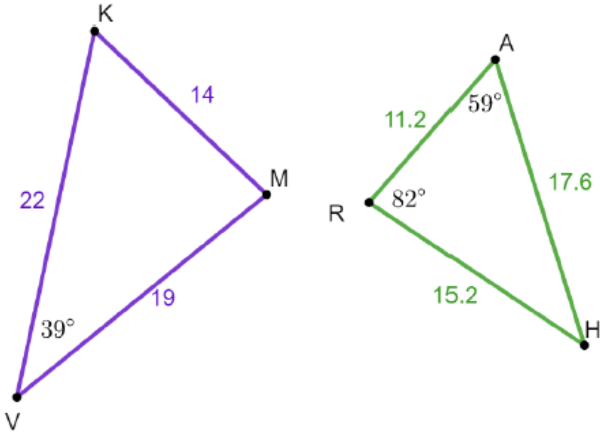
Corresponding Angles	Corresponding Sides
$\angle GBE \cong \angle CPK$	$\frac{PK}{BE} = \frac{KN}{EW}$
$\angle PKN \cong \angle BEW$	$\frac{CN}{GW} = \frac{KN}{EW}$
$\angle PCN \cong \angle BGW$	$\frac{CP}{GB} = \frac{NC}{WG}$
$\angle GWE \cong \angle CNK$	$\frac{KP}{EB} = \frac{PC}{BG}$

Think-Pair-Share
This section is intentionally designed for partner work. Consider reviewing group norms that may begin with independent thought, then lead into partnered discussion. Finally, follow up with class shared responses.

- *What is different between questions 1a and 1b?*
- *Does the word "congruent" have the same meaning as "proportional"?*
- *What is the difference between congruent and proportional?*

6.10.2 Collaboration: Similar Figures (continued)

2. ΔVKM and ΔHRA are shown. $KM = 14$, $MV = 19$, $VK = 22$, $RA = 11.2$, $AH = 17.6$, $HR = 15.2$, $m\angle KVM = 39°$, $m\angle HRA = 82°$, and $m\angle RAH = 59°$.



a. Using the given information, what can you conclude about ΔVKM and ΔHRA?
The side lengths of triangles VKM and HAR are proportional.

b. What similarity theorem or postulate could be used to prove the triangles are similar?
SSS Similarity

c. Write a similarity statement for ΔVKM and ΔHRA.
$\Delta VKM \sim \Delta HAR$

For question 2a, the order of vertices was purposely not aligned so that students cannot rely on that to determine corresponding parts.

For question 2b, students may also determine the $\angle RHA$ and then use SAS similarity. Consider having students who used different methods share with the whole group.

d. Complete the table.

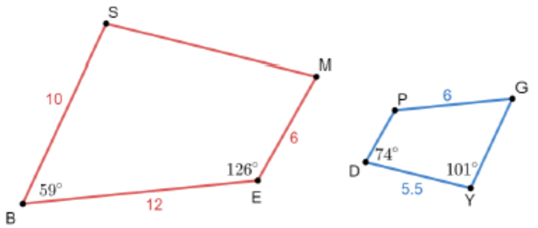
Angle Measures	Side Proportions
$m\angle VKM = 59^\circ$	$\frac{MK}{RA} = \frac{14}{11.2}$
$m\angle KMV = 82^\circ$	$\frac{MV}{RH} = \frac{19}{15.2}$
$m\angle RHA = 39^\circ$	$\frac{KV}{AH} = \frac{22}{17.6}$

e. What scale factor are the sides of ΔVKM multiplied by to find the corresponding sides of ΔHRA?
0.8 or $\frac{4}{5}$

f. If $MK = 21$, what should the length of $\overline{RA}$ be so that the triangles are similar?
16.8

If students respond to question 2e with 1.25, then they incorrectly named the scale factor of ΔHRA to ΔVKM instead of ΔVKM to ΔHRA.

3. $SMEB$ and $YDPG$ are shown. $SB = 10$, $BE = 12$, $EM = 6$, $DY = 5.5$, $PG = 6$, $m\angle SBE = 59°$, $m\angle BEM = 126°$, $m\angle PDY = 74°$, and $m\angle DYG = 101°$.



a. If $SMEB \sim YDPG$, determine the measure of each.
$m\angle BSM =$ <u>101°</u> $m\angle SME =$ <u>74°</u>
$m\angle DPG =$ <u>126°</u> $m\angle PGY =$ <u>59°</u>
$SM =$ <u>11</u> $PD =$ <u>3</u> $GY =$ <u>5</u>

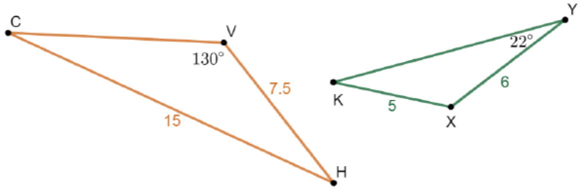
b. Explain how the similarity relationship, $SMEB \sim YDPG$, helped determine which sides of the quadrilaterals are proportional.
Answers will vary. The similarity statement tells me what angles are congruent. From this I can find corresponding sides that are proportional.

6.10.3 Your Turn!

Component Overview: *Students find unknown side lengths and angle measures using similar shapes and corresponding parts.*

6.10.3 Your Turn!

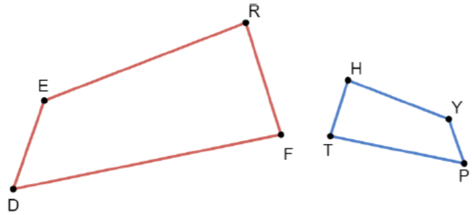
1. ΔCVH and ΔYXK are shown where $VH = 7.5$, $CH = 15$, $KX = 5$, $XY = 6$, $m\angle CVH = 130°$, and $m\angle XYK = 22°$.



   a. Given that $\Delta CVH \sim \Delta YXK$, determine the measure of each angle.
   $m\angle VCH = \underline{22°}$ $m\angle VHC = \underline{28°}$
   $m\angle YKX = \underline{28°}$ $m\angle YXK = \underline{130°}$

   b. Given that $\Delta CVH \sim \Delta YXK$, determine the measure of each side.
   $CV = \underline{9}$ $KY = \underline{10}$

2. $RFDE$ and $HTPY$ are shown, where $RE = 6.4$, $ED = 2.8$, $HY = 3.2$, $HT = 1.7$, $PT = 4$, $m\angle ERF = 86°$, $m\angle EDF = 59°$, and $m\angle HYP = 131°$.



   Assume: $RFDE \sim HTPY$

   a. Find the measure of $\angle DER$. Show your work or explain your thinking.
   $m\angle DER = 131°$ because the similarity statement tells me that $\angle DER$ corresponds to $\angle PYH$ and the $m\angle PYH = 131°$.

If students struggle with seeing the corresponding parts from the figure, consider scaffolding question 1 by first asking students to complete a list of all corresponding angles and sides by using the order of the vertices in the similarity statement. For example, use the letters in the similarity statement to determine the following correspondence.

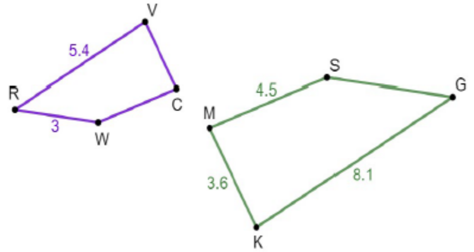
- *$\angle CVH$ corresponds to $\angle$_______.*
- *$\angle CHV$ corresponds to $\angle$_______.*
- *$\angle VCH$ corresponds to $\angle$_______.*
- *$\overline{CV}$ corresponds to ______.*
- *$\overline{CH}$ corresponds to ______.*
- *$\overline{HV}$ corresponds to ______.*

b. Determine each measure.
$m\angle HTP = \underline{84^\circ}$ $m\angle DFR = \underline{84^\circ}$
$m\angle THY = \underline{86^\circ}$ $m\angle YPT = \underline{59^\circ}$

c. What is the ratio of the corresponding sides of $RFDE$ to $HTPY$?
The ratio is $2:1$.

d. Determine each measure.
$DF = \underline{8}$ $RF = \underline{3.4}$ $YP = \underline{1.4}$

3. Kirstyn and Roberto were finding the missing side measures for $RVCW$ and $GKMS$. They were given $RVCW \sim GKMS$, $VR = 5.4$, $RW = 3$, $SM = 4.5$, $MK = 3.6$, and $KG = 8.1$.



Kirstyn's and Roberto's work to find the measure of $\overline{VC}$ is shown.

Kirstyn's Work	Roberto's Work
$\frac{8.1}{5.4} = 1.5$ $3.6 \times 1.5 = 5.4$ $VC = 5.4$	$\frac{8.1}{5.4} = 1.5$ $3.6 \div 1.5 = 2.4$ $VC = 2.4$

a. Who do you agree with?
I agree with Roberto.

b. Explain the error in the incorrect work.
Kirstyn multiplied the 1.5 *when she should have divided.*

c. Complete the statements.
$CW = \underline{3}$ $SG = \underline{4.5}$

If students don't remember the sum of the angles of a quadrilateral, spiral back to decomposing figures into triangles.

For students who need an extra challenge, task them with creating a problem with another polygon other than a quadrilateral.

6.10.4 Wrap-Up: Reflection

***Component Overview**: Students reflect on their progress throughout the lesson.*

6.10.4 Wrap-Up: Reflection

1. One thing I struggled with today was ...
   Answers will vary.
2. One thing I need to practice more is ...
   Answers will vary.
3. One thing I understand very well is ...
   Answers will vary.

Consider keeping a list of "One thing I struggled with" statements that continue reoccurring. If patterns emerge, consider reteaching these topics.

Individually praise students for the topics they understand. Consider inviting them to be an area expert on that topic for a future lesson.

6.11 Solving Problems Using Similarity – Part 1

Lesson Introduction

Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.	Learning Target	• I can solve mathematical problems involving similarity in two-dimensional figures.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.	N/A

Essential Questions	• How are similar figures used to find missing values? • How do you use scale factor to determine the length of a side?
Lesson Narrative	This lesson extends the previous lesson by using the relationships between angles and sides of similar figures to create equations. These equations are then solved to find the values of variables and missing angle and side measures. Student knowledge of scale factor is key in determining and writing the correct equation.

Component Summary				
	6.11.1 Warm-Up (~5 min.)	Applying similar triangles to model a building *(SE p. 354)*	6.11.3 Your Turn! (10-15 min.)	Determining measures using similar figures *(SE p. 356)*
	6.11.2 Collaboration (20-25 min.)	Determining measures using similar figures and error analysis *(SE p. 354)*	6.11.4 Wrap-Up (~5 min.)	Analyzing errors in sample work *(SE p. 358)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Similarity • Congruent • Proportional • Corresponding	• (Optional) Calculators	• If using “area experts” for 6.11.3 Your Turn!, create question cards for the expert to use while helping students.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may set up equations incorrectly by setting the wrong angles equal to each other.
Students may multiply by the wrong scale factor when determining lengths.</td><td>Consider utilizing “area experts” for 6.11.3 Your Turn!. While monitoring 6.11.2 Collaboration, listen and look for students who demonstrate understanding of the lesson learning targets. Assign each as the expert for one of the questions 1a through 1e. If other students have questions, they seek out the area expert as the first resource. The area expert should not give out answers, but instead assist students by looking at their work and asking guiding questions.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Stronger and Clearer Each Time
As students progress through this lesson, they may find opportunities, particularly during 6.11.2 Collaboration, to use the Stronger and Clearer Each Time routine to encourage students to develop responses that are stronger (often longer) with better justifications and clearer with more precise terms and linked, organized, complete sentences.</td><td>MA.K12.MTR1.1: Participate in Effortful Learning
6.11.3 Your Turn! offers students an opportunity to embrace the concept of effortful learning. Students can solve equations on their own or in small groups or even help others as they monitor peers for success.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson requires the use of solving multi-step equations. If students still struggle with this skill, consider using the following lessons found in the On-Ramp to Algebra 1:
Solving Two-Step Equations (Integers)
Solving Two-Step Equations (Rational Numbers)
Solving Multi-Step Equations</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

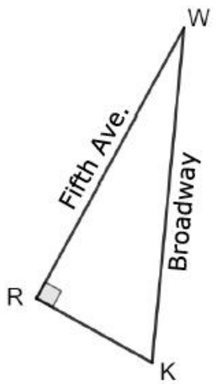
6.11.1 Warm-Up: The Flatiron Building

***Component Overview**: Students consider how similar figures might be used in the real world.*

6.11.1 Warm-Up: The Flatiron Building

The Flatiron Building, designed by Daniel Burnham and built in 1902, is a triangular 22-story building located in New York City. The building sits on a triangular block formed by 5th Avenue, Broadway, and East 22nd Street. The building got its name from an old-fashioned flat iron.

Although the ground base appears to be an isosceles triangle, the base of the Flatiron Building is actually a right triangle. Morgan used a map to draw a triangle that models the base of the building.



1. What other measurement should Morgan check to be sure her drawing is similar to the building?
   Answers will vary. Morgan should check $\angle RWK$ or $\angle WKR$ for congruence.

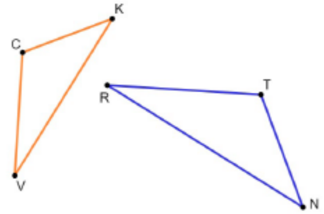
In what industries or specific situations might similar figures be useful?

6.11.2 Collaboration: Similar Figures

Component Overview: *Students will use the relationship between the similar figures to create equations needed to solve for unknown variables, angles, and side lengths. Close attention should be paid to what the scale factor represents and how it is used in the equations.*

6.11.2 Collaboration: Similar Figures

1. ΔCKV and ΔTNR are shown, where $\Delta CKV \sim \Delta TNR$.



a. Some measurements of the angles in ΔCKV and ΔTNR are given.
$m\angle TRN = 28°$, $m\angle VCK = (8x + 26)°$, and $m\angle RTN = (12x - 8)°$.

Determine each.
$x =$ 8.5 $m\angle CKV =$ 58°

Common Misconception
If students found the value of x to be 0.25 or 3, then they set the wrong angles equal to each other in the equation.

Some measurements of the sides of ΔCKV and ΔTNR are given.

$VK = 8$, $CK = 5z + 1.7$, $CV = 5.4$, $TR = 2y - 1.25$, $RN = 10$, and $TN = 5.25$.

b. What is the scale factor from ΔCKV to ΔTNR?
$\frac{5}{4}$

c. Determine the value of each variable.
$z =$ 0.5 $y =$ 4

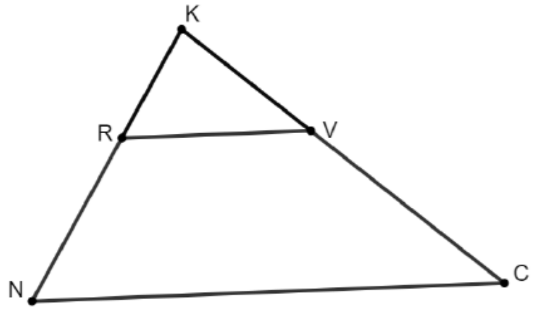
d. Determine the measure of each side.

$CK =$ 4.2 $TR =$ 6.75

Consider prompting struggling students, if needed.

- *Which triangle is the larger of the two?*
- *Is a side length of triangle CKV larger or smaller than the corresponding side length of triangle TNR?*
- *Should the scale factor that will be multiplied by the sides of triangle CKV be larger than 1 or between 0 and 1 to find the corresponding side length of TNR?*

2. ΔKVR and ΔKCN are shown, where $\Delta KVR \sim \Delta KCN$.



Some measurements of the sides are given.
$RV = 4.6$, $KR = 3$, $KV = 5x - 11$, $NC = 11.5$, and $KC = 3x + 1$.
Maria used the information to determine the length of $\overline{KC}$. Her work is shown.

Maria's Work	
First, Maria found the scale factor.	$\frac{NC}{RV} = \frac{11.5}{4.6} = 2.5$
Second, she used the scale factor to write an equation. Then she solved for x.	$5x - 11 = 2.5(3x + 1)$ $5x - 11 = 7.5x + 2.5$ $-2.5x = 13.5$ $x = -5.4$
Lastly, she substituted x into an equation.	$KC = 3(-5.4) + 1$ $KC = -15.2$

Students may struggle differentiating the two triangles. Consider having students draw two separate triangles labeling the smaller one triangle KRV and the larger one triangle KNC.

6.11.2 Collaboration: Similar Figures (continued)

Maria knows side lengths cannot be negative values, but she does not know what she did wrong.

a. Find and circle Maria's error.

b. Write a note to Maria explaining what her mistake was and how she can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

Consider helping students find the error by asking the following probing questions, as needed.

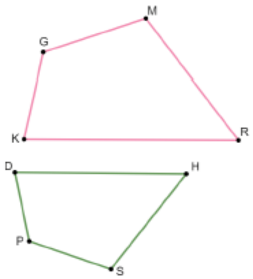
- *Does this scale factor represent a reduction or an enlargement?*
- *Which triangle side lengths will you multiply this scale factor by to determine the lengths of the corresponding side lengths in the similar triangle?*

Component Overview: *Students will practice finding unknown values in similar figures.*

6.11.3 Your Turn!

1. $GMRK$ and $PSHD$ are shown, where $GMRK \sim PSHD$.

   Some measurements of the sides of $GMRK$ and $PSHD$ are given.



$KG = 12$, $GM = 4x - 12$, $MR = 7k - 15$, $DP = 9.6$, $SP = 12.8$, $KR = 30$, and $SH = 4k - 4$.

a. Determine the value of each variable.
   $x =$ 7 $k =$ 5

b. Determine the value of each side.
   $GM =$ 16 $DH =$ 24
   $SH =$ 16

c. What is the perimeter of $GMRK$?
   78

Some measurements of the angles of $GMRK$ and $PSHD$ are given.
$m\angle KGM = (11w - 12)°$, $m\angle GMR = 110°$,
$m\angle GKR = (6z + 18)°$, $m\angle DPS = (8w + 24)°$, and
$m\angle PDH = (8z - 2)°$.

d. Determine the value of each variable.
   $w =$ 12 $z =$ 10

e. Determine the value of each angle.
   $m\angle KGM =$ $120°$ $m\angle DHS =$ $52°$

   $m\angle PDH =$ $78°$

Consider utilizing "area experts" for each of the questions 1a through 1e. If other students have questions, they seek out the area expert as the first resource. The area expert should not give out answers but instead assist students by looking at the other students' work and asking guiding questions.

Consider putting questions the expert might ask on cards to give to each expert. Consider having two experts for 1a, as more questions may arise there.

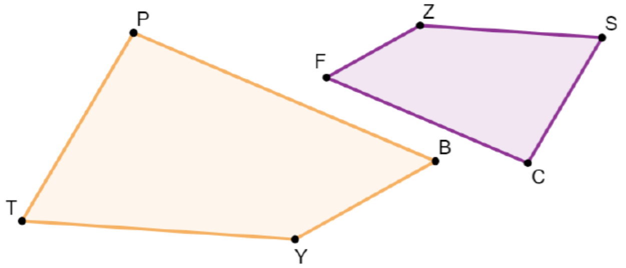
- *1a – What is the scale factor? If your scale factor is greater than 1, did you multiply it by the side length of the smaller figure? If your scale factor is between 0 and 1, did you multiply it by the side length of the larger figure? What steps did you use to solve the equation?*
- *1b – What value did you substitute in for the variable? Did you follow the order of operations?*
- *1c – What is the definition of perimeter? What are the lengths of each side?*
- *1d – Why did you set those expressions equal? What steps did you use to solve the equation?*
- *1e – What value did you substitute in for the variable? Did you follow the order of operations?*

6.11.4 Wrap-Up: Error Analysis

***Component Overview**: Students will analyze two work samples and determine if either are correct. Students must justify their responses. Then students find the length of a side of one of the similar triangles. Students should be allowed to access a calculator to complete 6.11.4 Wrap-Up.*

6.11.4 Wrap-Up: Error Analysis

1. Cheryl and Austin are finding the measure of side SC. They were given the following information about similar quadrilaterals $ZSCF$ and $YTPB$ that are shown. $ZSCF \sim YTPB$, $ZS = 15$, $SC = 5x - 8$, $PT = 3x + 6$, and $YT = 22.5$.



Cheryl's and Austin's equations are shown.

Cheryl's Work	Austin's Work
$15(5x - 8) = 22.5(3x + 6)$	$1.5(5x - 8) = 3x + 6$

a. Determine whose equation, if any, is set up incorrectly. Show your work or write an explanation.
Answers may vary. Cheryl's equation is incorrect. Cheryl should have written the equation $22.5(5x - 8) = 15(3x + 6)$ *because the proportion relating the side lengths of the quadrilaterals could have been written* $\frac{22.5}{15} = \frac{3x+6}{5x-8}$.

b. What is the length of $\overline{SC}$?
$SC = 12$

Use the responses in this activity to determine any remediation needs.

- *If students state Cheryl's work is correct, they may need help on setting up a proportion.*
- *If students state Austin is correct, they may need help understanding what a scale factor is and how it is used.*
- *If students miss question 1b, they may have set up or used the wrong equation or they may need help solving multistep equations.*

6.12 Solving Problems Using Similarity – Part 2

Lesson Introduction

Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.	Learning Target	• I can solve real-world problems involving similarity in two-dimensional figures.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.	N/A

Essential Question	
	• How can similar figures be used to find values in the real world?

Lesson Narrative	
	This lesson applies the concepts of similar figures to figures found in the real world. Students may need to refer to previous lessons to review key concepts of similar figures.

Component Summary				
	6.12.1 Warm-Up (~5 min.)	Reflect on solving real-world problems based on career video *(SE p. 359)*	6.12.3 Guided Practice (10-15 min.)	Determining the length of sides given similar rectangular moving boxes *(SE p. 360)*
	6.12.2 Guided Instruction (10-15 min.)	Determining the length of sides given similar birdhouses *(SE p. 359)*	6.12.4 Try It (10-15 min.)	Determining the length of sides given similar gardens *(SE p. 361)*
	6.12.5 Wrap-Up (~5 min.)	Summarizing the lesson by writing a note to an absent friend *(SE p. 362)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> • Similarity • Congruent • Proportional • Corresponding	• Career Profile Video • (Optional) Highlighters	• 6.12.1 Warm-Up contains a video to accompany the question prompts.

Teacher Tips	Common Misconceptions	Things to Consider
	• Students may use the scale factor incorrectly when writing equations to solve problems. • Students may confuse when to find the area versus the perimeter when solving real-world problems.	• Consider inviting a local architect, builder, designer, or someone from any other profession that uses scaled drawings to speak to the class (virtually or in person).
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk This routine can be implemented for 6.12.2 Guided Instruction. Post each question one at a time for students to quietly think about and solve. When they have signaled that they are ready, select students to share their strategies with the class. Have students share a variety of strategies.	**MA.K12.MTR.7.1: Real-World Contexts** This lesson connects the mathematical concept of similar triangles to everyday experiences. Using corresponding parts, students answer questions about the side lengths, angle measures, perimeters, and areas of given figures.
Differentiate Instruction	Students will be exercising their comprehension of the topics by supplying background knowledge in each component. It would be valuable to prompt students to explore patterns that they may notice throughout each component. Students should be able to identify the critical features and what steps are the most important to being able to solve each question. Consider having students highlight or underline the critical information in each question so it stands out.	

Lesson Notes:

6.12.1 Warm-Up: Biomedical Career

***Component Overview**: Students watch the video and consider where creativity, math, and science can be used to solve problems.*

6.12.1 Warm-Up: Biomedical Career

Liz Williams and Kate Lomas co-founded the Australian-based biomedical technology company, Hemideina. In Kate's studies she found that the insect wētā has an amazing hearing system. Through her studies she invented an implant that can help those with hearing loss live a life without limits.

1. List at least one advantage that Kate may have had as an artist that helped her create her invention.
   Answers will vary. Her interest in wildlife photography led her to the study of animal behavior and a degree in biology.

Both Liz and Kate express excitement knowing that that their skills in science can be used to solve real-world problems.

2. What is one real-world problem that you think creativity, math, and science could be used to solve?
   Answers will very.
   I think creativity, math, and science can be used to solve the worldwide pollution problem.

Display the "Career Profile – Biomedical Career" video for the entire class. After students watch this 2-3 minute video, prompt the students to read and answer the questions in 6.12.1 Warm-Up independently.

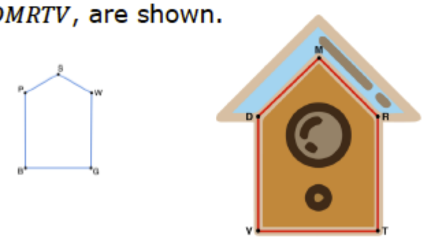
Do you know someone who is in the biomedical field? Consider inviting a guest speaker to join your class (virtually or in person). Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

6.12.2 Guided Instruction: Similar Figures in the Real World

Component Overview*: Given a blueprint of a birdhouse and an image of the birdhouse along with the scale factor from the blueprint to the birdhouse, students will solve for unknown variables and side lengths.*

6.12.2 Guided Instruction: Similar Figures in the Real World

1. Haley built a birdhouse by first drawing a blueprint. She then constructed the birdhouse using the dimensions from the blueprint. Her blueprint, $PSWGB$, and birdhouse, $DMRTV$, are shown.



To build the birdhouse, Haley multiplied each of the dimensions from the blueprint by 4 to get the dimensions of the face of the birdhouse. She used equations to represent the sides.

$PS = (y + 1.2)$ inches (in.), $PB = (3x - 1.5)$ in., $BG = (2w - 6.4)$ in., $VT = (4w - 5.6)$ in., $DV = (4x + 10)$ in., and $DM = (6.2y + 2.6)$ in.

a. What is the value of x?
$x = 2$

b. What is the length of $\overline{DV}$? Include units.
$DV = 18$ *in.*

c. What is the value of y?
$y = 1$

d. What is the length of $\overline{PS}$? Include units.
$PS = 2.2$ *in.*

e. What is the value of w?
$w = 5$

f. What is the length of $\overline{VT}$? Include units.
$VT = 14.4$ *in.*

g. If $DV = RT$ and $DM = MR$, what is the perimeter, in inches, of the face of the birdhouse?
The perimeter is 68 *in.*

Algebra Talk
Consider implementing the instructional routine Algebra Talk for 6.12.2 Guided Instruction.

- *Post one question on the board at a time for students to think about and work individually.*
- *Students indicate when they are done by holding their fist at their chest. If they answered the question using one method, they hold up one finger. If they can solve using another method, they hold up two fingers.*
- *Have one student give their answer and how they found it.*
- *Have another student repeat what the first student said.*
- *Ask "Who thought about this a different way?" and record that approach.*

Repeat this process for each question.

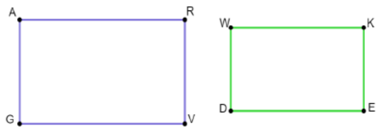
Consider offering students an opportunity to design a blueprint of a real-life object that could be built. Make sure to offer students the flexibility to create a piece of art or a useful gadget.

6.12.3 Guided Practice: Similar Figures in the Real World

***Component Overview**: Students will solve for unknown values in similar figures related to real-life problems.*

6.12.3 Guided Practice: Similar Figures in the Real World

1. Moving-R-Us creates boxes in the shape of a rectangular prism. The company uses a computer program to calculate similar size boxes by using equations. The diagram shows the rectangular bases, $ARVG$ and $WKED$, of two different size boxes, where $WD = 2$ inches (in.), $AR = (7w - 30)$ in., $RV = 2.5$ in., and $WK = (6w - 26)$ in.



   a. What is the value of w?
   $w = 5$

   b. What is the area of $ARVG$? Include units.
   $A = 12.5$ *in*^{2}

   c. What is the perimeter of $WKED$? Include units.
   Perimeter $= 12$ *in.*

Moving-R-Us created a third similar rectangular box. The base, $MHST$, is shown, where $MH = 6.25$ in.

   d. What is the length of $\overline{MT}$? Include units.
   $MT = 3.125$ *in.*

   e. What is the perimeter, in inches, of $MHST$?
   Perimeter $= 18.75$ *in.*

Consider probing students with the following questions after they have read the question stem.

- *In similar figures, what is the relationship of the angle measures?*
- *In similar figures, what is the relationship of the side lengths?*
- *How do you determine the relationship between the corresponding side lengths of the similar figures?*

Common Misconceptions

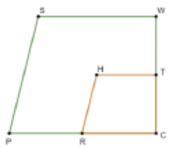
- *If students answered question 1a with 4.18 inches, they either multiplied 0.8, the scale factor of rectangle $ARVG$ to rectangle $WKED$, times WK instead of AR OR they multiplied 1.25, the scale factor of rectangle $WKED$ to rectangle $ARVG$, times AR instead of WK.*
- *If students answered question 1b with 15 inches, they found the perimeter instead of the area.*
- *If students answered question 1c with 8 in.^{2}, they found the area instead of the perimeter.*

6.12.4 Try It: Solving in Real-World Context

***Component Overview**: Students will solve for unknown values in similar figures related to real-life problems.*

6.12.4 Try It: Solving in Real-World Context

1. Gloria, a landscape engineer, created a trapezoidal-shaped garden with a similar trapezoidal-shaped garden inside. A diagram of the gardens, $SWCP$ and $HTCR$, are shown, where $PC = (3y + 4)$ feet (ft), $RC = (y + 3)$ ft, $SP = 8.2$ ft, $SW = (7x - 13)$ ft, $HT = (2x - 2)$ ft, and $HR = 4.1$ ft.



   a. What is the value of x?
   $x = 3$

   b. What is the length of side SW? Include units.
   $SW = 8$ ft.

   c. What is the value of y?
   $y = 2$

   d. What is the length of side RC? Include units.
   $RC = 5$ ft.

   e. If $SW = WC$, what is the perimeter, in feet, of the larger garden?
   Perimeter $= 34.2$ ft.

   After Gloria created the gardens, she decided to check the angle measures of each corner. She had the following information written down: $m\angle SWC = (12z + 6)°$, $m\angle HTC = (7z + 41)°$, $m\angle TCR = 90°$, $m\angle SPC = (5w + 11)°$, and $m\angle HRC = (4w + 24)°$.

   f. What is the value of z?
   $z = 7$

   g. What is the value of w?
   $w = 13$

   h. What is the measure of $\angle RHT$?
   $m\angle RHT = 104°$

Consider prompting students to draw the overlapping trapezoids as two separate figures so that they can visualize how they appear when not overlapping. Students can then use the copies to label the sides and angles.

If students don't remember the sum of the angles of a trapezoid, ask questions that spiral back to decomposing figures into triangles.

6.12.5 Wrap-Up: Cool Down

Component Overview: *Students will reflect on the lesson by writing a note to a friend about what happened in class.*

6.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to let them know what they missed in today's lesson.
   Answers will vary.
   Today we used the scale factor to determine missing side lengths of similar figures found in the real world. We also found the measures of corresponding angles in similar figures.

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons based on what students are taking away from the lesson.

6.13 Similarity in the Real World

Lesson Introduction

<table>
<tr><td>Benchmark Focus</td><td colspan="2">MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.</td><td>Learning Target</td><td colspan="2">I can solve real-world problems involving similarity in two-dimensional figures.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="3">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="3">MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="5">How can similar triangles be used to solve real-world problems?
How can similar figures be used to plan large-scale structures?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">This lesson is a culminating activity for the unit. Students will apply what they learned about similar figures to work through real-world scenarios to solve problems to find missing variables, angles, and side lengths.</td></tr>
<tr><td rowspan="2">Component Summary</td><td>6.13.1 Warm-Up
(~5 min.)</td><td>Reflecting on the usefulness of math
(SE p. 363)</td><td>6.13.2 Activity
(35-40 min.)</td><td colspan="2">Solving real-world problems using similar figures
(SE p. 364)</td></tr>
<tr><td>6.13.3 Wrap-Up
(~5 min.)</td><td colspan="4">Ticket Out the Door analyzing student work to determine a scale factor and find a segment length
(SE p. 365)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms:</u> N/A

<u>Additional Terms</u>:
Similarity
Scale Factor
Proportional
Corresponding
Congruent</td><td colspan="2">Growth Mindset Video
(Optional) Highlighters</td><td>6.13.1 Warm-Up contains a video to accompany the question prompts.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>

- Students may only find the value of the variable without answering the question asked.

</td><td>

- Consider these options for implementing the 6.13.2 Activity.
  - Gallery Walk
  - Trivia game, such as Jeopardy-style or similar, where everybody can answer each question simultaneously
  - Task cards
  - Whiteboard group challenge

</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>

Take Turns

For 6.13.2 Activity, set students up to take turns in the work of the activity by having one person in the group talk through the problem while another group member writes. At each new problem, the two roles swap to other group members. Having one person talk while one person writes requires the talker to give clear and logical directions and requires the writer and other group members to listen carefully, check for understanding, and evaluate the accuracy of their peers' reasoning.

</td><td>

MA.K12.MTR.7.1: Real-World Contexts

Students are expected to navigate a variety of real-world applications of similar figures for 6.13.2 Activity. Each of these questions allows students the opportunity to relate work done throughout this unit to the real world.

</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Students will be exercising their comprehension of the topics by supplying background knowledge for 6.13.2 Activity. It would be valuable to prompt students to explore patterns that they may notice throughout each question when they have completed them. Students should be able to identify the critical features of each question and what steps are the most important to be able to solve each question.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

6.13.1 Warm-Up: With Math I CAN

Component Overview: *Students watch a growth mindset video and then answer three questions independently.*

6.13.1 Warm-Up: With Math I CAN

1. What struggles have you had in mathematics?
   Answers will vary. In math, I have struggled with fractions and negatives.

2. Students often ask teachers, "When will I ever use this?" What is one thing in mathematics you've wondered if you would ever use?
   Answers will vary. I have wondered if I will ever use variables or proofs.

3. Sometimes it isn't that a job requires you to prove a theorem, but to use the logical thinking you learned in proving a theorem. Describe a few careers that you're interested in and what can you do with math.
   Answers will vary. I am interested in going into the medical field. I might use math there to measure and mix doses. I might go into automotive. I could use math there to calculate air and fuel ratios.

Display the "With Math I Can" video for the entire class. After students watch this 1:41 minute video, prompt the students to read and answer the questions in 6.13.1 Warm-Up independently.

6.13.2 Activity: Real-World Problem Gallery Walk

Component Overview: *Students answer each of six provided problems applying what they have learned in the unit to solve real-world problems.*

6.13.2 Activity: Real-World Problem Gallery Walk

Complete each problem by recording your work and answer in the spaces provided.

Problem	Work	Answer
Problem A	$HR = 15$ $\frac{HM}{HR} = \frac{6}{15} = \frac{2}{5}$ $0.9 = \frac{2}{5}x$ $x = 2.25$	*The height of the tennis racket is* 2.25 m.
Problem B	$\frac{36}{12} = \frac{CP}{SC} = 3$ $3(4w - 5) = 3w + 3$ $w = 2$ $PN = 3 \cdot 2 + 3 = 9$	*The maximum depth for swimmers is* 9 *feet.*
Problem C	$\frac{855}{427.5} = 2$ $CM = 2 \cdot 527.5 = 1055$	*The lake is* 1055 *feet across.*
Problem D	$\frac{74}{18.5} = 4$ $4(2x - 6.5) = 3x + 9$ $x = 7$ Perimeter = $PT + BT + HB + HP$ Perimeter = $74 + 32 + (15 \cdot 4) + (3 \cdot 7 + 9)$	*The perimeter of quadrilateral HBTP is* 196 *feet.*
Problem E	$0.5x - 5 = \frac{1}{562} \cdot (22x + 39)$ $x = 11$ $HA = 22 \cdot 11 + 39 = 281$ Perimeter $= 5 \cdot 281 = 1405$	*The Pentagon has a perimeter of* 1405 m.
Problem F	$4(1.5y - 5) = 3y - 5$ $y = 5$ $HP = 3 \cdot 5 - 5 = 10$	*The length of* $\overline{HP}$ *is* 10 ft.

Consider implementing one of the following for this activity.

- *Gallery Walk – Post the problems around the room. Consider printing two sets of each problem to help spread students out. Students will work on one problem at a time before moving to the next problem. Students will record their work and answers in the table.*
- *Task Cards – Make cards of each problem that are then passed around the groups. Students will record their work and answers in the table.*
- *Game – Jeopardy or other game where each problem is displayed for all to see and work on simultaneously.*
- *Whiteboard Group Challenge – Divide the class into groups of three or four. Give each group member a different colored marker and a mini whiteboard. Display one problem at a time. Groups work together to answer the problem, but each member MUST work out the problem on their own whiteboard with their colored marker. When ready, call out for the students with a particular colored marker to stand up, showing their board. All groups whose member has the correct work and answer gets a point.*

Common Misconception
Students may need a reminder not to just solve for variables but to also make sure they are fully answering each question completely.

Consider demonstrating how to use highlighters to organize information. Highlighters can be used to note important information in a problem or to highlight the corresponding angle or side length in the image one color. This provides a visual entry-point.

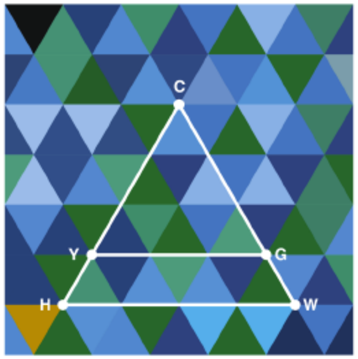
6.13.3 Wrap-Up: Ticket Out the Door

***Component Overview**: Students analyze two work samples to determine who correctly calculated the scale factor between two similar triangles. Students then solve for a side length using the scale factor.*

6.13.3 Wrap-Up: Ticket Out the Door

1. Beth and Trevor are trying to find the lengths of the triangles on the quilt.

   Beth and Trevor know that $\Delta CYG \sim \Delta CHW$, $CG = 21$ inches (in), $YG = (2.5x + 1.5)$ in, $GW = 7$ in, and $HW = (6x - 14)$ in. Their work to find the scale factor is shown.



Beth's Work	Trevor's Work
$\frac{CG}{GW} = \frac{21}{7}$ Scale Factor is 3	$\frac{CG}{CW} = \frac{21}{28}$ Scale Factor is 0.75

   a. Whose work do you agree with?
   I agree with Trevor.

   b. What is the length of $\overline{HW}$?
   $HW = 22$ in.

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.

Remind students to check the validity of their answers by taking a step back and asking, "Does my answer make sense?" Example observation: If I am looking for a side length longer than the one given, my scale factor must be greater than 1.